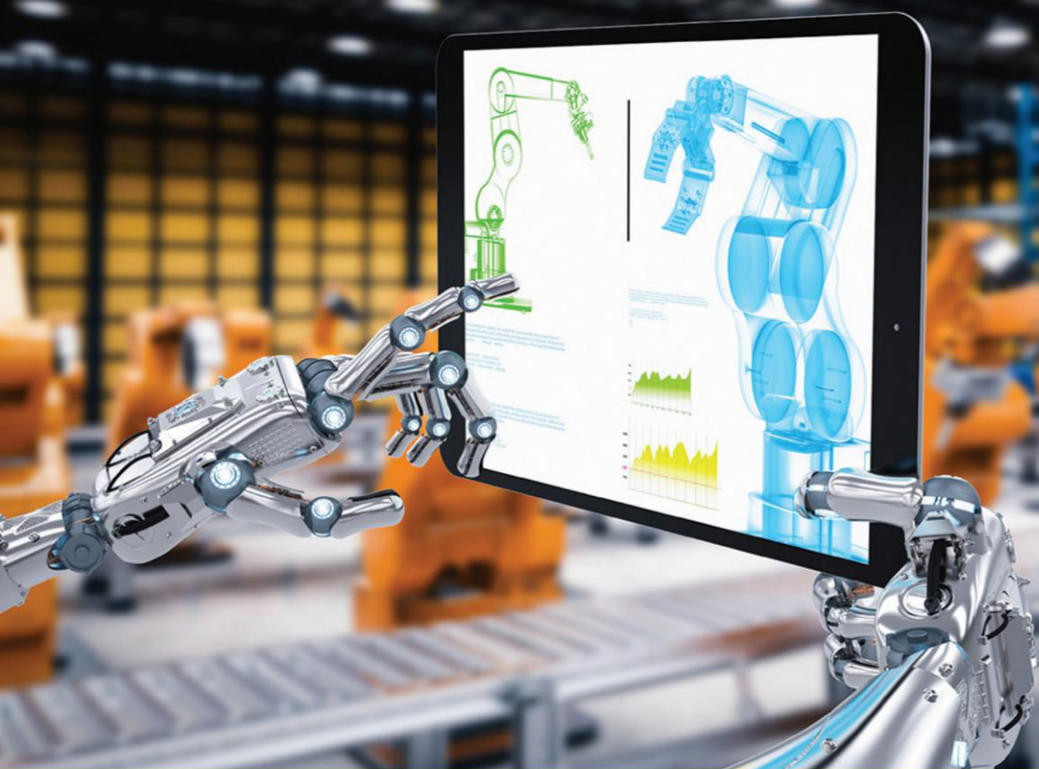


Digital Twin Driven Smart Manufacturing

$$M_{DT} = (PE, VE, S_s, DD, CN)$$



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PREFACE

With the rapid developments of new technologies, such as cloud computing, Internet of Things (IoT), big data analytics, and artificial intelligence (AI), the smart manufacturing era has arrived. Accordingly, various national manufacturing development strategies have been put forward, such as Industry 4.0, Industrial Internet, manufacturing based on cyber–physical system (CPS), and Made in China 2025. It is clear that achieving cyber–physical fusion is a crucial step for the implementation of smart manufacturing. Digital twin (DT) is introduced as a pragmatic way for the cyber–physical fusion. It helps to develop a smarter manufacturing system with higher efficiency and reliability.

In this book, the authors apply the DT to build high-fidelity digital mirror models for manufacturing resources, enhance cyber–physical interactions, and fuse data from both the cyber and physical worlds. The DT is also combined with the new technologies to better accomplish complex tasks. It is expected that this research work could contribute to the pervasive applications of smart manufacturing.

This book is a summary of the authors' studies on DT and its applications in manufacturing. It has 12 chapters, which are classified into three parts. Part 1 includes Chapter 1, Background and Concept of Digital Twin; Chapter 2, Applications of Digital Twin; and Chapter 3, Five-Dimension Digital Twin Modeling and Its Key Technologies; mainly introducing the background and connotation of the DT. Chapter 1 investigates the development background and concept of the DT, then discusses its values from seven aspects. Chapter 2 reviews the state-of-the-art for the DT in both academia and industry, and describes observations from these works. Chapter 3 analyzes new requirements for the DT and proposes an innovative five-dimension DT model. The modeling methods and key technologies are also researched. Based on this, part 2, including Chapter 4, Digital Twin Shop-Floor; Chapter 5, Equipment Energy Consumption Management in Digital Twin Shop-Floor; Chapter 6, Cyber–Physical Fusion in Digital Twin Shop-Floor; and Chapter 7, Digital Twin Driven Prognostics and Health Management; is developed to integrate the DT with manufacturing activities seamlessly. Chapter 4 puts forward the concept of digital twin shop-floor (DTS) and studies its implementation methods and key technologies. Chapter 5 explores the energy consumption

management of equipment in the DTS and highlights potential advantages. Chapter 6 discusses cyber—physical fusion in the DTS from four aspects and studies the related criteria and technologies. Chapter 7 proposes a DT driven prognostics and health management method for complex equipment. In part 3, the DT is fused with the new technologies to achieve greater efficiency in manufacturing operations. Chapter 8, Digital Twin and Cloud, Fog, Edge Computing, describes the combination of the DT with cloud, fog, and edge computing to form a hierarchical structure. Chapter 9, Digital Twin and Big Data, compares the DT with big data and integrates the two concepts to achieve greater intelligence. Chapter 10, Digital Twin and Services, discusses servitization methods for the DT. Chapter 11, Digital Twin and Virtual Reality and Augmented Reality/Mixed Reality, explores the integration of DT with virtual reality and augmented reality/mixed reality to provide more immersive and interactive services to the users. Finally, the DT is discussed as a focused application of the CPS in Chapter 12, Digital Twin, Cyber—Physical System and Internet of Things, and the roles of IoT in the DT are highlighted.

The authors would like to acknowledge the invaluable cooperation and suggestions from many collaborators, both in China and other countries, on their research works on digital twin in smart manufacturing.

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PART 1

**Background and
Connotation**

CHAPTER 1

Background and Concept of Digital Twin

With the developments of new generation information technologies (New IT), such as cloud computing, Internet of Things (IoT), big data and artificial intelligence (AI), a smart manufacturing era is emerging. Accordingly, various national manufacturing development strategies have been put forward, such as Industry 4.0, the Industrial Internet, manufacturing based on cyber—physical system(CPS), and Made in China 2025. It is clear that achieving cyber—physical fusion is a crucial step for the implementation of smart manufacturing. Although CPS provides the framework and mechanism for seamless interaction between physical and cyber components, it mainly belongs to the academic category and focuses on the verification of scientific principles rather than the practical optimization of engineering applications. Therefore, digital twin (DT) is introduced as a pragmatic way for cyber—physical fusion. In this chapter, the background and history of DT are introduced. The different understandings of the DT concept from researchers in academia and practitioners in industry are next reviewed. Some concepts related to DT are also introduced to provide a deeper insight. Finally, the potential benefits of DT are concluded.

1.1 BACKGROUND OF THE DEVELOPMENT OF DIGITAL TWIN

In the past, physical space played the main role in industry. In those days, physical assets close in distance were organized by humans to handle design and manufacturing tasks. However, due to limited personal capabilities and geographical constraints, high efficiency was hard to achieve. Until the 20th century, technologies such as computers, simulation tools, the Internet, and wireless networks have helped to create a parallel virtual space that could virtualize physical assets, organize them with the aid of computing facilities, and enable the cooperation with assets remotely. This has provided a possibility to conduct plans and operations more

efficiently and effectively. Nowadays, along with the developments of new generation information technologies (New IT), such as cloud computing, the Internet of Things (IoT), big data, and artificial intelligence (AI), the roles of the virtual space are becoming increasingly important and interactions between the physical and virtual spaces are more active than ever before. Therefore, seamless integration and fusion between the two spaces will be an inevitable trend, which will create a new potential to improve the current situations and technologies in the fields of design, manufacturing, service, etc.

Various countries are converging on this trend as the next industrial revolution and have proposed related national strategies, such as Industry 4.0, the Industrial Internet, and Made in China 2025. Although the strategies are proposed under different environments, their common objective is to capture the great opportunity brought by the seamless integration and fusion of the physical and virtual spaces for improving the national industry. As cyber–physical system (CPS) is an architecture to converge the physical and virtual spaces, they have attracted a large amount of attention from industrialists, researchers, and practitioners in recent years. CPS aims at embedding communication and computing capacities into physical assets to allow the physical space to be monitored, coordinated, and controlled by the virtual side in real-time to realize a seamless coupling. Many systems related to CPS have emerged rapidly in different fields, such as the cyber–physical production system (CPPS) [1], cloud-based CPS [2], cyber–physical–social system [3], and CPS-based smart agent system [4]. Just as the Internet has transformed how humans interact with each other through interconnecting computer networks, CPS will transform how humans interact with entities through integrating the physical and virtual spaces [5].

Although CPS provides the framework and mechanism for the integration between physical and virtual spaces, it is mainly conceptual aiming at inspiring new scientific foundations and technologies, rather than practical solutions to directly guide the development of engineered systems. As digital twin (DT) is a focused application of the CPS and provides more practical values and implementation details, it is introduced as a pragmatic way for seamless integration and fusion. With respect to the DT, for each physical object, there exists a virtual mirror model with abilities of analyzing, evaluating, optimizing and predicting, etc. The two parts interact with each other and remain synchronous in forming a closed loop. In addition, data from both the physical and virtual sides can be

obtained and fused to generate more comprehensive information. In fact, the DT can be considered as a paradigm of CPS, as it consists of the most important components of CPS (i.e., the physical object, the virtual counterpart, and the connection of the two parts) and becomes more concrete. To date, an increasing number of organizations and companies are aware of the values of the DT, investing a large amount of resources into relevant research and deployment. In future, greater capabilities of the DT will be explored and it will play a growing role in industry for the integration and fusion between the physical and virtual spaces.

1.2 HISTORY OF DIGITAL TWIN

The concept of “twin” was generated from NASA’s Apollo program, which built two real identical space vehicles. One of them was launched into the air space to perform the mission, while the other stayed on Earth, allowing engineers to mirror the conditions of the launched one [6]. According to the Whitepaper written by Grieves [7], the DT was first proposed at his executive course on product lifecycle management (PLM). With technical improvements, the DT was introduced into the aerospace industry by National Aeronautics and Space Administration (NASA) and U.S. Air Force [8]. Accordingly, the space vehicle remaining on Earth could be replaced by a digital mirror model to provide more insights through high-fidelity simulation. Nowadays, the DT has been applied to more fields and has become a much pursued topic.

As shown in Fig. 1.1, the short development history of DT can be divided into three stages.

In the first stage, the concept of DT was proposed by Grieves in 2003 according to the Whitepaper written by him [7]. It was defined in three dimensions, including a physical entity, a digital counterpart, and a connection that ties the two parts together [7]. In 2005, Grieves put forward

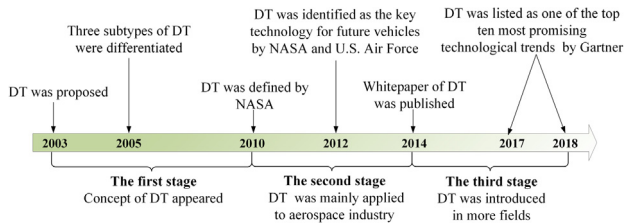


Figure 1.1 History of DT.

another idea that the DT could be classified into three subtypes, including DT prototype, DT instance, and DT aggregate [9]. However, due to technical and cognitive limitations, there were very few related reports in the following 5 years. Fortunately, during this period, the New IT has emerged and developed, which laid the foundations for the future development of the DT.

In 2010, NASA illustrated the definition and function of DT for space vehicles in detail in the *Draft Modeling, Simulation, Information Technology and Processing Roadmap* [10]. In 2011, the U.S. Air Force explored the application of DT in the structural health management of aircrafts [11]. In 2012, NASA and the U.S. Air Force jointly published a paper about the DT, which stated the DT was the key technology for future vehicles [8]. After that, the number of research studies on the DT in aerospace has increased steadily. For example, Tuegel proposed the Airframe DT for design and maintenance, and discussed the challenges for its development [12]. Allaire et al. researched a dynamic data-driven application system, which was described as the implementation foundation of the DT for aerospace vehicles [13]. Reifsnider and Majumdar presented a multidisciplinary physics-based methodology for the DT in fleet management [14].

In 2014, the Whitepaper [7] on DT was published and the three-dimensional structure of DT was widely publicized. Subsequently, the DT was introduced into more fields beyond the aerospace industry, such as automotive [15], oil and gas [16], and healthcare and medicine [17]. In recent years, many established organizations have attached great importance to the DT. Both in 2017 and 2018, Gartner regarded the DT as one of the top 10 most promising technological trends in the next decade [18,19]. In 2017, Lockheed Martin Space Systems Company listed the DT as the top of six prominent technologies for future defense and aerospace [20]. Smart Manufacturing Association of China Association for Science and Technology stated that the DT was one of the top 10 scientific and technological advances for smart manufacturing in the world [21]. Based on the current trend, it can be expected that the DT will experience a rapid development in the next few years.

In China, the DT has attracted wide attention. The authors of this book published the first paper on digital twin shop-floor (DTS) [22] in January 2017 in the *Chinese Journal of Computer Integrated Manufacturing Systems*, and further explored the key technologies, mechanisms, implementation methods, cyber-physical fusion, etc., for the DTS [23,24]. They also published a paper [25] to discuss the DT-driven product design, manufacturing, and

service in PLM, which was first online in March 2017 in the *International Journal of Advanced Manufacturing Technology*. Then in July 2017, the first conference on “Digital Twin and Smart Manufacturing Services” was held at Beihang University, Beijing, China. Over 70 experts and scholars from more than 20 universities in China have participated. Based on discussions at the conference, the participants afterwards published a paper “Digital twin and its potential application exploration” [26] together in the *Chinese Journal of Computer Integrated Manufacturing Systems*, which has generated wide public interest in China. The second conference, which will be held at Zhengzhou University of Light Industry in July 2018, is expected to be an academic feast on the DT.

1.3 CONCEPT OF DIGITAL TWIN

1.3.1 Theoretical Definition of Digital Twin

The concept of DT dates back to 2003 when Grieves first presented it in his PLM course. However, at that time, the DT only had a three-dimensional conceptual framework, including a physical object, a virtual counterpart, and a connection, without concrete description and explanation. In 2010, NASA released a detailed definition of DT for a space vehicle as “an integrated multiphysics, multiscale simulation of a vehicle or system that uses the best available physical models, sensor updates, fleet history, etc., to mirror the life of its corresponding flying twin” [10]. This definition was well received and accepted. Based on it, researchers from different universities and institutes also proposed their own understanding of the DT. The concepts in different research fields are diverse in details, as shown in Table 1.1.

In most definitions, the DT is considered as a virtual representation that interacts with the physical object throughout its lifecycle and provides intelligence for evaluation, optimization, prediction, etc. They focus on both the physical and virtual sides as well as the connection, which are the essential elements in the DT three-dimension framework [7].

Recently, Tao et al. from Beihang University proposed an extended five-dimension definition for the DT, adding DT data and services to the three-dimension version [46]. Compared with the previous concepts, besides the physical–virtual interaction, the newly proposed definition can fuse data from both the physical and virtual aspects using DT data for more comprehensive and accurate information capture. It can also encapsulate the functions of the DT (e.g., detection, judgment, and prediction)

Table 1.1 Theoretical concept of DT

Research field	Universities/institutes	Theoretical concept
PHM	U.S. Air Force Research Laboratory University of California—Los Angeles University of Illinois at Urbana—Champaign University of South Carolina	“An ultrarealistic model of an as-built and maintained aircraft that is explicitly tied to the materials and manufacturing specifications, controls, and processes used to build and maintain a specific airframe” [12,27]
	University of Cincinnati	“Integrating ultrahigh fidelity simulation with an on-board health management system, maintenance history, and historical vehicle and fleet data to mirror the life of a specific flying physical twin to enable significant gains in safety and reliability” [14]
	Belarusian State University of Informatics and Radioelectronics	“A digital model of the real machine that operates in the cloud platform and simulates the health condition with an integrated knowledge from both data-driven analytical algorithms as well as other available physical knowledge” [28]
	Vanderbilt University GE Global Research Center	“A digital replica of real physical installation, which can check the consistency for monitoring data, perform data mining to detect existing and forecast upcoming problems, and use AI knowledge engine to support effective business decisions” [29]
		“A digital model that flies virtually through the same load history as the actual aircraft wing, integrates various uncertainty sources over the entire life of aircraft wing and heterogeneous information, reduces the uncertainty in model parameters, tracks the time- dependent system states using measurement data, and predicts the evolution of damage states if no data is available” [30]
	University of British Columbia Iowa State University Department of National Defence Canada	“A living model that continually adapts to changes in the environment or operation using real-time sensory data and can forecast the future of the corresponding physical assets for predictive maintenance” [31]

(Continued)

Table 1.1 (Continued)

Research field	Universities/institutes	Theoretical concept
Production	University of Stuttgart	“A digital representation that contains all the states and functions of a physical asset and has possibility to collaborate with other digital twins to achieve a holistic intelligence that allows for decentralized selfcontrol” [32]
	Politecnico di Milano	“A virtual and computerized counterpart of a physical system that can exploit a real-time synchronization of the sensed data coming from the field and is deeply linked with Industry 4.0” [33]
	Chalmers University of Technology Fraunhofer-Chalmers Centre for Industrial Mathematics	“A digital copy of a product or a production system, going across the design, preproduction, and production phases and performing real-time optimization” [34]
	Reutlingen University	“A digital copy of a real factory, machine, worker, etc., which is created and can be independently expanded, automatically updated as well as being globally available in real-time” [35]
	Beijing Institute of Technology	“A dynamic model in the virtual world that is fully consistent with its corresponding physical entity in the real world and can simulate its physical counterpart’s characteristics, behavior, life, and performance in a timely fashion” [36]
	University of Applied Sciences of Southern Switzerland	“A digital avatar encompassing CPS data and intelligence, representing structure, semantics, and behavior of the associated CPS, and providing services to mesh the virtual and physical worlds” [37]
	The Pennsylvania State University Indian Institute of Technology	“A rigorous validation for additive manufacturing process, predicting the most important variables that affect the metallurgical structure and properties of the components, and replacing expensive, time-consuming physical experiments with rapid, inexpensive numerical experiments” [38]
PLM	Polytechnic University of Madrid and AIRBUS Group	“A product equivalent digital counterpart that exists along the product lifecycle from conception and design to usage and servicing, knows the product past, current and possible future states, and facilitates the

(Continued)

Table 1.1 (Continued)

Research field	Universities/ institutes	Theoretical concept
Design		development of product related intelligent services” [39]
	Friedrich-Alexander-Universität Erlangen-Nürnberg	“A bidirectional relation between a physical artifact and the set of its virtual models, enabling the efficient execution of product design, manufacturing, servicing, and various other activities throughout the product lifecycle” [40]
	University Paris-Sud	
	Ruhr University of Bochum	“Having a high semantic content and considering both virtual product models as well as feedback data from the physical product along its whole lifecycle” [41]
	Federal University of Rio Grande do Sul	“A set of models from different stages of product lifecycle, such as the system models, functional models, 3D geometric models, multiphysics models, manufacturing models, and usage models, which are kept interacting with each other” [42]
	Technische Universität Berlin	“Consisting of a unique instance of the universal digital master model of an asset, its individual digital shadow, and an intelligent linkage (algorithm, simulation model, correlation, etc.) of the two elements above” [43]
	Fraunhofer Institute Production Systems and Design Technology	
	Guangdong University of Technology	“Realistic product and production process models linking enormous amounts of data to fast simulation and allowing the early and efficient assessment of the consequences, performance, quality of the design decisions on products and production line” [44]
	University of Ottawa	“The cyber layer of CPS, which evolves independently and keeps close integration with the physical layer” [45]

as services for unified management and on-demand usage. The five-dimension DT has been applied to manufacturing [22,23] and prognostics and health management (PHM) for complex equipment [46] with high efficiency.

1.3.2 Digital Twin in the Views of Enterprises

Recently, many enterprises have also been attracted by the DT. In their views, the DT has a series of benefits, such as providing visibility of business, accelerating product development, optimizing the operation process, and improving service quality. Hence, they have actively combined the DT with their companies and products to grasp the great opportunity for promotion of their businesses. The understanding of the DT from enterprises is presented in [Table 1.2](#) and some products/tools related to the DT concept are also presented.

1.3.3 Cores of Digital Twin: Models, Data, Connections, and Services

Although there are various understandings of the DT among academic researchers and industrial practitioners, models, data, connections, and services always play the most important roles in the DT. The four parts can be considered as the core of the DT, and their roles are introduced, respectively.

1. *Models.* An important characteristic of the DT is that for each physical object, a digital companion exists in the virtual world to work with it together across the lifecycle. In essence, the digital companion is composed of a set of models that allow the digitalized physical object to be viewed in three dimensions on the computer. The models are very critical in the DT, since they developed the initial concept “twin” into “digital twin” and provided more deep insights through simulation. Combined with data from sensors, information systems, persons, etc., the models mainly have the following four kinds of functions. First, they can reproduce the real properties, behaviors, and rules of the physical counterpart with high fidelity and thus form an exact digital duplicate that records all changes of the physical one. Second, they can operate autonomously in the virtual space to generate a series of simulated behaviors, which can be considered as the “ideal behaviors” to guide the operation of the physical object. Third, they have the abilities to predict problems on the physical side even before the occurrence of these problems and to develop preventive strategies. Lastly, the models can validate the performances of a product or a system before they are completed.
2. *Data.* Since big data analytics are available for dealing with large and diverse data sets, valuable information can be mined efficiently from

Table 1.2 DT in eyes of enterprises

Company	Industrial concept	Related products/ tools
Siemens	“Including product digital twins for efficient design of new products, production digital twins for manufacturing and production planning, and performance digital twins for capturing, analyzing, and acting on operational data” [47]	Siemens PLM Software
General Electric Company (GE)	“Providing a software representation of a physical asset based on Predix Platform and enabling companies to better understand, predict, and optimize the performance of each unique asset” [48]	Predix platform
Parametric Technology Corporation (PTC)	“A digital representation of a specific asset in the field, including current and past configuration states, taking into account serialized parts, software versions, options, and variants” [49]	PTC Creo Simulate and other analytics tools
Dassault	“A virtual equivalent to a physical product, which can improve manufacturing excellence by allowing people across the enterprise to better collaborate and achieve continuous process improvement” [50]	3D experience platform
Oracle	“An important concept that is going to be strategic to business operations as IoT deployments proliferate through organization” [51]	Oracle IoT cloud
ANSYS	“Combining all the organization’s digital information on a specific product and merging physics-based understanding with analytics” [52]	CAE tools
International Business Machines Corporation (IBM)	“A virtual representation of a physical object or system across its lifecycle, using real-time data to enable understanding, learning, and reasoning” [53]	IBM Watson IoT platform

(Continued)

Table 1.2 (Continued)

Company	Industrial concept	Related products/ tools
System Applications and Products (SAP)	“A live digital representation (or software model) of a connected physical object” [54]	SAP Leonardo platform
Altair	“A capability with which product performance is predicted, optimized, tracked, and measured throughout the product lifecycle” [55]	CAE tools
Microsoft	“Visualizing the physical world, being intelligent, collaborative, interactive and immersive, and providing a method to simulate electronic, mechanical, and combined system outcomes” [56]	Azure IoT Hub Microsoft HoloLens
TIBCO Software	“A software representation of a device that can create efficiencies across the entire product lifecycle” [57]	Project Flogo and TIBCO Graph Database
TwinThread	“A digital representation of any physical asset, including all information about the asset current and historical running conditions” [58]	Software solution
Bsquare	“Digital representations of real-time configuration and state information for physical devices” [59]	Bsquare IoT
Sight Machine	“Offering sets of analytical models that mirror the entire production process, encompassing machines, lines, plants, or supply chains” [60]	Sight Machine Platform
Simulating Critical Infrastructures (SIM-CI)	“A digital copy of a city allowing us to accurately mimic its vital infrastructures” [61]	DT cities platform
Det Norske Veritas and Germanischer Lloyd (DNV GL)	“A digital, virtual representation of an asset, maintained throughout the lifecycle and easily accessible at any time” [62]	DNV GL—Software
PACCAR	“A virtual version of an engine based on sensor data from the real-world versions to manage the maintenance and repair of engines” [15]	DataV system

(Continued)

Table 1.2 (Continued)

Company	Industrial concept	Related products/ tools
Deloitte	“An evolving digital profile of the historical and current behavior of a physical object or process that helps optimize business performance” [63]	IoT solution
Intellectsoft	“A real-time digital representation of a physical object that continuously monitors changes in environment and reports back the updated state in the form of measurements and pictures” [64]	AR solution in construction
Infosys	“Virtual replications of physical products, systems, and processes that are indistinguishable from their real counterparts” [65]	Infosys NiaTM platform
Autodesk	“Spanning both the factory and product, and making use of augmented reality technologies borrowed from media and entertainment software line as well as capabilities from SeeControl, an IoT cloud services platform provider” [66]	Reality capture technology and design software

the data. Based on this, the data can be considered as a driver that provides intelligence to make the constructed DT operate continuously. Data of the DT come from both the physical and virtual spaces, such as product lifecycle data from physical entities, simulated data from digital models, operation data from information systems, and related knowledge. They can comprehensively drive the operations of the DT. For example, in the DT, digital model construction can be driven by rules and constraints mined from data of entities, decisions in the related information systems can be driven by simulated data from the digital models, and operations of the entities can be driven by the pre-defined orders and plans from the models and systems. Without the data, the DT cannot start working, let alone provide further analysis and optimization. With real-time data being generated continuously, more valuable information will be accumulated for the DT.

3. *Connections.* Seamless connections are critical to enable every element in the DT (e.g., entity, model, and information system/tool) interacts with each other. The connections can be classified into three categories: connections within physical space, connections within virtual space, and connections across physical and virtual spaces. Through these connections, entities in the physical space can be interconnected for data exchange, while models and information systems would be linked to form an integrated information flow. In addition, the entities can be deeply intertwined with the corresponding virtual components to achieve a stable closed loop for both sensing and control, which keeps the physical and virtual spaces synchronous and consistent. Based on the connections, interactions among elements in the DT are enabled that can support collaboration to achieve iterative optimizations.
4. *Services.* As users of the DT may have little professional knowledge of related fields, it is essential to encapsulate functions provided by the DT (e.g., evaluation, optimization, prediction, and validation) into standard services for easy and convenient usage. The services are the final presentations of the DT, which are generally expressed in standard formats to describe their inputs, outputs, basic information, function, and quality with user-friendly interfaces. They can be invoked individually or in combination with others according to the demands. For the users, the services are black boxes that can be used without any knowledge of internal mechanisms. They only need to offer the input parameters and demands (e.g., cost, time, quality) to the services and will obtain the results that they require (e.g., evaluation results, optimized parameters, and predictive states). This simplifies the usage process significantly, which is beneficial to expand the application of the DT toward different user groups.

1.4 DIGITAL TWIN AND RELATED CONCEPTS

1.4.1 Digital Twin and Physical/Virtual Space

The DT goes across two spaces, that is, the physical space and the virtual space. It can bridge the two spaces and blurs the boundary between them.

The physical space is a three-dimension space and exists in substance. It contains many entities (e.g., persons, apparatus, infrastructures, and hardware) with physical existence properties such as shape, size, color, structure, mass, speed, etc. Each entity occupies a location, behaves

according to physical laws, and deals with an uncertain environment. The entities have practical functions and can be organized in an orderly way to complete certain tasks considering constraints on time, cost, quality, etc. Data from the entities are analog data with smoothly varying values.

With the development of information technologies, such as the computer technology, modeling and simulation, communication technology, the Internet, cloud computing, IoT, etc., the virtual space has developed and plays a growing role in people's life. It converts analog data from the physical space into discrete digital signals that can be easily stored, processed, and displayed. For each entity, the virtual space builds a three-dimension digital mirror model, which is similar to the physical counterpart not only in geometric appearance, but also in physical properties and behaviors. With virtual reality (VR) and augmented reality (AR), the virtual space becomes immersive, which means it can simulate the physical space through senses and perception, thus realizing full body immersion. Through integrating data from multiple sources, such as human experience, sensor data, and domain knowledge, the virtual space can provide simulation, prediction, optimization, and verification for the physical entities during their lifecycles.

The DT contains both the entity from the physical space and the digital mirror model from the virtual space. Each entity in the DT has a virtual companion that operates and evolves with it all the time. On one hand, the model is built and calibrated to accurately reproduce the properties and behaviors of the entity during its running. On the other hand, the entity is controlled to approximate the predefinitions of the model to the maximum extent. It forms the bidirectional maps between the physical and virtual spaces, which makes the two spaces synchronous and consistent. In addition, as the DT can obtain data from the entity and the virtual counterpart, it can fuse the data from both the physical and virtual spaces to capture more comprehensive and accurate information. The DT can be considered as an efficient way to bridge the two spaces and further achieve cyber—physical fusion.

1.4.2 Digital Twin and Virtual Prototype

A virtual prototype is a three-dimension virtual model that replaces a physical prototype to test and evaluate a product in a computational environment via computer-aided tools, such as CAD and CAE [67]. The virtual prototype can be used as a virtual substitute to detect faults and

predict performances of the physical product early in the design stage, and allows easy modification and manipulation once mistakes or faults are captured. When the virtual prototype is finished, it can be sent to customers to obtain feedback before the physical product is available. Therefore, the virtual prototype can help avoid costly mistakes during production, shorten the design cycle of products, and improve customer involvement, thus leading to fast and cost-effective product development.

The virtual prototype can be considered as the foundation of the DT. They have some similarities as follows: (1) Both of them construct three-dimension virtual models to replace the corresponding physical products, so that activities in the physical space (e.g., product test, evaluation, and validation) are performed in the virtual space to reduce costs in both time and money. (2) Compared with traditional design methods using a physical prototype, their virtual models can produce more insights in different scales to optimize the product in the design stage. (3) Customers can be involved in the design stage to provide experience and comments for product optimization through interacting with the models.

However, the virtual prototype is not identical to the DT. The DT has advantages over the virtual prototype as follows: (1) The virtual prototype mainly serves the product design stage for evaluation and validation. Whereas, the virtual model in the DT accompanies the physical counterpart throughout its lifecycle, from creation to disposal. As the DT integrates abundant actual data of the product in different stages (e.g., manufacturing, sale, and maintenance), it can take into account all that may happen in the physical space and conduct a more comprehensive proofing in the design stage to eliminate potential failures. Furthermore, with the product lifecycle data, the DT has the ability to spark innovation in design. (2) There is little connection between the virtual prototype and the physical product. Whereas, the virtual model in the DT always stays connected with the product during the lifecycle to reflect the actual states and essential insights in real-time, which can feed designers valuable information to improve the product in time for a quick adaptation to the market. (3) The virtual prototype only provides the ideal product that is expected, but the DT can present both the ideal and the actual products. In the DT, the ideal product model is constructed in the design stage, and the actual product model is formed gradually after the design through integrating the actual product data generated in the manufacturing, operation, maintenance, disposal, etc. Through comparing and analyzing the two kinds of product models, their differences can be found intuitively and eliminated.

1.4.3 Digital Twin and PLM

PLM is “managing products across their lifecycles,” and it was defined by Stark as “the business activity of managing, in the most effective way, a company’s products all the way across their lifecycles; from the very first idea for a product all the way through until it is retired and disposed of” [68]. It enables a company to grow revenues by improving innovation, reducing time-to-market for new products, and providing superb service for existing products [68].

In PLM, managing big data generated from the product lifecycle is the most crucial step, which can offer insights and spark creation of fresh products, and provide valuable information to guide the product manufacturing and service. As mentioned in a PLM special report published by Raconteur, if PLM is the lifeblood of innovation, then data is the lifeblood of PLM [69]. Nowadays, although vast amounts of data throughout a product’s lifecycle can be obtained with the aid of advanced sensory technologies, data in different stages is still isolated, fragmented, and stagnant, which leads to low data-use efficiency. Hence, many companies aim at turning PLM data into meaningful insights and potential new revenue streams [69].

The DT can tightly integrate data in the product lifecycle and continuously produce meaningful information for companies. That is one of the reasons why the industry is excited about it. The DT constructs a high-fidelity digital mirror model to run with the physical product together. In different stages of the lifecycle, the virtual duplicate is used to record and combine the real-time data with different natures of the physical product (e.g., picture, tabular, web, video, etc.), and transforms the data into a unified view continuously to eliminate information silos. With the integrated data, the virtual model can present a 360-degree view of the product, and accurately simulate, optimize, predict, and validate the manufacturing processes, usage scenarios, environmental conditions, and other variables for the product in its lifetime. In addition, with the aid of the DT, the companies will be more likely to integrate data into different projects, and motivate innovation and knowledge-sharing.

1.4.4 Digital Twin and Digital Asset/Enterprise/Industry

The DT can be applied to different levels, including single asset, entire enterprise, and even industry, to enhance the existing functions.

A digital asset is a virtual representation in binary formats for the corresponding physical asset (e.g., product, machine, and infrastructure), including digital reports, audible contents, motion pictures, and other relevant digital data. The digital asset is introduced to manage the physical asset in a more efficient way and the appearance of the DT can further enhance the function. (1) With the DT, each physical asset owns one unique digital replica in highly realistic three-dimension form to describe and analyze its performance thoroughly, which can provide real-time insights, early warnings, predictions, and ideas for optimization. (2) The DT can help validate operations of the asset even before it starts to perform, which reduces the potential failures in the actual execution. (3) The DT can integrate data throughout the lifecycle of the asset and spark innovations.

In the digital age, enterprise has moved toward virtual environments, and thus the term “digital enterprise” has appeared. It transforms the enterprises to digital structures by means of information and communication technologies to give a unique opportunity for planning and controlling the businesses [70]. The DT can be applied to all lines of business in digital enterprise and the advantages include (1) enhancing the ability to make proactive and data-driven decisions, increasing efficiency, and avoiding potential issues [71]; (2) shortening new product development cycles and reducing the time to market; and (3) making collaboration between different departments more efficient and effective through providing digital mirror models that can integrate heterogeneous data.

The entire industry, covering the value chain from the product design to production and then to retail, has been gradually digitalized. With the DT, a large amount of data produced in the industry will have an increasing share. As the DT can provide virtual models with unified data and encapsulated services with standardized functions, it can also support the industry convergence, which will redefine industry boundaries by shifting the focus from individual products to cross-industry value experiences [72]. In addition, it can help save natural resources in design, manufacturing, logistics, and usage through better addressing the customers’ needs.

1.4.5 Digital Twin and Digital Thread

“Digital thread” originated in the aircraft industry and was defined by Kraft from the U.S. Air Force as “an extensible, configurable and agency enterprise-level analytical framework that seamlessly expedites the

controlled interplay of authoritative data, information, and knowledge in the enterprise data—information—knowledge systems, based on the Digital System Model template, to inform decision makers throughout a system's lifecycle by providing the capability to access, integrate and transform disparate data into actionable information.” Herein the “Digital System Model” refers to a digital representation of a weapons system [73]. The primal motivation of the digital thread is improving the performance of the future military programs by means of digital technologies.

The DT was developed from early airframe models preserved in the digital thread, aiming at applying the high-fidelity modeling to the manufacturing, operation, and maintenance of the aircraft structural elements to reduce the military spending [74]. The DT is enabled by the digital thread, since all data used in the DT (e.g., models, sensor data, and knowledge) for assessment, analysis, update, etc., are captured from the thread [74].

In recent years, the DT and the digital thread have been gaining attention beyond the aerospace industry. They have been introduced to the product lifecycle management in the civil field to provide the best possible information for design, manufacturing, and service. Toward wider industrial applications, a widespread definition of the digital thread is “a communication framework that allows a connected data flow and integrated view of the product's data throughout its lifecycle across traditionally siloed functional perspectives” [75]. It can converge the digital connectivity goals into a single framework and can deliver the right information to the right place at the right time. Therefore, many large companies are focusing on the digital thread and trying to apply it to fully digitize their supply chain, manufacturing process, and service for dealing with the increasingly competitive pressure. For example, Siemens holds that creating the digital thread can closely integrate each step of the product lifecycle, which dramatically lowers production costs and manufacturing timelines and improves the quality of the product [76]. At GE, the digital thread is considered as a seamless digital connection that bridges the gap between data from different systems and stages for closed-loop optimization [77]. STEP Tools, Inc. tends to build the digital thread for manufacturing to make everyone in the enterprise see how they affect each other in real-time for better decision making, operation, and maintenance [78].

Combined with the digital thread, the DT can obtain the best available data from the entire lifecycle for high-quality mirror and simulation.

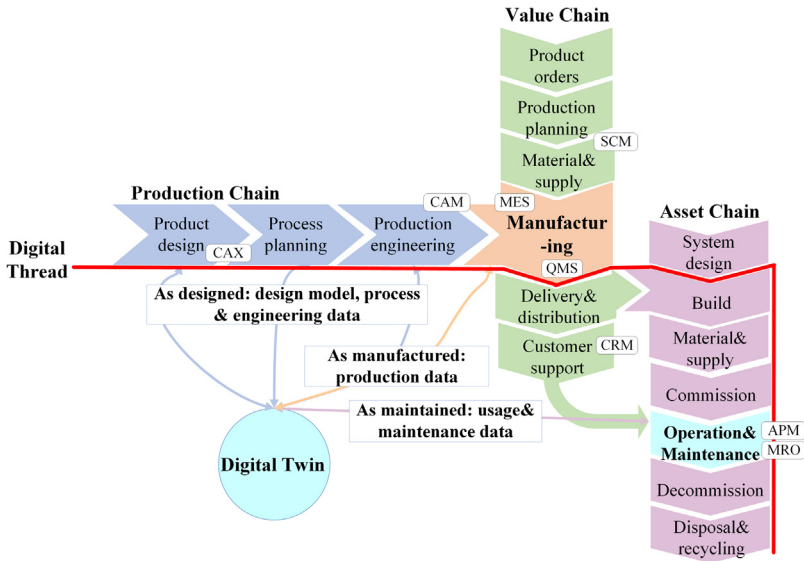


Figure 1.2 DT and digital thread [79].

As shown in Fig. 1.2, the digital thread goes through the product lifecycle and keeps interactive with the DT to drive its operation. The data drawn from the thread come from different stages of the product chain, value chain, and asset chain as well as various information systems, including the design model, process and engineering data, production data and maintenance data, etc. The disparate data are linked and integrated, and can be continuously injected into the DT according to demand. Driven by these data, the DT performs analysis, optimization, and prediction for the physical asset and generates an amount of simulated data, which are then fed back to the thread.

1.4.6 Digital Twin and Digital Shadow

Digital shadow is a data profile that couples with the corresponding entity throughout its lifecycle, and carries all the data and knowledge to reflect the individual's shape and historical, current, and expected future status [43]. Data in the digital shadow are not scattered but stored together in a single electronic document and are actively handled by a dedicated software service or software agent [80]. That makes the data integrated and processed uniformly and efficiently to generate meaningful information. The primary purpose of the digital shadow is supporting decision making to increase the utilization and efficiency of physical resources for a more

sustainable world [80]. In the digital shadow, digital security and digital risks are always crucial problems.

As the DT is a virtual representation carrying data of the physical counterpart throughout the entire lifecycle, it is similar to the concept of the digital shadow. However, the two terms still have differences and the DT is superior to the digital shadow on the following aspects: (1) The DT can provide a high-fidelity digital mirror model to describe the entity intuitively and thoroughly. (2) Based on the model, the physical processes and activities can be validated before execution, which reduces the risk of failure. (3) The model operates in synchronization with the entity and can provide a comparison between the actual and the simulated performances to capture their difference, which is valuable for evaluation, optimization, and prediction. (4) The data in the DT are not only from the physical world, but also from the virtual models, and some data are derived from fusion of data from the two worlds, such as synthesis, statistics, association, clustering, evolution, regression, and generalization. Hence, in the DT, the data is more abundant to generate more accurate and comprehensive information.

1.5 VALUE OF DIGITAL TWIN

DT's value is summarized as follows:

1.5.1 Increasing Visibility

The DT offers high-fidelity three-dimension models, which are updated in real-time to keep consistent with the corresponding physical counterparts. It allows personnel to monitor products or systems in a more direct and clear way, and thus increases visibility.

1.5.2 Reducing Time to Market

As the market changes rapidly, users are always eager for new products. To deal with the market demand, on one hand, the DT can provide designers with a chance to develop insights about how a product behaves even before it is completed thereby eliminating potential failures, and can thus reduce the time to market. On the other hand, the DT can generate a highly realistic virtual substitute before the physical product is finished, which enables the user experience in advance.

1.5.3 Keeping Optimal Operation

Virtual models in the DT are kept connected to the physical counterparts in the physical world. This physical–virtual connection offers a possibility to analyze how the physical entity performs under different conditions in real-time, and thus makes in-time adjustments to ensure it works exactly as planned to keep optimal operation.

1.5.4 Reducing Energy Consumption

In a single DT, as the states of the physical asset can be analyzed in real-time, the degraded components can be replaced in a timely manner to avoid additional energy consumption. In addition, multiple DTs can be connected to enable each DT to negotiate with one another. It makes the corresponding physical counterpart start or close at different periods in an orderly way to pursue a lower total energy consumption.

1.5.5 Reducing Maintenance Cost

Modern assets generate a mass of data that can be studied and assessed to determine when a future problem might occur. The DT can take advantage of these data, and combine data coming from both the physical asset and virtual model to predict a breakdown or maintenance event in advance to largely reduce downtime and maintenance costs.

1.5.6 Increasing User Engagement

For new product design, the users can interact with the high-fidelity virtual models and feedback comments to optimize the design schemes. For existing products, the DT can record and analyze product behaviors in real-time, which can reflect the users' habits to some extent. Based on the data, improvement and innovation on products can be inspired to better serve the users.

1.5.7 Fusing Information Technologies

The DT can fuse different information technologies, including machine learning, simulation, modeling, IoT, cloud computing, big data, etc., to accomplish complex tasks better. For example, the DT possesses the ability to predict what is going to occur through fusing sensor data, simulation results, machine learning algorithms, etc., and the DT can carry out simulations in real-time in combination with cloud computing to obtain powerful computing capacity.

1.6 SUMMARY

As DT is an emerging pragmatic technology for the seamless integration and fusion between the physical and virtual spaces, it has attracted much attention recently. This chapter introduces the background of the DT and explains its importance. The historical development of the DT from 2003 to today is reviewed and the various concepts of the DT from both academia and industry are presented. Based on this, the cores of the DT are concluded as models, data, connections, and services, which play the most significant roles in the DT. Then the related concepts of the DT are introduced, including the physical/virtual space, virtual prototype, PLM, digital asset/enterprise/industry, digital thread, and digital shadow. Finally, the benefits of the DT are discussed. In summary, this chapter intends to help the readers gain a preliminary understanding of the DT, and is the foundation for the study of the following chapters.

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CHAPTER 2

Applications of Digital Twin

As digital twin (DT) is considered as an efficient way to realize the fusion between physical and virtual spaces, both academic researchers and industrial practitioners have applied it in various fields. This chapter summarizes the existing applications of the DT from two perspectives. First, scientific literature about DT applications in different stages of a product's lifecycle is collected and investigated. Second, industrial applications/potential applications of the DT are explored, which are grouped into 11 domains: aerospace, electric power generation, automotive, oil and gas, healthcare and medicine, maritime/shipping, etc. Some of the important exemplary applications are discussed to offer deeper understanding of the DT in industry. As the DT holds great promise to promote the industry development, the future market for the DT is predicted and challenges for DT applications are discussed.

2.1 DIGITAL TWIN IN PRODUCT LIFECYCLE

As shown in Fig. 2.1, the entire product lifecycle can be divided into three stages: design stage, production stage, and service stage. Each stage can be subdivided into more detailed phases. Accordingly, to explore the academic research of the digital twin (DT) in product lifecycle thoroughly, the related works can be classified into the following categories: (1) DT in design stage, (2) DT in production stage, (3) DT in service stage, and (4) DT across multiple stages [104]. Functions and advantages of the DT in each category are explored separately. Based on analysis and statistics of the existing research and published papers, two conclusions are obtained: first, production and prognostics and health management (PHM) are the most popular fields for application of the DT; and second, the DT has attracted the greatest attention in the United States, China, and Europe.

2.1.1 Digital Twin in Design Stage

As the design stage can determine the overall structures and specifications of a product or a system and thereby determine the subsequent stages, it

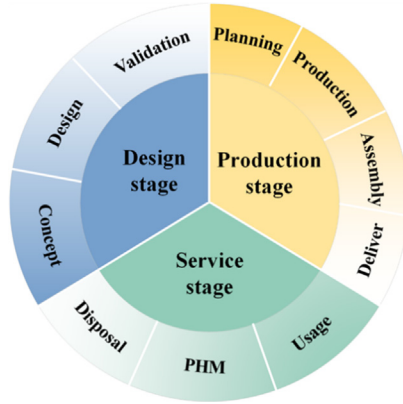


Figure 2.1 Product lifecycle.

has an important meaning in the entire product lifecycle. A good design not only can show advantages in planning, but also has good manufacturability. To improve the traditional design method, current research introduces the DT in the design stage. Some related works are reviewed in this section.

Tao et al. [1] believed the emergence of the DT would change the entire current product design paradigm, including conceptual design, detailed design, and virtual verification. At present, there exist some weaknesses in big data-driven product design processes, cloud manufacturing-based processes, and some other design processes. The DT is expected to overcome these weaknesses, because it can reflect as-designed and as-built products at the same time and eliminate their differences.

Liu et al. [2] proposed a DT-driven methodology for rapid individualized design of an automated flow-shop manufacturing system, which applies physics-based system modeling and distributed semiphysical simulation to generate a digital design of the system at preproduction stage. The authors also proposed an iterative design framework based on the DT for the iterative optimization between static design and dynamic execution, and explored key enabling technologies for the framework [3]. Based on this work, how the production would perform in the real world could be simulated in advance and optimization guides for the design could be provided.

Ferguson et al. [4] demonstrated how the US-based Mechanical Solutions, Inc. (MSI) used a simulation package from Siemens Product

Lifecycle Management (PLM) Software (i.e., STAR-CCM + software) to build DT models for water pumps and to address design challenges. Via this software's powerful simulation capabilities, designers could easily and efficiently predict the performance of the pumps and avoid damaging effects.

Stark et al. [5] introduced a new architecture based on the DT for modularized design of cyber–physical production systems (CPPS). Through integrating existing production resources in a construction kit, an evaluation method for new manufacturing processes was developed to help identify needs for new machines and cyber–physical system (CPS) units. Furthermore, with a virtual prototype, the architecture of the CPPS was validated and optimized.

Damiani et al. [6] applied DT-based simulation to design a real plant that produces anchoring plates for electric motor brake discs. AnyLogic software, which describes layouts, products, operators, machines, and packaging workings, was used to construct the plant and to simulate the operation process. It aims at achieving the lowest level of work in process (WIP), thus reducing the cost of storing.

Mavris et al. [7] explored a design method for an aircraft based on the DT and digital thread, which could efficiently deal with challenges in traditional design methods, including lacking physical data, uncertainty, integrating of multiple disciplines, and data and knowledge management.

With high-fidelity simulation provided by the DT, the design for a product or a production system can be validated to eliminate the potential failures before actual execution. What is more, as the DT can reflect as-designed and as-built products synchronously, it can eliminate differences between the two.

2.1.2 Digital Twin in Production Stage

Production is a complex process, during which material, energy, and information flow exist synchronously and interact with each other. At this stage, researchers have made a lot of efforts, including increasing flexibility of production processes, reducing energy consumption, improving product quality, etc. Recently, some related research has combined the DT with production and obtained performance improvement.

Rosen et al. [8] discussed how the DT could be applied in supporting autonomous production systems. As the DT integrates all available data and knowledge seamlessly, it could combine related data to perform

accurate simulation-based forecasting for the production, which allows high-level flexibility and automaticity.

Tao et al. [9] proposed the concept of digital twin shop-floor (DTS), and explored the conceptual model, interaction mechanism, characteristics, and key technologies for the DTS. Moreover, they discussed the operation mechanism and fusion method for each composition of the DTS to give more insights [10,11]. Based on the above work, an application on equipment energy consumption management in the DTS was presented and potential advantages were analyzed [12].

Coronado et al. [13] designed a framework of a shop-floor DT for production control and optimization. In the framework, raw data collection for physical assets is enabled using MTConnect, a mobile device application, and hypertext transfer protocol. Based on the collected data, the shop-floor DT in virtual space was created to serve the managers, providers, and clients, who take actions that affect the factory. The proposed framework was finally validated in a manufacturing research center, achieving good performance.

Sun et al. [14] proposed two DT models for an assembly line to improve its throughput. A throughput improvement DT model was designed to analyze real-time status of the line and provide optimized schemes for improving the line throughput, while a throughput prediction DT model was developed to predict the future throughput with low model simplicity and high accuracy.

Uhlemann et al. [15] explored the modeling method of the DT for a CPPS, mainly focusing on the concept. Collection and integration of multimodal data were discussed, and a concept for the realization of the CPPS by the DT was presented. Furthermore, they compared the traditional value stream mapping using manual data with the DT combining real-time data to illustrate the advantages of the DT in production [16].

Zhang et al. [17] discussed an architecture of DT-driven smart shop-floor, which included a physical shop-floor layer, network layer, data layer, cyber layer, and application layer. Then in the architecture, a dynamic resource allocation optimization method was designed to reduce the task completion time.

Senington et al. [18] proposed a linked data approach to link raw data from machines to simulation models of the DT for the purposes of analysis and optimization of the manufacturing process. The authors believed that the proposed approach would play an important role in Industry 4.0, as it could support decentralized data storage and processing, modularity

of software for smart factories, interoperability of heterogeneous data sources, and virtualization of factories.

Leng et al. [19] presented a DT-driven manufacturing cyber–physical system (MCPS) for parallel control of shop-floors. By establishing cyber–physical connection via decentralized DT models, various manufacturing resources could be formed as a dynamic autonomous system to create personalized products. The framework, key technologies, and implementation of the MCPS were discussed, respectively.

Vachálek et al. [20] optimized a physical production line based on the DT. As the DT collected real-time data from the actual production process, it could evaluate the production in real-time and satisfy the real-time requirements to simulation parameters. In addition, it has the ability to keep the actual process working as expected.

Banerjee and coworkers [21] described a mechanism to extract and infer knowledge from large-scale production line data based on the DT, which enhances manufacturing process management with reasoning capabilities. As the DT integrates data from manufacturing, maintenance, operations, and operating environments, it could reflect the key behaviors of each asset, system, or process, thus providing intelligence for the mechanism.

Weyer et al. [22] predicted that the DT would represent the next generation simulation, and it would play a critical role in developing CPPS. As the DT was characterized by the synchronization between physical and virtual spaces, it enables human operators to monitor the production process and to make timely adjustments. The methods about how to work with the DT were also discussed.

Parrott and Warshaw [23] discussed the DT model of a manufacturing process from an integrated and holistic aspect and described the DT-enabled closed loop from physical space to virtual space and back to the physical space again.

Ameri and Sabbagh [24] considered the DT as a digital factory that could represent the technological capabilities of manufacturing facilities. The authors described the creation method for the digital factory, including capability extraction, supply chain configuration, and factory digitalization, and applied it to a machine shop.

Brenner and Hummel [25] reported that the DT would lead to a new paradigm of creating a digital shop-floor. They analyzed the hardware and software requirements for implementing the DT in the European School of Business (ESB) Logistics Learning Factory to realize smooth interactions among person, machine, and product.

Zhuang et al. [26] explored a DT-based smart production management and control framework for the complex product assembly shop-floor and discussed the related technologies. The proposed framework was applied to a satellite assembly shop-floor.

Knapp et al. [27] developed a DT for the additive manufacturing process to predict cooling rate, temperature gradient, microhardness, velocity distribution, and solidification parameters. It was shown that the usage of the DT has led to more accurate predictions of cooling and melting rate than the level set method, volume of fluid, and heat conduction models on the basis of the finite element method (FEM).

From these studies, it can be seen that as the DT offers a bridge to link the physical and virtual spaces together, it can make the virtual space mirror the physical practical situations in a timely manner and control behaviors of the physical objects in real-time. Based on these findings, the DT has the abilities to optimize and elevate the production process to a higher level of effectiveness and flexibility.

2.1.3 Digital Twin in Service Stage

In the service stage, PHM is a crucial step that is used to monitor the health of a product, perform diagnosis and prognosis, and provide design rules for maintenance strategies. However, traditional works on the PHM are primarily driven by the physical product and its data, with little connection to its virtual model, which affects the efficiency of the PHM. To change this situation, some researchers have introduced the DT into this field.

Glaessgen and Stargel [28] introduced a DT-based prognosis method for future National Aeronautics and Space Administration (NASA) and U. S. Air Force vehicles, which could tightly combine historical data, fleet data, and sensor data to generate insights for fault prediction. It has the ability to improve conventional methods that are inadequate to meet the increasing needs for real-time monitoring and accurate prediction.

Tuegel et al. [29] applied the DT to predicting the remaining structural life of an aircraft considering multiphysics modeling, multiscale damage modeling, integration of structural FEM and damage models, uncertainty quantification, big database manipulation, and high-resolution structural analysis. They claimed that the DT could enable better management of the aircraft service life.

Zakrajsek and Mall [30] built a DT model to predict the touchdown wear and the probability of failure for aircraft tires. It was shown that the DT model could demonstrate certain superiorities over the traditional models in predicting the probability of failure for the varying sink rate, yaw angle, and speed.

Li et al. [31,32] built a DT model based on a dynamic Bayesian network for the diagnosis and prognosis of an aircraft wing fatigue crack growth. The method integrates physics models and various aleatory and epistemic uncertainty sources in crack growth prediction, and has produced accurate prediction results.

Gockel et al. [33] discussed challenges for implementing an ultrarealistic and cradle-to-grave DT model for an airframe to forecast its structural life. They concluded seven technology gaps that need to be addressed in the future for realizing this envisioned process.

Reifsnider and Majumdar [34] introduced a method to build high-fidelity DT models for fleet management. The method was superior to the traditional methods because it no longer requires damage initiations to perform the fault diagnosis and shows a high sensitivity to fracture development.

Seshadri and Krishnamurthy [35] developed a damage characterization structural health management method for damage diagnostics and prognostics for aircraft structures. The method uses a genetic algorithm-based optimization procedure to predict damage location, size, and orientation and shows high efficiency and accuracy.

Kraft and Kuntzagk [36] built DT models with three layers to support deterioration and failure analysis for engines, and to predict remaining life for their critical parts. The authors validated the method in CFM56-5C, a mature commercial two-spool high bypass engine installed on the Airbus A340-300 and the experiment results presented major improvements.

Lee et al. [37,38] built twin models for manufacturing machines to integrate sensory data, record historical data, and manage hundreds of similar physical machines, to further assess machine conditions, identify the failure mode, and provide selfawareness and selfprediction through simulation.

Tao et al. [39] explored a DT-driven PHM method for complex equipment. A five-dimension modeling method for the DT of the equipment was constructed, and then based on the model, a three-stage workflow was developed to guide the PHM activities. The method was finally validated on a wind turbine.

Bitterwolf et al. [40] trained and evaluated a supervised machine learning method based on the DT for the automated fault detection. The method was validated with a motor control model and could reduce costs associated with unnecessary maintenance or premature breakdowns.

Motivated by the DT concept, Cerrone et al. [41] presented the as-manufactured geometry to predict crack paths on a material test specimen. A specimen DT was modeled and simulated in a commercial finite element software package to deal with the ambiguity problem of the crack paths.

Magargle et al. [42] used a simulation-based DT combining with physics-based modeling techniques for heat monitoring and predictive maintenance for an automotive braking system. The method displayed immense superiority in collecting sensor information to train learning algorithms, which could contribute to accurate diagnosis of the controller in the system and failure detection of the tone ring, a speed sensor for wheel speed measurement.

Liu et al. [43] proposed a system architecture of DT ecosystem, in which the DT fused multisource data, including historical information, technical data, maintenance report, manufacturing report, asset features, etc. to generate knowledge for the proactive maintenance, thus reducing costs and downtimes.

Cai et al. [44] constructed a DT virtual machine based on sensor data integration and information fusion to help engineers conduct diagnosis and prognosis through checking the DT model when a machine breaks down.

Hochhalter et al. [45] combined the DT with sensory materials to overcome shortcomings of traditional methods, which are overly dependent on the empirical data and hence could not respond to uncertainties in a timely manner. Using a nonstandard test specimen as an example, it was proven that the new approach could result in more accurate predictions of repair, inspection, and replacement.

Gabor et al. [46] developed a DT-based architecture that includes four tiers: physical necessity, machine–environment interface, immediate reaction, and planned reaction, to predict the behaviors of CPS during its operation.

Yun et al. [47] proposed a novel architecture of large-scale DT platform including distributed DT cooperation framework, flexible data-centric communication middleware, and a platform-based DT application to develop a reliable advanced driver assistance system.

For these works, on one hand, the DT-based simulation can provide more insights for diagnosis and prognosis. On the other hand, abundant models and simulated data can be obtained and integrated with physical data to generate more comprehensive and valuable information for health condition detection and analysis. Therefore, combined with the DT, the PHM has the potential to be improved to achieve higher efficiency and accuracy.

2.1.4 Digital Twin Across Multiple Stages

In the PLM, due to different processes and characteristics, data from different stages are always isolated, fragmented, and stagnant, which hinders the efficient management of the entire lifecycle. As the DT can integrate data from the lifetime of a product seamlessly, it has been applied into the PLM by some researchers.

Tuegel [48] proposed an airframe DT concept for airframe designing and maintenance, which can reduce model uncertainty with increasing service experience. As modeling of the DT is a core step in the method, some technical challenges in developing the DT model were discussed in detail.

Tao et al. [49] put forward a DT-driven product design, manufacturing, and service method to support the product across its lifetime. The frameworks for the DT applications in the three stages were presented and further validated in bicycle design, drive shaft manufacturing, and power transformer service, respectively. Tao et al. [50] also explored potential applications of the DT in different stages across the product lifetime.

Liau et al. [51] introduced the DT into stages of the injection molding industry, including mold design, mold making, and the injection molding process. The DT helped the entire injection molding process move toward real-time process optimization, process monitoring, and prediction of quality defects through building a closed loop between physical and virtual spaces.

Söderberg et al. [52] developed a DT to secure good geometrical quality in final products. The DT supported locating scheme optimization and statistical variation simulation in the design stage, inspection preparation in preproduction stage, as well as virtual trimming, joining sequence optimization, root cause analysis, etc. in the production stage. They further specified the related strategies in another paper [53].

Talkhestani et al. [54] developed an anchor point method to synchronize cross-domain mechatronic data models of mechatronic components

in the DT during the lifecycle of existing production systems, which could detect changes in the production systems to reduce the occurrence of expensive errors.

Boschert et al. [55] believed that the DT could provide continuous values across the lifecycle of the physical entity and bridge the gaps between different stages. It could shorten the design period by using historical lifecycle experience in design, make accurate simulations in various conditions and analyze root causes of failures in production, and detect faults and give solutions for maintenance in service.

Grieves and Vickers [56] showed how the DT worked across the product lifecycle, including the creation, production, operation, and disposal stages. They deemed that the DT was helpful for transforming clients' thoughts, knowing systems deeply, checking accuracy of predictions, etc. Obstacles and opportunities of the DT were discussed as well.

Schleich et al. [57] proposed a comprehensive reference model based on "skin model shapes" as the DT of a physical product in design and manufacturing stages. The model abstraction, representation, and operations were presented and applied to the geometrical variations management for the product.

Gregory et al. [58] believed that the DT has great significance for self-aware vehicles, as it could provide information on structure states and vehicle dynamics to assess vehicle states and generate both real-time and offline diagnosis and prognosis for structural health to improve maintenance efficiency.

Bohlin et al. [59] launched a study on necessary infrastructure, components, and data flows in the DT for geometry assurance for parts. A closed loop was described with the DT to explain what the parts experienced and relevant methods were also introduced.

Konstantinov et al. [60] believed that the DT was consistent with CPS to some extent and developed a set of virtual engineering tools to support the manufacturing system lifecycle description. The authors used an electric machine assembly station as a case study to verify the efficiency of the presented tools.

Zhuang et al. [61] established an architecture for a product DT; discussed implementation methods for the DT in design, production, and service stages, respectively; presented its potential functions; and finally pointed out the future development trends.

Yu et al. [62] proposed that a product DT could continuously integrate and perfect data and knowledge from the product lifecycle, and

hence realize data management, tracking, and maintaining for product configuration management and improve visualization and transparency during management.

Tao et al. [63] introduced methods of modeling and simulation, dynamic prediction, and evaluation for a complex product during its life-cycle based on the digital thread and the DT. The methods realized the dual mapping between physical and virtual spaces and implemented a closed loop of the product lifecycle.

With the DT, data generated from the product lifecycle can be recorded and integrated into a high-fidelity digital model with a unified encapsulation. Via powerful computing and analysis abilities of the DT, valuable information and knowledge can be mined continuously, forming continuous value streams to support activities in the PLM.

2.1.5 Observations

Based on the above papers, the following observations are obtained:

2.1.5.1 *Production and PHM Are the Most Popular Applied Fields for the DT*

As shown in Fig. 2.2, papers on the production and service (mainly referring to PHM) stages account for the largest proportion, while papers on the design stage are relatively few. It is clear that production and PHM are the most popular applied fields for the DT. There are two reasons to explain the phenomenon.

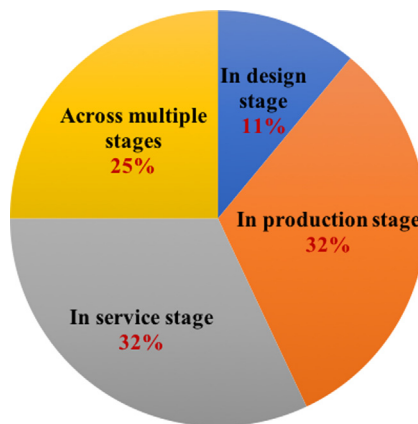


Figure 2.2 Distribution of papers in different stages of PLM.

First, for PHM, as it was the earliest field in which the DT was applied, related works and technologies have been developed over a longer period of time and some exciting results have been obtained, which has inspired more researchers to work in the field. Besides, as NASA and the U.S. Air Force have introduced the DT into the PHM for future aircrafts, the DT is considered as one of the most promising technologies in the field, and thereby has attracted a lot of attention. However, current research on the PHM still has some limitations. For example, the application areas mainly focus on high-value equipment because of the DT's high costs. There will be broader applications as DT technology matures.

Second, for production, as the era of smart manufacturing is coming, many researchers are concentrating on how to realize smart operations in the production process. As the DT has the ability to bridge physical and virtual spaces seamlessly and maintain the two spaces consistently and synchronously, it shows a great potential in real-time process control and optimization, as well as accurate prediction. Hence, researchers focusing on production feel excited about the DT and have been making efforts to deploy it in their research. In the future, with the advancement of technologies and infrastructures in factories, the DT will be further popularized and adopted in production.

2.1.5.2 DT Has Attracted the Most Attention in the United States, China, and Europe

In Fig. 2.3, the first authors of DT related papers from different countries are cited. It can be seen that authors from the United States, China, and Europe represent the largest proportion. This observation can be explained as follows.

The United States, China, and Europe are the world powerhouses in manufacturing. They have proposed strategies at their respective national levels, including Industrial Internet, CPS or CyberManufacturing, Made in China 2025, and Industry 4.0, to encourage and support related researchers and practitioners to explore implementation ways of smart manufacturing. As the DT appears as a pragmatic technology for seamless cyber–physical fusion, which is the core of future manufacturing, researchers in the United States, China, and Europe have enthusiastically placed great emphasis on DT-related studies ahead of other countries. They are currently in the leading position in DT literature.

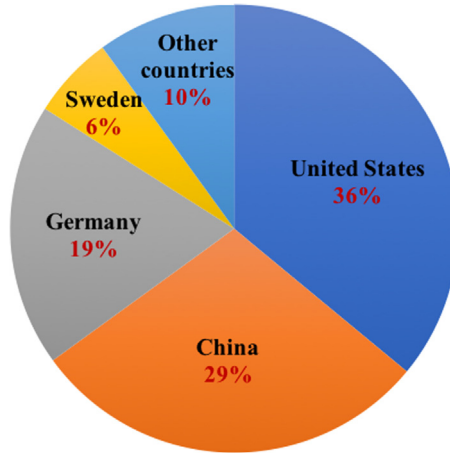


Figure 2.3 Distribution of papers published in different countries.

2.2 DIGITAL TWIN IN INDUSTRIAL APPLICATIONS

DT has been widely applied in industry [104]. In Table 2.1, major industrial applications/potential applications of the DT in 11 domains are listed for easy review. In the following sections, some important examples in each domain are discussed.

2.2.1 Digital Twin in Aerospace

Aerospace is the earliest industrial domain adopting the DT, which was initially launched by NASA and the U.S. Air Force. Subsequently, many large companies, for example, Northrop Grumman, Airbus, Boeing, and General Electric Company (GE), also combined DT technology with their related businesses, including the aircraft maintenance, production, assembly, safety and security management, etc. Some examples are presented as follows.

To predict malfunctions and design maintenance strategies for an aircraft in space, NASA [64] has built a DT to mirror the actual conditions of the aircraft. Through feeding the DT with real-time data, it can optimize performance of the aircraft, predict potential faults in advance, and help engineers on Earth better understand the faults and provide efficient solutions. At present, NASA also attempts to use the DT to ensure crews' safety when the aircraft is out of order.

The U.S. Air Force Research Laboratory (AFRL) [65] has released a plan, that is, Spiral 1, for the development of the DT in aerospace. AFRL

Table 2.1 Major industrial applications/potential applications of DT

Domains	Applications/potential applications
Aerospace	Prognosis and maintenance for spacecraft Catastrophic malfunction analysis for spacecraft Assembly line monitoring Decision optimization for assembly line Safety and security management
Electric power generation	Prognosis and maintenance for wind turbines Health management for electric power plants Power grid planning, operation, and maintenance
Automotive	Design validation Downtime prediction and maintenance Fuel efficiency optimization Car performance test under various conditions
Oil and gas	Remote monitoring and control Asset management Production optimization Equipment failure detection Maintenance planning
Healthcare and medicine	Health monitoring Cardiac research Personalized medicine Medical resource assignment Staff scheduling
Maritime/shipping	Lifecycle management for a ship Choosing best solution for ship building Reducing fuel consumption Fault prediction and maintenance
City management	Real-time monitoring Planning and decision making optimization Support formulation of policy Heating, ventilation, and air conditioning (HVAC) system control
Agriculture	Monitoring olive fly occurrence and spread Identifying diseases, pests, and pesticides for crops Health monitoring of domestic animals Tracking movements of animals Tracking farm machines
Construction	Progress monitoring Work schedule and budget adjustment Resource allocation and waste tracking Safety monitoring for workers Quality assessment for constructions Improving usage rate for equipment
Environmental protection	Water resource management Optimization of forest use and management
Security and emergency	Mitigating existing risks or hazards Security network protection Catastrophic disaster prevention

has signed a \$20 million commercial contract with GE and Northrop Grumman to carry out related research that integrates the most advanced technologies and aims at identifying the differences between entities and models accurately.

Northrop Grumman [66] has applied the DT to the final assembly line for the F-35 fighter to improve manufacturing efficiency and reduce cost. It set up a digital thread-based infrastructure to support decision making on unsatisfactory products, and has improved multiple engineering processes. For example, based on the infrastructure, the decision-making time for processing machining defects of air inlets for F-35 was shortened by 33%, which contributed to Northrop Grumman winning the Defense Manufacturing Technology Achievement Award from the U.S. Department of Defense in 2016.

In 2011, Airbus [66–67] adopted an intelligent location solution from Ubisense (UBI) that connects entities in the A350XWB final assembly line in real-time, makes industrial processes and equipment applications transparent, and grasps distributions of equipment in the plant. Since then, UBI's solutions have continuously been introduced into the assembly lines and aircraft projects in the Airbus, involving the A330, A380, and A400M. Based on this, the Airbus has further deployed RFID on key tools, materials, and parts and finally established a DT for the assembly line that could predict bottlenecks based on models and optimize operation performance.

In terms of DT applications, aerospace is a relatively mature domain and a good amount of research has been conducted. The DT plays a significant role in this domain, as it can perform real-time prediction and high-fidelity validation in the virtual space, which improves the reliability of the aircraft, reduces accidents, and conserves resources.

2.2.2 Digital Twin in Electric Power Generation

The electric power industry is closely related to industrial development and people's daily lives, covering the generation, transmission, distribution, and sale of electric power. It is a basic industry for a country and the DT has been introduced to this industry by some companies.

GE [68,69] has built a digital wind farm to redefine the future of wind power. Through collecting real-time data (e.g., weather, component messages, service reports) continuously, a DT can be built for each wind turbine to optimize equipment maintenance strategy, improve reliability, and

increase annual energy production. A 20% gain in efficiency is expected to be achieved based on the application of the DT. To help customers realize it, GE provides an integrated hardware and software solution comprising wind turbine products and a set of software applications built on the Predix software platform.

Siemens [70] provided a solution for the Finnish power system, which attempted to create a DT for the power grid to be used for planning, operating, and maintaining the facilities. The key benefits brought by the digital grid include: (1) transforming most manual works on simulation to automated works, (2) improving the utilization of data, (3) standardizing interfaces for data, and (4) providing huge possibilities for improving decision making based on big data that are linked to the digital grid.

Beijing BKC Technology Co., Ltd. (BKC) [71], a solution provider for health management of electric power plants in China, has applied DT technology to its businesses. It developed a set of smart solutions, including the visual management for the entire electric power plant, three-dimension online monitoring system for steam turbines, interactive virtual simulation for training, underground pipe network visualization, etc. Through providing visualization information and analysis tools, BKC is helping customers to improve plant operation efficiency, reduce costs, and save energy consumption.

It is clear that the DT has the ability to keep the power grid, electrical power plant, and essential equipment running with higher reliability, which is important for ensuring smooth operations of factories and enterprises, and ensuring normal daily life for people.

2.2.3 Digital Twin in Automotive

The automotive industry involves different stages across the entire lifetime of a motor vehicle, such as design, production, and maintenance. At present, as car structures become more complex, requirements for high-precision testing and maintenance are increasingly higher. Some companies, such as Bsquare, PACCAR, ANSYS, have made efforts on DT applications in the automotive industry.

An Internet of Things (IoT) software company, Bsquare [72,73], established DT models for truck engines or other parts to create repair scenarios for the truck maker PACCAR. Through making full use of real-time data collected using sensors and information in the DataV system, an engine model under specific conditions can be created to support

truck maintenance. Based on this, gains in reducing maintenance time can be obtained in a short time, and a 20% reduction of maintenance time can be achieved.

GE [74] deems that the digital thread and the DT can be used to track a locomotive throughout its lifecycle in the future. With the digital thread and the DT, how a locomotive is designed, configured, built, operated, and serviced can be digitally tracked. Particularly, as health conditions of each component and changes in related variables can be obtained in real-time, engineers can optimize operations of the locomotive from fuel efficiency to unplanned downtime.

International Business Machines Corporation (IBM) [75] believes that the DT will bring huge value for vehicles within a few years. Engineers can construct a DT before building the real vehicle to simulate how it will perform under different conditions, how it will interact with different drivers, what its will be vulnerabilities, etc. They can also see what is happening at any given time, and find out when, why, and how breakdowns occur to reduce the costs and risks of unplanned downtime.

As the DT shows obvious superiority in conditions monitoring, operation optimization, and fault prediction in the automotive industry, it can be considered as a promising technology for the domain.

2.2.4 Digital Twin in Oil and Gas

As oil and gas are usually mined from remote areas with severe environmental conditions, there are many issues in process management. Hence, some companies have introduced the DT into this domain and expect to improve management efficiency.

ARC Advisory Group [76] estimated the unplanned downtime could lead to the loss of process industry revenue and profitability up to \$1 trillion per year. Fortunately, it also pointed out that the emergence of the DT would bring a tremendous opportunity to improve oil and gas maintenance and operations, which would change this situation thoroughly. Benefits from the DT mainly include: (1) helping operating teams better understand risks, (2) helping the creation of an executable plan and work schedule, and (3) helping to identify and manage changes in design.

CIMS RAFFLES [77] integrated the China International Marine Containers (CIMC) Blue Whale #1, a semisubmersible drilling platform, with advanced technologies such as augmented reality (AR)/virtual reality (VR), IoT, virtualization, and data integration to form the DT for the

platform and to establish related management systems. It has realized the visualization display, research and development process display, operation and maintenance management, and has provided an efficient remote means for data collection and state monitoring.

British Petroleum (BP) [78,79] has used the DT to model physical projects, such as new oil fields and associated infrastructure, which is helpful for maintenance planning, equipment identification for decommissioning, equipment installation planning, as well as asset management. The BP has applied the DT-based technologies to a facility in Alaska.

The emergence of the DT provides an efficient way to realize remote monitoring and control, downtime prediction, and risk reduction for related infrastructure, equipment, and process in the oil and gas industry, which can greatly improve the management efficiency.

2.2.5 Digital Twin in Healthcare and Medicine

In the healthcare industry, related companies and organizations provide medical services, fabricate medical equipment, and develop pharmaceuticals to maintain and improve health conditions of people. It is one of the world's largest industries and is closely related to everyone. The DT can be applied to the healthcare industry to accelerate its digitization course.

GE Healthcare Camden Group [80] has turned the DT of a hospital into reality. Through advanced data monitoring and processing, the DT can contribute to bed planning, staff scheduling, operating room allocation, etc., which can maximize care for patients.

Dassault [81,82] conducted the Living Heart Project (LHP) to build a DT for the human heart via biotech sensors and scanning technology. The DT is a personalized full-dimensional model of a heart with electrical and muscular properties, which can mimic the behavior of the real heart. It can support not only various manipulations, such as inserting pacemakers, reversing chambers, cutting any cross-section, and running hypotheticals, but also provide virtual analysis of the heart to deliver care to a cardiac patient before the sickness starts.

Researchers at the University of Amsterdam [83] attempted to build a DT as a digital clone of a human, which can mimic human behaviors accurately, such as breathing, walking, even the fracture of a leg and disease development. The digital copy will contribute significantly to personalized medicine, which thereby will change traditional treatments based on what is best on average for a large group of patients. It also leads to

faster and more efficient introduction of new medication and reduces the use of animals in medical research.

In the healthcare industry, the DT has a bright prospect in health management for human beings, in that it can improve existing medical services and infrastructure to create better patient care on one hand, and develop innovative ways to deal with difficult illness cases on the other hand.

2.2.6 Digital Twin in Maritime/Shipping

The maritime industry has a history over hundreds of years. It involves sea transportation, ocean exploitation, and national defense construction, and has a strong driving effect on the import and export trade. At present, some companies in the industry plan to improve related businesses through advanced technologies. Accordingly, the DT has attracted substantial attention.

GE [84] attempts to work with Military Sealift Command (MSC), a leading provider of ocean transportation for the U.S. Navy and the Department of Defense, to improve performance and mission readiness. To do this, GE will enable high-speed data sampling for critical marine equipment and build a DT for the equipment. Enabled by the Predix platform, real-time data from the physical equipment can be compared with simulated data from the virtual model to detect the difference, which may represent a performance degradation leading to a potential failure. Based on the difference, operators can identify and solve the corresponding problem before it occurs, thus increasing equipment reliability and availability, and reducing maintenance cost.

DNV GL [85] attempts to establish a “virtual sister ship” for the physical one to reduce costs, improve efficiency, and boost safety during the vessel’s lifecycle through enabling collaboration among designers, builders, operators, and others. It has analyzed values brought by the DT from the perspectives of different stakeholders in the maritime industry [86]:

1. *For ship owners:* supporting visualization of ships and subsystems, data qualification and analytics, optimization of performance, internal and external communication, autonomous operations, and decommissioning.
2. *For equipment manufacturers:* providing tools for system integration, performance demonstration, system quality assurance, and additional services.

3. *For authorities*: offering a systematic framework to automatically produce information and high-quality reports.
4. *Universities*: supporting related research and education.
5. *Maritime academies*: training future and existing maritime personnel.
6. *Consultancies*: providing business opportunities.

With the DT, the traditional marine industry will benefit from advanced digital technologies, and predictive analytics for the critical marine equipment and PLM for the entire ship will be application spots in the future.

2.2.7 Digital Twin in City Management

Cities are areas of human settlement, with high population density, complete infrastructure, and buildings. To improve the urban environment and people's quality of life, the DT has been applied to this domain.

The government of Singapore [87] has partnered with Dassault to build a DT to describe everything in the city, from bus stops to buildings, for testing bold new ideas. With the digital copy, massive benefits can be obtained, including: (1) providing a bird's eye view of the city, (2) enabling virtual experiments to optimize related measures, (3) optimizing long-term planning and decision making, and (4) supporting large-scale simulations.

IBM Watson [88] has shown how the DT can be used in a cognitive building to control the heating, ventilation, and air conditioning (HVAC) systems and monitor indoor climate conditions. In the building, there are 600 room sensors (e.g., temperature readings, light sensors, and occupancy monitoring), 250 smart meters, and 169 HVAC machines. Based on the collected data, the DT of the building can be created to help engineers regulate energy consumptions and detect potential use cases, and to provide technicians with guided services, maintenance procedures, and knowledge of climate control.

Cityzenith [89] built a "5D smart world platform" for city management. It integrated digital images with available data sets and overlaid IoT networks with contextual data schema to create a virtual representation for an entire city. With the platform, the infrastructure development process can be digitized and a more integrated digital lifecycle of the city can be enabled. At present, the platform has been used in over 100 cities, including Chicago, London, Doha, etc.

It is obvious that the DT is a valuable technology in city management. With high-fidelity virtualization and simulation provided by the DT,

urban management can be optimized, and more smart cities with intelligence in operation, analysis, and prediction will be built.

2.2.8 Digital Twin in Agriculture

Agriculture is a crucial sector in a country that produces food, fiber, medicinal plants, and other products to sustain and enhance life, through cultivating animals and plants. At present, this sector is under intense pressure due to the growing population in the world. Fortunately, the emergence of the DT may help change the current situation.

Verdouw and Kruize [90] from Wageningen University and Research believe that the DT can act as a central means to manage farms and has the ability to revolutionize agriculture. They have explored potential DT applications in agriculture with different topics, which provides guidelines for related companies. The potential applications mainly include: (1) remotely monitoring dairy cattle to detect their health conditions, (2) recording and identifying plant pests and diseases, (3) monitoring feed silos of livestock farms and optimizing replenishment, (4) tracking machinery fleet in real-time and monitoring energy consumption of crops, (5) monitoring olive fly occurrence and spread, and (6) monitoring apiaries to identify diseases, pest infection, pesticide exposure, and toxicity.

Microsoft [91] also deems that the DT has the power to accelerate agriculture-related businesses, including farm supply, production, harvest, packaging and distribution, and sales and marketing. It will provide solutions for customers to seize the vast opportunities ahead and support sustainability of the agriculture industry.

At present, practical cases showing DT's use in the agricultural sector are relatively few. However, as the DT shows a great potential in remote monitoring, virtualization, predictive analytics, etc., it will be an indispensable technology for the agriculture industry in the future.

2.2.9 Digital Twin in Construction

Construction is the process of erecting buildings or infrastructure facilities, including the stages of design, building, decorating, and management. It provides people with spaces for accommodation and entertainment, and greatly relates to economy, culture, and life. Some companies have tried to introduce the DT into this domain to build cheaper, greener, and more durable buildings.

VEERUM [92] provides DT solutions for large construction sites to predict and resolve issues in virtual space before they impact costs and schedules in reality. Steve Fisher, VEERUM CEO, says, “creating a digital duplicate of a physical construction site enables project teams to be more certain in their decision-making and deliver projects more efficiently, from design to construction”. For example, as the DT provides ongoing progress monitoring of soil type and quantities moved, project teams can be aware of the project progress and surroundings, and make timely necessary changes.

Intellectsoft [93] is exploring the DT applications on construction sites. For the construction, using the DT means people can work with as-built and as-designed models synchronously, which allows detection of the difference between the two parts daily and hourly, to find out the potential problems early in the project. As the DT provides automatic resource allocation monitoring and waste tracking, it also enables a predictive and lean approach for resource management. It can help managers know inappropriate behaviors of workers early to prevent the occurrence of hazards.

PricewaterhouseCoopers (PWC) [94] discussed the digital transformation in the construction industry. It pointed out that the DT could be applied in all kinds of buildings. For example, in an office building setting, the DT can show the number and position of people at any time, which may feed the management system with valuable data for predictive analytics and optimization.

In construction, using the DT can help combine the physical and virtual spaces together to build cheaper, greener, and more durable buildings and to provide more efficient property management for a completed building.

2.2.10 Digital Twin in Environmental Protection

Protecting the natural environment can benefit both the environment and human beings. However, due to the pressures of overconsumption, population, and technology, the environment is deteriorating gradually, which has been impacting social progress and human survival. Nowadays, works from some companies indicate that the DT can play an important role in environmental protection.

Metsä Group, Tieto, and CTRL Reality [95] have developed a “virtual forest,” which can be considered as a part of the concept “digital

forest twin.” The virtual forest can run on VR headsets, mobile devices, and web browsers. With it, users can visit the forest in virtual space to see impacts of different forest management activities, estimate income and cost of each activity, examine properties of a single tree, and predict performances of the forest in the future, which helps to develop sustainable forest management.

Powered by Predix, GE [96] can provide software solutions for water management. It combines disparate data sets with algorithms to identify potential failures on pipelines, enables faster reactive maintenance and increases proactive maintenance to reduce overflow frequency in storm and sewer management, and uses the DT technology to create a powerful digital replica of a customer’s plant to improve the water plant efficiency.

Siemens [97] will focus on the digitalization of water and waste water plants. It provides solutions to cover the entire lifecycle of plants, from planning to operation and maintenance based on the DT, to support water pipes monitoring, comprehensive energy management, and early detection of potential vulnerability and threats, etc.

In environmental protection, the DT has massive value to virtualize related resources (such as water and forests) and improve management efficiencies, thus reducing waste of resources and supporting sustainability of the environment.

2.2.11 Digital Twin in Security and Emergency

Security and emergency measures, which are critical to the protection of assets, infrastructure, and private information against malicious attacks, are commonly involved in various industries. The DT can be introduced to the domain to achieve lower cost and better performance.

VIAKOO [98] has discussed how the DT can protect the security of a company. In the traditional situation, when a company’s devices or machines fail, the company has to provide a maintenance team with an access to its internal system to collect private data for failure identification. This is often a security risk. As the DT has the ability to integrate a large amount of data from the lifecycle of a physical asset and generate valuable information through computing, it can automatically inform the company of potential failures and related maintenance strategies. Therefore, the DT allows the company to keep maintenance in-house without enlisting a permanent team, and thus can avoid the potential security risk.

System Applications and Products (SAP) [99] deems that the DT can prevent disasters through providing PHM services to equipment in harsh environments, such as subsea equipment. On one hand, the DT can replace or complement complex and costly physical inspections with more cost-efficient digital inspections. On the other hand, it can simulate the conditions of the equipment accurately to mimic what is exactly happening and what will happen, to optimize decision making. With the DT, the security of the equipment and related staff can be guaranteed.

In the future, with the DT, the security of physical assets, information, and even human beings will be further enhanced, and the emergency measures will also be optimized.

2.2.12 Observations

At present, the DT has been introduced to various industries. Among them, applications of the DT in aerospace are relatively mature. The reasons mainly include three aspects.

1. Aerospace employs and integrates the most advanced technologies in industry (e.g., high-fidelity modeling and simulation, lifecycle data integration and fusion, and connection and interconnection), which makes it a good candidate for the introduction of DT.
2. As DT applications in aerospace have been developed over a longer period of time than other industries, more researchers and practitioners have dedicated their efforts to the domain, leading to substantial improvements and progress.
3. Aerospace engineering is a special sector. As it is closely related to national defense, it has a high requirement for security and confidentiality, but a relatively low requirement for services. Hence, the initial three-dimension DT concept can exactly satisfy the requirements of this sector. However, for other sectors, it is a crucial step to encapsulate functions provided by the DT into standard services for easy and convenient use, as well as for a more enjoyable user experience. For the new demand, the initial DT concept shows shortcomings that could hinder its acceptance.

As the DT is an emerging technology, developments on related models, technologies, tools, infrastructures, and standards are still lacking. In the future, with the continuous improvement and perfection of the DT, more mature applications can be realized in different industries.

2.3 FUTURE MARKET FOR DIGITAL TWIN

As DT represents the next surge of digitalization and can be considered as the bridge between physical and virtual spaces in Industry 4.0, it has promising commercial prospects in the future. Some organizations have conducted market research for it and provided related data support. Gartner predicted that “by 2021, half of the large industrial companies would use DTs which would result in a 10% improvement in effectiveness for those organizations” [100]. “Markets and Markets” reported that “the DT market was valued at USD1.82 billion in 2016 and is expected to reach USD15.66 billion by 2023, at a compound annual growth rate (CAGR) of 37.87% during the forecast period” [101]. International Data Corporation (IDC) predicted that “by 2020, 30% of G2000 companies would be using data from DTs of IoT-connected products and assets to improve product innovation success rates and organizational productivity, achieving gains of up to 25%” [102]. “Global market insights” pointed out that “as the DT can offer a virtual outlook of any product, at any stage of lifetime, it is expected to be adopted at a significant rate over the forecast period” [103].

The DT market will cover increasingly more fields in the future and penetrate every aspect of society, involving education, transportation, environment, healthcare, food, etc. Hence, companies in various domains have chances to gain benefits from the DT. They can combine the DT with their products to complete product upgrading, and provide better services for customers. The DT can also be applied to company management for better operation through optimizing businesses, conserving energy, reducing time to market, sparking innovation, etc. In each respect, the DT can bring huge economic benefits for the companies and further raise their competition capability.

From the above analysis and data, the authors believe that the DT market will be promising in the coming years and companies that follow the DT trend in a timely fashion can seize the opportunity for rapid development in their fields and reach a favorable position in the market.

2.4 CHALLENGES OF DIGITAL TWIN APPLICATIONS

Challenges of DT applications are summarized in the following seven aspects.

2.4.1 Cognitive and Technical Level of People

First, as the DT is an emerging technology, most people lack a comprehensive understanding and cannot identify its values clearly, which leads to an oversight of the DT. Second, due to psychological and cultural boundaries, people may be limited by old stereotypes and prefer to stay in comfort zones with familiar ways than to be challenged by new approaches. Third, considering the differences in technical levels of operators, some of them do not have sufficient skills (e.g., modeling, simulation and data fusion) to work with the DT.

2.4.2 Technology and Infrastructure

To realize DT applications, many technologies will be involved, such as high-fidelity modeling and simulation, real-time data collection and fusion, flexible data architecture construction, continuous two-way interaction, etc. Some of the technologies have not yet been fully developed and related commercial products are lacking, which has put up barriers to the popularization and application of the DT. In addition, infrastructure (e.g., Internet connectivity, broadband speed, and sensor performance) also need to be improved to serve the corresponding technologies efficiently.

2.4.3 Support Tools

Although many researchers and practitioners feel excited about the DT, when it comes to DT applications, they are not clear about related support tools. Although some big companies (e.g., Siemens, PTC and GE, Microsoft) have launched their respective software and hardware to support the DT applications, these tools are still fragmented and isolated, and have not been adopted widely. Hence, for users, a set of integrated and universal tools are urgently needed.

2.4.4 Standards and Specifications

For DT applications, there is a lack of standard specifications, which should be jointly developed by related researchers, suppliers, and users and should formally describe the standards for models, interfaces, protocols, data, etc. For example, heterogeneous data from different sources need to be involved and processed, but as a standardized format for the data is lacking, it is difficult to complete the data exchange, integration, and fusion to further feed the DT with valuable information.

2.4.5 Cost Control and Management

At present, constructing a complete DT is a costly project. On the one side, building high-fidelity models to precisely and thoroughly simulate entities or processes will be a time-consuming and labor-intensive process. On the other side, realizing real-time collection and processing for a large amount of data needs the support of high-performance hardware and software. Therefore, in most situations, the DT will play an important role in the management of high-value, major products, equipment, or processes. For users, before deploying the DT, taking full consideration of the balance between costs and benefits brought by the DT is necessary but challenging.

2.4.6 Cyber Security and Intellectual Property Rights

Assets and businesses of enterprises can be digitalized and virtualized via the DT. Hence, cyber security is important to protect sensitive information and ensure the success of DT applications. Ample measures must be taken to ensure security, including improving security systems to deal with emerging threats, protecting networked systems in production and office space, and resisting malicious attacks on sensory devices. In addition, with developments of the DT, a set of innovative invention patents will be generated predictably. To establish a fair and orderly investment environment and market for the DT, protecting intellectual property rights efficiently is mandatory but challenging.

2.4.7 Insufficient Development of Digital Twin

As the DT is an emerging technology with a short history, it is still in the infant stage and has many issues, such as confusion in concepts, insufficiency in tools and technologies, and shortage of talents, which bring difficulties to the DT applications. In addition, as a new technology, the DT has uncertainties in its future development and may bring risks in multiple aspects, such as products, markets, businesses, investments, etc. The insufficient development of DT has thus far hindered its progress to some extent.

2.5 SUMMARY

This chapter reviews the DT applications from both academic and industrial perspectives. The conclusion can be drawn easily that the DT can be

used in the entire product lifecycle and various industries, showing highly promising values in monitoring, control, optimization, and prediction of any operations. However, as the DT is an emerging technology, there remain many challenges on technologies, tools, infrastructures, standards, etc. Hence, substantial development is still required for the adoption and application of the DT.

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CHAPTER 3

Five-Dimension Digital Twin Modeling and Its Key Technologies

Digital twin (DT) modeling is the basis for implementing DT in practice. As the concept of DT becomes better developed, many researchers and practitioners have been working on the construction of a DT. However, the initial three-dimension DT proposed by Grieves, including the physical entity, virtual entity, and connection, cannot accommodate the new requirements from applications, technology, modeling objects, etc. In this situation, an extended five-dimension DT, adding DT data and services, is proposed and explored in this chapter to meet the new demands. Based on this, the DT is further classified into three levels toward different applications: unit level, system level, and system of systems (SoS) level. Key technologies for the five-dimension DT are also discussed. Finally, eight rules for guiding future modeling works on the DT are proposed.

3.1 TRADITIONAL THREE-DIMENSION DIGITAL TWIN

The traditional three-dimension DT was first proposed in 2003 by Grieves in his product lifecycle management (PLM) course at the University of Michigan [1]. This can be considered as the origin of the DT. In this section, the three-dimension model is introduced and existing works on its modeling are reviewed.

3.1.1 Three-Dimension Digital Twin

As shown in Fig. 3.1, the three-dimension DT contains three parts [1]: (1) a physical entity in physical space, (2) a virtual entity in virtual space, and (3) a connection of data and information that ties the physical and virtual entities together.

The physical entity, such as an aircraft, radar, and related infrastructure, exists substantially with real functions and capabilities. It can complete specific missions in the physical space and produce actual outputs. The

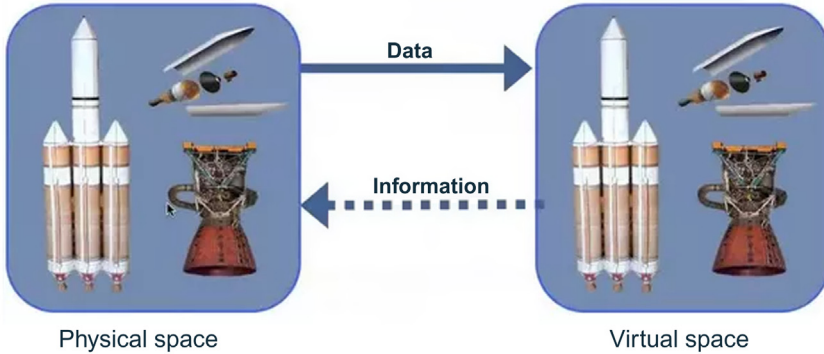


Figure 3.1 Three-dimension DT concept model [1].

virtual entity consists of a set of models to describe the physical counterpart from different perspectives, including geometrical dimensions, physical properties and behaviors, etc. It is a digital mirror that accompanies the physical entity during its lifetime, characterized by real-time synchronization, faithful mapping, and high fidelity. Interactions between the two parts are enabled via the connection. On one side, real-time data from the physical entity are collected and transmitted to the virtual replica for model updating and calibration, and on the other side, valuable information generated from virtual simulations is fed back to the physical space to guide and optimize the corresponding physical entity. The DT forms a closed loop from the physical space to the virtual space and back to the physical space again.

3.1.2 Existing Works on Digital Twin Modeling

Existing works that can support the modeling of the three-dimension DT are briefly reviewed in the following respects:

1. *With respect to physical entity:* To build and maintain the physical entity, works that cover the entire lifespan of the entity, from design to disposal, are needed. In the early stage, these works are mainly conducted via human activity, with low accuracy and efficiency. With technical improvements in computers, communication, automation, mechatronics, etc., activities such as product design [2,3], production [4,5], and prognostics and health management (PHM) [6] are optimized and improved greatly, which enables the construction of more complicated and precise entities. In addition, with the developments of Internet of Things (IoT) technology, smart devices such as wireless sensor

networks (WSNs) [7], radio frequency identification (RFID) [8], embedded systems [9], etc. are becoming more widely available for the physical entities for real-time sensing and control.

2. *With respect to virtual entity:* To reproduce the physical entity, different types of models are built in the virtual space. With respect to geometric models, parameters including shapes, sizes, and assembly relations are modeled to simulate assembly processes [10], machining processes [11], logistics [12] and spatial layouts [13], etc. Combined with virtual reality (VR) and augmented reality (AR), these models can create an immersive environment similar to the physical space for simulation. Based on the geometric models, some researchers further explore the modeling of physical factors (e.g., force, temperature, and vibration), and then conduct a physical parameters changing simulation [14], process plan evaluation [15], and reliability evaluation [16], etc. Through adding the physical parameters, the virtual entity approximates the physical counterpart not only in sense, but also in actual properties. To mimic behaviors of the physical entity under different conditions, methods such as neural network [17], Bayesian [18], finite state machine [19], and Markov chain [20] can be used to build linear or nonlinear relationships between the parameters and the behaviors. Via the behavior models, internal structures of the physical entity can be grasped and some performances can be predicted in advance. Models for rules can also be constructed through processing a large amount of historical data of the physical entity, based on machine learning algorithms [21]. The rule models provide criteria and rules for the physical entity optimization.
3. *With respect to connection:* It is critical to build a connection between the physical and virtual entities for data exchange. Related research mainly covers the real-time communication mechanism [22,23], protocol standardization [24], integration with intranet [25,26], etc. Standardized technologies and devices, such as MTConnect [27], middleware technology [28], and Industrial Internet Hub are also discussed to shield the heterogeneity of protocols and interfaces [29].

Based on the existing modeling foundations and works, some researchers have tried to construct the three-dimension DT. For examples, Majumdar et al. [30] studied synergistic material behaviors for DT modeling based on a multiphysical modeling framework. Smarslok et al. [31] proposed a DT framework for error quantification and confidence assessment, which also included a set of metrics to measure the fidelity of DT

models. DebRoy et al. [32] outlined the technical requirements for building a DT for a 3D printing machine, including the process of heat transfer modeling, solidification modeling, property prediction, residual stress modeling, and distortion modeling. Yun et al. [33] proposed a universal DT platform adopting international standards, which would support the functions of communication, cosimulation, and gateway. Schroeder et al. [34] used a three-stage method based on AutomationML to model attributes related to the DT and enable data exchange between different systems.

The above works have contributed much to the construction of three-dimension DT. However, with the rapid developments of technologies and the increasing growth of demands from applications, new requirements have been put forward on the DT, which are analyzed in the following sections.

3.2 NEW REQUIREMENTS ON DIGITAL TWIN

New requirements on the DT are mainly reflected in the following four aspects: application, technology, modeling object, and modeling method.

3.2.1 From Application Aspect: Requiring Wider Application

When the DT was first introduced, it mainly focused on military products in aerospace engineering, which is a particularly closed industry with restricted access to the Internet. In this situation, the three-dimension DT can basically deal with demands from aerospace.

However, in recent years, as the DT has been introduced to wider application fields, such as automotive, construction, agriculture, health-care, and medicine, it becomes more acceptable in daily products, which are developed in an open and user-oriented environment. Accordingly, new requirements are put forward on the DT, such as high-quality service, user demand mining, and short time to market. Therefore, to adapt to the needs from wider applications, a more general model for the DT has to be built.

3.2.2 From Technology Aspect: Requiring to Embrace New IT

New generation information technologies (New IT), such as IoT, cloud computing, big data, and artificial intelligence (AI), play important roles in industry, enabling smart interconnections and operations. Specifically,

data generated in the physical space can be first accessed and transferred to the virtual space using IoT, then stored and processed via cloud, and finally transformed into valuable information with big data and AI. The information will be encapsulated into services that are then given back to the physical space for on-demand use. It can be seen that New IT has built a closed loop between the physical and virtual spaces.

On the other hand, the DT aims at realizing greater cyber–physical fusion between the two spaces, thus achieving better performances in various industrial applications. Since it is obvious that the closed loop enabled by the New IT lays a major foundation for the fusion, embracing the New IT is an essential step for the implementation of DT. However, due to deficiencies of the traditional three-dimension model, the DT cannot fully take advantage of the New IT to serve itself to the best extent. Therefore, requirements for the seamless integration with New IT are put forward.

3.2.3 From Modeling Object Aspect: Requiring Data and Services

At present, data is an extremely important element for the modeling object, as it contains abundant information that is useful for modeling, simulation, optimization, and prediction. With the developments of New IT, there will be tremendous increases in the volume, richness, and fidelity of data from both the physical and virtual spaces. To obtain more accurate and comprehensive information, the modeling object requires the DT to integrate and fuse the data from the different spaces, such as operational data and environmental data from the physical space, and simulated results and predictive states from the virtual space. However, the three-dimension DT cannot satisfy this requirement.

Service is another important element for the modeling object. Since current users of the DT come from various industries with different technical levels, most of them do not have enough professional knowledge to work with the DT, which becomes an obstacle to the effective interactions between the DT and users. To solve the problem, it is imperative to encapsulate functions of the DT into standardized services with friendly interfaces for easy and on-demand usage, which enables convenient user operations. However, the three-dimension DT does not pay enough attention to servitization of its functions.

Therefore, from the modeling object aspect, requirements on data and services are put forward on the DT.

3.2.4 From Modeling Method Aspect: Requiring High-Fidelity Virtual Modeling

In the DT, to describe the physical entity accurately, a virtual entity that can synchronously integrate multiple variables, scales, and abilities to reproduce a complete mirror for the physical counterpart is required. Although different types of models have been built, as reviewed in [Section 3.1.2](#), some issues still exist. Compared with the works on geometrical modeling and physical modeling, there are relatively few works on behavior modeling and rules modeling. Different types of models need to be integrated and fused to describe the physical entity from multiple perspectives comprehensively. Therefore, a modeling method to produce a complete high-fidelity virtual mirror for the physical entity is required.

3.3 EXTENDED FIVE-DIMENSION DIGITAL TWIN

To satisfy these new requirements on DT modeling, an extended five-dimension DT [\[35\]](#) is proposed, adding DT data and services. It is designed for a wider range of applications and to support data fusion and on-demand usage. Besides, a high-fidelity modeling method is developed to reproduce geometries, physical properties, behaviors, and rules of the physical entity completely and thoroughly.

3.3.1 Five-Dimension Digital Twin

The traditional three-dimension DT defined by Grieves [\[1\]](#) is extended to a five-dimension model, as shown in expression (3.1) [\[35\]](#),

$$M_{DT} = (PE, VE, Ss, DD, CN) \quad (3.1)$$

where PE refers to the physical entity, actually existing in the physical space; VE represents the virtual entity, consisting of a set of models; Ss stands for the services for both PE and VE; DD is the DT data; and CN is the connection that ties different parts of the DT together. According to expression (3.1), the five-dimension DT concept model is built as shown in [Fig. 3.2 \[35\]](#).

Compared with the previous three-dimension definition, the proposed one is more efficient to embrace the broader perspectives of the DT. The newly added “DT data” (i.e., DD) can fuse all related data from both physical and virtual aspects, and then provide more accurate and comprehensive information through data processing. It can be considered as the

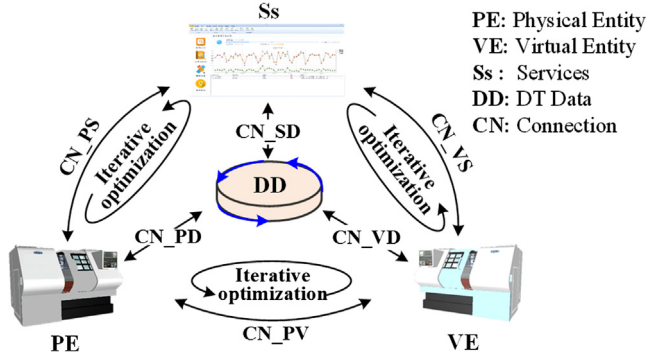


Figure 3.2 Five-dimension DT concept model.

driver of the DT [36]. The “services” (i.e., Ss) enables the various functions in the DT to be standardized and encapsulated, which can help implement easy and on-demand usage. In addition, the Ss also improves the utilization of the functions of the DT, as the corresponding services can be reused individually or in combination with others for different demands.

If the five-dimension DT is compared to a person, its components (i.e., PE, VE, Ss, DD, and CN) can be considered as important tissues or organs. PE plays a role as the “skeleton,” which forms the supporting structure of the DT. VE is the “heart,” as it “pumps” simulation results/strategies to other components. Ss is the “sense organ,” which interacts with users directly. DD is the “blood,” which feeds the DT with valuable information continuously. Accordingly, CN is the “blood vessel,” carrying the data to different components of the DT. Specific modeling methods for each component are described in the following sections.

3.3.2 Physical Entity

Generally, PE consists of various subsystems and sensory devices. The subsystems include the dynamic system, control system, hydraulic system, etc., which can be combined to complete specific tasks. The sensors collect real-time states of the subsystems (e.g., vibration, force, torque, and speed) and their working conditions (e.g., environment parameters, loads, and control orders). To construct and maintain the PE, besides works on design, manufacturing, PHM, etc., real-time data collection based on advanced sensory technologies is also required [35].

3.3.3 Virtual Entity

As a faithful mirror image of the PE, VE is represented by expression (3.2), where G_v , P_v , B_v , and R_v stand for the geometry model, physics model, behavior model, and rule model, respectively [35].

$$VE = (G_v, P_v, B_v, R_v) \quad (3.2)$$

G_v is stereoscopic and describes the geometric parameters of the physical entity, such as shapes, sizes, and assembly relations. G_v can be considered as a basis, providing a geometric container for the other models. The common CAD tools for G_v include SolidWorks, 3D MAX, ProE, AutoCAD, etc.

P_v is given the physical properties (e.g., speed, wear, and force) based on G_v and can reflect the physical phenomena of the entity, such as deformation, delamination, fracture, and corrosion. The finite element method (FEM) is widely applied in this level.

B_v describes the responding mechanisms and behaviors of the entity, such as the state transition, performance degradation and coordination, under the driving factors (e.g., control orders) or disturbances (e.g., human interferences) from the external world. Methods such as the finite state machine, Markov chain, and ontology-based models can be applied to describing the responsive processes.

R_v is a set of rules extracted from the historical data of the PE. These rules (e.g., associations, constraints, and deductions) enable the VE to have the abilities of judgment, evaluation, optimization, and/or prediction. To build R_v , machine learning algorithms such as neural networks, support vector regression, and random forests can be used.

After modeling, with model integration technologies, G_v , P_v , B_v , and R_v are coupled in both functions and structures to form a complete mapping (i.e., VE) for the PE. Then the accuracy and sensitivity of the VE can be tested using verification, validation, and accreditation (VV&A). Although the VE has the ability to mirror the PE faithfully, the degree of fidelity can be adjusted in different applications. For example, for the collision test of the PE, the VE can be built just in the geometric level and physical properties level, and other properties can be omitted.

3.3.4 Services

Ss includes services for both the PE and the VE. It tries to make the PE work as expected through real-time regulation, and sustains high fidelity

of the VE with the PE through model parameters calibration [35]. For the PE, the Ss mainly includes the monitoring service, PHM service, state prediction service, energy consumption optimization service, etc. For the VE, the Ss mainly consists of construction service, calibration service, and test service for models. Each service in the Ss is encapsulated as a quintuple model as in expression (3.3) [35], describing the function, input, output, quality, and state, respectively.

$$Ss = (\text{Function, Input, Output, Quality, State}) \quad (3.3)$$

3.3.5 Digital Twin Data

Digital twin data (DD) is denoted as expression (3.4) [35] with five parts. D_p is the data from the PE, mainly including the operation states and working conditions. D_v refers to the data from the VE and consists of model parameters and model operation data. D_s is the data from the Ss, describing the services encapsulation, combination, invocation, etc. D_k represents the domain knowledge, which is mined from the collected data or obtained from the existing databases and systems. D_f denotes the fusion of D_p , D_v , D_s , and D_k , which can be obtained through data fusion algorithms, such as neural network, D-S algorithm, and Kalman filter.

$$DD = (D_p, D_v, D_s, D_k, D_f) \quad (3.4)$$

3.3.6 Connection

Connection (CN) includes six parts expressed as follows [35]:

$$CN = (CN_SD, CN_PD, CN_VD, CN_PS, CN_VS, CN_PV) \quad (3.5)$$

where CN_SD, CN_PD, CN_VD, CN_PS, CN_VS, and CN_PV denote the connection between Ss and DD, PE and DD, VE, and DD, PE and Ss, VE and Ss, and PE and VE, respectively. Each of them is bidirectional and the delivered data of each connection (denoted as CN_XX) are modeled as follows [35]:

$$CN_XX = (\text{Data sources, Unit, Value, Scope, Sampling interval}) \quad (3.6)$$

3.4 APPLICATION-ORIENTED THREE-LEVEL DIGITAL TWINS

In the physical space, entities can be classified into three levels: a unit, a system, and a system of systems (SoS). Taking a shop-floor as an

example, a single person, a machine, a tool, or a product in it can be seen as a unit, which is the smallest element participating in production. These units can work together to form a system, that is, a production line, in which the units interact with each other and share a common production goal. The entire shop-floor can be considered as a SoS, which is a more complex system, consisting of multiple production lines and different flows (e.g., information flow, energy flow, and material flow). Accordingly, to describe the above entities in different levels, the DTs fall into three types, including the unit-level DT, system-level DT, and SoS-level DT [38].

From the perspective of DT applications, the classification of DTs is also necessary. For example, for the monitoring, fault prediction, and maintenance of a single piece of equipment, the unit-level DT is enough to address the corresponding issues. But for applications that need to deal with different units, such as scheduling, progress control, and product quality control, the system-level DT is required. Furthermore, if the optimization and coordination of the entire shop-floor is required, the SoS-level DT needs to be used to cope with the complex relations, combinations, and associations.

Therefore, in the following sections, the authors discuss the three levels of the DT, respectively, that is, the unit-level DT, system-level DT, and SoS-level DT.

3.4.1 Unit-Level Digital Twin

A unit-level DT is a minimum but independent individual, which cannot be further divided. It contains a basic closed loop of data between the physical and virtual spaces. With the abilities of sensing and computing, the unit-level DT supports state perception, computational analysis, and selfcontrol.

The unit-level DT can be built based on the five-dimension DT, involving the PE, VE, Ss, DD, and CN. In the unit cell, the PE refers to a single entity (e.g., a person, a machine, or a product). The VE consists of models with geometry, physical properties, behaviors, and rules to describe the single entity thoroughly. The Ss mainly provides services for the unit-level PE and VE, such as the monitoring service and maintenance service for a single machine. The DD only contains data of the unit, involving both physical and virtual aspects. The CN connects the above components together.

3.4.2 System-Level Digital Twin

The unit-level DT can be connected with each other through certain relationships to form a system-level DT, which is a regularly interactive or interdependent group of units. The system-level DT is characterized by selforganization, selfconfiguration, selfdetermination, and selfoptimization through interconnection of units, data interoperation, collaborative control, etc.

In the system-level DT, connections between different PEs, VEs, Ss, and DD in unit-level DTs need to be created to enable the data exchange processes, and thus form a collaborative network with functionality and capacity. Meanwhile, besides existing services for the units, new services will be added to serve the entire system, such as scheduling service, progress control service, and process optimization service toward multiple units. Data from different units should also be integrated to create a bigger data set, which can fully fill the information gaps among the units.

3.4.3 System of Systems-Level Digital Twin

The system-level DTs can be combined to create a SoS-level DT, which offers stronger functionality and performance than a simple sum of the constituent systems. The SoS-level DT is characterized by comprehensive information perception, in-depth analysis, precise decision-making and global optimization, and is driven by data aggregation, distributed computing, model interoperation, etc.

The SoS-level DT can be constructed through linking key unit-level DTs in each system level, thus forming a more complex system that allows for synergism of data and intelligence. As the SoS-level DT enables the integration and interoperability of system-level DTs, it can pursue optimizations and predictions from a global perspective, given global resources and constraints. In this level, both services and data are much more abundant than that of the system-level DT, due to the increasing physical and virtual entities, as well as a large amount of dependencies, relationships, and interactions among them.

3.5 KEY TECHNOLOGIES FOR DIGITAL TWIN MODELING

As shown in [Fig. 3.3](#), based on the five-dimension DT, key technologies for modeling of each dimension (i.e., PE, VE, Ss, DD, and CN) are presented.

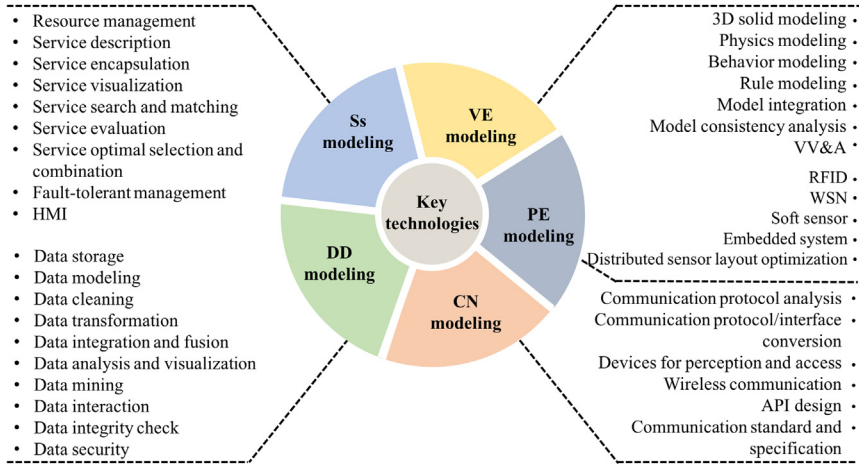


Figure 3.3 Key technologies for five-dimension DT modeling.

3.5.1 Key Technologies for Physical Entity Modeling

To build and maintain the PE, besides technologies in design, manufacturing, PHM, etc., real-time data collection is also essential as it enables state capture for the PE during its lifetime. Some relevant technologies are listed as follows:

- *RFID*: It can be used to track and identify the PE.
- *WSN*: It consists of spatially distributed autonomous sensors, which can collect the PE's operational states and working conditions, and organize the collected data at a central location.
- *Soft sensor*: It uses knowledge and algorithms to calculate the PE's partial parameters, which are difficult to measure with hardware, thus reducing measurement cost and enlarging measurable scope.
- *Embedded system*: It can be embedded as a part of the PE for monitoring and control.
- *Distributed sensor layout optimization*: It can reduce redundant information, improve utilization rate of sensors, and keep the stability of collected data.

3.5.2 Key Technologies for Virtual Entity Modeling

Since models in the VE should be built in different aspects, and then integrated to form a complete mirror of the PE, key technologies for VE modeling mainly include the following aspects.

- *Three-dimension solid modeling*: It describes the geometric appearance of the PE with a high sense of realism and immersion.
- *Physics modeling*: It reflects the physical details of the PE through multi-physics simulation, involving simulations for fluid, structure, aeroacoustics, thermodynamics, etc.
- *Behavior modeling*: Behavior features of the PE need to be extracted under different conditions, and modeled by mathematical equations, graphs, networks, etc.
- *Rule modeling*: Rules should be mined from a large amount of historical data and knowledge through machine learning algorithms.
- *Model integration*: Different models are kept interactive and integrated in both functions and structures, forming a unified representation.
- *Model consistency analysis*: On one hand, it keeps the PE and the VE consistent, which means the difference between the two parts is at a tolerable scope. On the other hand, it makes related parameters in different models of the VE compatible, which avoids a contradictory description for the PE.
- *VV&A*: It conducts verification, validation, and accreditation activities for the models, which keeps precision, sensitivity, and robustness of the VE.

3.5.3 Key Technologies for Services Modeling

Key technologies to create and maintain services in the Ss mainly include the following aspects. Technical details have been explored in the books *Theory and Practice: Optimal Resource Service Allocation in Manufacturing Grid* [36] and *Resource Services Management in Manufacturing Grid System* [37].

- *Resource management*: It efficiently collects, classifies, and organizes resources in the DT, such as model resources, algorithm resources, data resources, and tool resources.
- *Service description*: It transforms resources into services through digital descriptions, which describe the basic information, functionality, and nonfunctionality of the services.
- *Service encapsulation*: It encapsulates services with standard inputs and outputs.
- *Service visualization*: It creates a software-based representation of services, with a high-quality visual display.
- *Service search and matching*: It searches for suitable services based on similarity matching algorithms.

- *Service evaluation*: Services are evaluated using indicators of time, cost, reliability, function similarity, security, maintainability, satisfaction, etc.
- *Service optimal selection and combination*: Optimal services are selected from candidates and combined to form composite services for corresponding demands from the PE and the VE.
- *Fault-tolerant management*: It tests and eliminates potential failures caused by services, demands, networks, applications, etc.
- *Human-machine interface (HMI)*: It enables direct interactions between services and users.

3.5.4 Key Technologies for Digital Twin Data Modeling

As the DD includes data from both physical and virtual sides, it has characteristics of multisources, heterogeneity, and massiveness. According to the characteristics, the following key technologies are mainly involved for DD modeling.

- *Data storage*: It stores data from the PE, VE, and Ss in distributed nodes, and keeps the data available and accessible.
- *Data modeling*: It builds an abstract framework to describe the data structures, operations, and constraints.
- *Data cleaning*: It deals with “dirty” data, such as duplicate data, incorrect data, incomplete data, and delayed data.
- *Data transformation*: It transforms different data formats (e.g., text, table, picture, video) into a unified form.
- *Data integration and fusion*: Data from the PE, VE, and Ss are integrated and fused to generate more consistent, accurate, and comprehensive information.
- *Data analysis and visualization*: Based on the integrated and fused data, data analysis can be conducted in different dimensions (e.g., time-domain analysis and frequency-domain analysis) and the analysis result can be visualized.
- *Data mining*: It can discover patterns in large data sets through statistical methods or machine learning algorithms.
- *Data interaction*: It allows data to be shared between the PE, VE, and Ss.
- *Data integrity check*: It keeps data consistent, correct, effective, and compatible.
- *Data security*: It protects privacy data and prevents unauthorized access to the DD.

3.5.5 Key Technologies for Connection Modeling

As the CN enables interactions among PE, VE, Ss, and DD, communication technologies are critical. Some key technologies are listed as follows:

- *Communication protocol analysis*: It allows two or more entities in the DT to transmit information to each other.
- *Communication protocol/interface conversion*: It transforms various communication protocols/interfaces into a unit form, simplifying the complexity of the communication system.
- *Devices for perception and access*: It develops a standard device for a unified conversion of protocols and interfaces.
- *Wireless communication*: It can connect entities in the DT wirelessly, thus improving flexibility in data transmission.
- *Application Programming Interface (API) design*: It realizes the communication between different software systems and models in the virtual space.
- *Communication standard and specification*: It provides standard and specification for communications among PE, VE, Ss, and DD.

3.6 EIGHT RULES FOR DIGITAL TWIN MODELING

3.6.1 Data and Knowledge Based

Data and knowledge are foundations for the DT modeling. On one side, the existing knowledge provides basic rules that should be followed during the modeling process. On the other side, real-time data need to be injected into the DT continuously, to feed the DT with new knowledge and to calibrate the initial virtual models, services, and data structures continuously.

3.6.2 Modularization

DT modeling is a complex project, which involves heterogeneous hardware, multifunctional models, multisource data, various services, etc. To improve the modeling efficiency, modularization is effective as it can separate and recombine compositions of the DT with benefits of flexibility and reusability. For example, dividing the DT into three levels (i.e., unit-, system-, and SoS-level) is a kind of method for modularization, which makes the DT decomposable and recomposable toward different applications.

3.6.3 Light Weight

To keep high fidelity with the physical entity and provide previous services, the DT consists of multiple models with complicated structures and mechanisms, which creates an obstacle for real-time operation and simulation. Therefore, a light weight model is essential for the DT to achieve a shorter transmission time, faster running speed, and better real-time performance.

3.6.4 Hierarchy

In the DT, if all the resources including data, knowledge, models, services, etc. are centrally stored and processed, it will cause a large burden to the transmission, computation usage, and maintenance. Accordingly, a hierarchical structure should be introduced to the DT. By using the edge-fog cloud structure, resources with different characteristics in the DT can be distributed in different layers for processing and analysis, thus satisfying the requirements on timeliness and efficiency.

3.6.5 Standardization

Service software, databases, and models in the DT may use different encoding strategies, interfaces, and communication protocols, which leads to difficulties in module integration, unified data exchange, information identification and maintenance, etc. It also hinders stable interaction and cooperation among different entities in the DT. Therefore, standardization should be focused on solving the above problems.

3.6.6 Servitization

As users of the DT may have little professional knowledge in the related fields, such as modeling, simulation, data integration, and fusion, it is essential to encapsulate functions provided by the DT (e.g., evaluation, optimization, prediction, and validation) into standard services for easy and convenient usage. To realize that, servitization is an essential step in the DT modeling.

3.6.7 Openness and Scalability

The DT should be open to integrate with various resources (e.g., models, services, and tools) from the external environment, which makes the DT have better compatibility to work with different systems, and enhance functions of the DT through taking advantage of capabilities from other

platforms. Besides, the DT should also be scalable, which enables functional extension through adding new models.

3.6.8 Robustness

As the DT is mainly applied to the industrial environment, which is characterized by uncertainty, dynamism, and complexity, it should be built with good robustness to deal with unpredictable changes. For example, when the working conditions of the physical equipment vary, the DT should have the ability to alter its initial configuration to adapt to the current situation.

3.7 SUMMARY

With growing social demands and unceasing technical developments, new requirements are put forward for the traditional three-dimension DT, including requirements from applications, technologies, modeling objects, and modeling methods. To meet the new needs, this chapter proposes an innovative DT model that extends the traditional three-dimension DT to the five-dimension DT, and discusses modeling methods for each dimension. Based on this, the DT is further classified into three levels toward different applications, that is, unit level, system level, and SoS level, respectively. Key technologies and eight rules are also proposed to provide guidelines for future works on DT modeling.

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PART 2

Digital Twin Driven Smart Manufacturing

CHAPTER 4

Digital Twin Shop-Floor

One of the specific challenges to achieve smart manufacturing is to converge the manufacturing physical space and the virtual space, for realizing smart operations in the manufacturing process. In this context, the shop-floor, as a basic unit of manufacturing, should achieve the interaction and fusion between physical and virtual spaces, which is not only the imperative demand of smart manufacturing but also an evolving trend. Accordingly, the novel concept of digital twin shop-floor (DTS) based on digital twin (DT) is introduced. In this chapter, the evolution path of shop-floor is analyzed first, then the concept of DTS and its four key components are discussed, including physical shop-floor (PS), virtual shop-floor (VS), shop-floor service system (SSS), and shop-floor digital twin data (SDTD). The operation process and interaction mechanism for DTS are studied and the characteristics, key technologies, as well as challenges ahead are investigated.

4.1 EVOLUTION PATH OF SHOP-FLOOR

Nowadays, various countries have proposed different manufacturing strategies (e.g., Industry 4.0, Industrial Internet, Made in China 2025, service-oriented manufacturing [1,2], and cloud manufacturing [3]) to prepare for the next industrial revolution. Although these strategies are under different environments, their common objective is to achieve smart manufacturing that satisfies the demands of socialization [4], personalization, servitization, intelligence, and greenization. They put forward requirements for reaching smart interconnection and interoperability between the physical space and the virtual space. To meet these requirements, realizing seamless integration and fusion of the two spaces is crucial. Since the shop-floor is the basic unit of manufacturing, which generates finished products for customers, achieving integration and fusion of the shop-floor becomes an imperative [6].

From the perspective of development, achieving the integration and fusion of the physical and virtual spaces also conforms to the evolution trend of the shop-floor itself. As shown in Fig. 4.1, the evolution path of

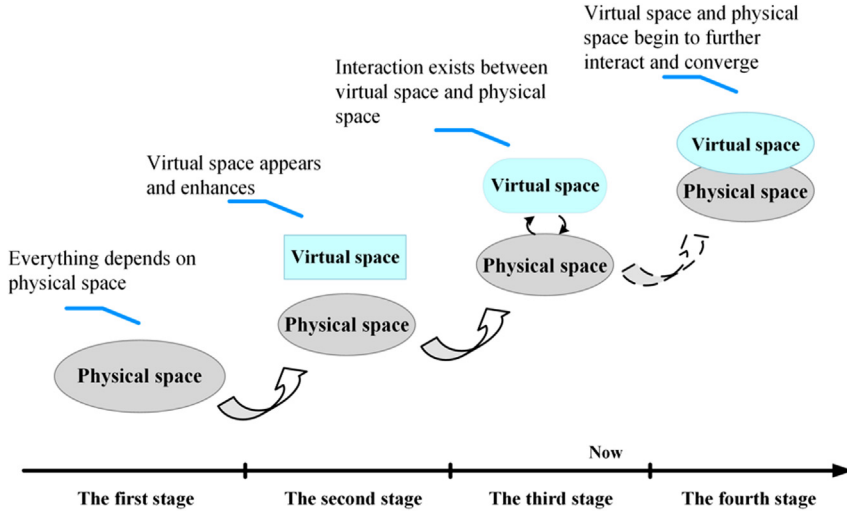


Figure 4.1 Evolution path of the shop-floor [5].

the shop-floor can be classified into four stages. In the first stage, due to the lack of effective information means, the production activities on the shop-floor depended on the physical space completely, leading to low efficiency, accuracy, and transparency. Then with the developments of information technologies, computer-aided systems began to be applied in production. But as the connection methods were unavailable, there was little interaction between the virtual space and the physical one. Nowadays, the shop-floor has moved into the third stage, where interaction is enabled by advanced communication technologies. In the future, with the developments of new generation information technologies (New IT), the virtual space will gradually play an equally important role with the physical one, while the continuous two-way interaction and the fusion of data from both the physical and virtual spaces will also be enhanced [6].

The evolution path can be illustrated by the development processes of production resource management, production activity planning, and production process control on the shop-floor, which have been researched by the authors in their previous work [5].

4.1.1 Production Resource Management

Production resources mainly include the personnel, equipment, materials, semifinished products and end products, etc. Production resource

management is the process of allocating, organizing, and coordinating the production resources reasonably through collecting, processing, and analyzing their information to finally ensure they operate as planned. In the beginning, it mainly relied on manual operations. Related data were recorded manually and stored in paper form, which made it difficult for information delivery and analysis. At that time, only manual means in the physical space existed. With the development of information technologies, various information systems were introduced into the virtual space for the management of human resources, equipment, inventories, tools, and products. It brought the shop-floor to the second stage and greatly improved management efficiency. However, as data for these systems could only be input by hand, the real-time states of the production resources could not be captured and the error rate of data was high. Nowadays, with the development of the Internet of Things (IoT), radio frequency identification (RFID) labels, smart sensory devices, and bar codes are deployed on the tools, materials, and work-in-process for real-time data collection and online data entry. Benefiting from this, the shop-floor has evolved into the third stage, where accurate data synchronization between the production resources and information systems are implemented and interactions exist. However, the continuous interactions are still insufficient in this stage and the fused data, which can provide more abundant information on the production resources, seldom attracts attention.

4.1.2 Production Activity Planning

Production activity refers to the process of combining the production resources in accordance with certain relationships and rules for jointly accomplishing a production task. Production activity planning refers to the process of organizing, planning, and arranging production activities with comprehensive consideration of the limits of all the resources and their capacities, under the premise that the plan satisfies the requirements and constraints on time, cost, quality, etc. In the first stage, when the information technology was not available, production activity planning was developed manually based on personal experience. The drawbacks were low efficiency and accuracy as well as high dependence on competence and experience of the plan makers. In the second stage, information technologies were introduced, making it possible to implement a production activity plan on computers based on mathematical models. Relying on the powerful processing capacity of the computer, this new way could

consider different factors at the same time and planned production activity by integrating multiple schemes, and thus could obtain a more scientific and optimized plan. It can greatly improve efficiency and decrease dependence on humans, and is adaptable to the increasingly expanded production scale. However, production activity planning was usually limited to the preproduction stage and the static plan tended to neglect the disturbance factors in actual production. In recent years, with the emergence of interaction between the virtual space and the physical space, production activity planning has been performed in a dynamic way and has entered the third stage. This stage can dynamically track the implementation of the plan through real-time collection of data of the personnel, equipment, and materials and can adjust the original plan in a timely manner in reaction to the disturbance. Although the production activity plan has better ability to resist disturbance, it is still insufficient in foreseeability and robustness due to the lack of high-fidelity prediction and fused data that can depict the activity comprehensively from different perspectives.

4.1.3 Production Process Control

Production process control refers to monitoring and controlling the production resources and the derived production activities during the production process to satisfy the quality demand (e.g., precision), customer demand (e.g., delivery date), and company demand (e.g., profit), etc. In the early stage, the monitoring usually relied on human senses and the problems or deficiencies occurring in the production process were judged by personal experience. This approach was too subjective and could not identify problems in a timely manner, which resulted in consequences such as inferior control, low precision, and large deviation. In the second stage, with the development of information technology, the above problems were partially solved. First, the digital instruments and apparatus were adopted to greatly improve the detection precision. Then computers were used for statistics, data analysis, and decision making based on the detected data, making the control strategies more scientific and accurate. Meanwhile, automatic actuators could improve the control precision compared with the manual operations. However, in this stage, due to the lack of interaction between the physical and virtual spaces, there were severe delays in the process control. With the development of IoT and other advanced communication technologies, the ability to collect real-time data in the production process has been enhanced significantly and

the manufacturing execution system can deal with real-time events in a timely manner in combination with real-time data. Based on this, real-time control is realized in practice and hence the abovementioned problem has been solved to some extent. Nevertheless, due to the lack of accurate simulation and validation in the virtual space as well as the fused data, the current production process control is still insufficient in accuracy, overall optimization, and intelligence.

In summary, the shop-floor has moved into the third stage to date, and the integration and fusion between the physical and virtual spaces are imperative for the future. In the third stage, the function of virtual space is still limited. It mainly works as a container and processor for the data from the physical space, lacking its own data and evolution, as well as continuous interactions with the physical space. Meanwhile, the existing works in shop-floor mainly focus on the collection, storage, testing, processing, and control of the data obtained from entities, but not those data generated by virtual models, especially the fused data converging both the physical and virtual sides. Therefore, for the further fusion in the fourth stage, stronger capacities in the virtual space, continuous interactions between the physical and virtual spaces, and fused data are still required [6].

Recently, digital twin (DT) as an efficient method for cyber–physical fusion has attracted increasing attention. On the other hand, New IT has been developed considerably, which enables DT to be applied on the shop-floor to generate a new paradigm for the future. For example, IoT can provide ubiquitous sensing ability to collect data from different elements, businesses, and processes of the shop-floor. Cloud computing can provide powerful computing capability, which lays the foundation for building and operating sophisticated models. With big data and artificial intelligence (AI) technology, large amounts of data can be fused and mined, and intelligence can be given to entities, models, and systems that can support autonomous negotiation and cooperation between the physical and virtual spaces [6].

In this situation, based on the DT, the concept of digital twin shop-floor (DTS) [5,6] is proposed to achieve cyber–physical fusion for the shop-floor. The DTS presents the shop-floor in dual visions, the physical and the virtual. It keeps these two sides consistent with and optimized by each other. Data from either the physical or the virtual side as well as the fused data are provided to drive the production [6]. In this chapter, the evolution path of the shop-floor is explored first, and the conceptual

model and the specific operation mechanism for the DTS are discussed. Then the implementing methods of its four components, namely, physical shop-floor (PS), virtual shop-floor (VS), shop-floor service system (SSS), and shop-floor digital twin data (SDTD), are studied, which provides a deeper insight into the DTS. Finally, key technologies, characteristics, and challenges for the DTS are explored as guidelines for future studies.

4.2 RELATED WORKS

Current research on the shop-floor has made significant efforts to enhance the interactions between the physical and virtual spaces, and strengthen the abilities of the virtual space. However, some deficiencies still exist in the following aspects, which have been studied in the authors' previous work [6]: (1) data collection, (2) data processing, (3) information system construction, (4) virtual model construction, and (5) exploration for new modes of production.

4.2.1 Data Collection

Fieldbus and Industrial Ethernet are prevalently employed on the shop-floor for the wired connection of sensors, actuators, and controllers. Research mainly covers the real-time communication mechanism [7,8], protocol standardization [9], integration with intranet [10,11], etc. With the development of IoT, wireless sensor networks (WSNs) and RFID are applied on the shop-floor to collect the real-time states wirelessly, which increases the flexibility of data acquisition [12–14]. Meanwhile, standardized technologies and devices are discussed to shield the heterogeneity of protocols and interfaces, such as MTConnect [15], middleware technology [16], and coherent and semantic encapsulation [17]. Although these works enable real-time data collection, they are mainly designed for the data from the actual production in the physical space, seldom focusing on the simulated data from the virtual space.

4.2.2 Data Processing

Various data processing methods have been applied on the shop-floor to transform raw data into valuable information for further analysis and decision making. The common processing methods include data regression [18,19], data filtering [20], multisensor data fusion [21], data dimension reduction [22,23], data classification [24], etc. However, the methods

mainly process data from the physical space, and few methods focus on the fusion of data from both the physical and virtual spaces, which may generate a more comprehensive and accurate result.

4.2.3 Information System Construction

Various information systems have been built to serve the production process. For instance, Zhong et al. [25] built an RFID-enabled real-time manufacturing execution system, which provides more practical planning and scheduling decisions for the shop-floor. Luo and Kuo [26] implemented a robot arm system to realize the proper grasping of target objects. Fang et al. [27] explored an event-driven shop-floor work-in-progress management platform to monitor and control the dynamic production. A real-time discrete event-based monitoring system was developed for the event construction, state prediction, and disturbance detection for the RFID-enabled shop-floor [28]. Benefiting by the real-time data, these systems can make in-time reactions to the events occurring on the shop-floor and generate real-time strategies for optimization. However, these systems are mainly fed by the data from the physical space; the simulated data and the fused data are seldom considered.

4.2.4 Virtual Model Construction

With the development of virtual manufacturing, a virtual environment in which models replace the physical entities for evaluation, verification, and optimization is established. For the shop-floor, a quantity of research has focused on geometry modeling of the production resources for the layout optimization [29], machining process control [30], and logistics planning [31]. Based on this, some other research has further explored the modeling of physical properties (e.g., wear, force, and torque), then simulated the physical parameter change process [32], and evaluated the process plan [33]. However, there is a lack of models that can describe the behaviors and rules of the shop-floor, especially models that could seamlessly fuse the geometric parameters, physical properties, behaviors, and rules to form a complete digital mirror of the shop-floor.

4.2.5 Exploration of New Modes for Production

With the development of New IT, some researchers have introduced these technologies onto the shop-floor to explore new production modes. For examples, Monostori [34] discussed the concept of cyber–physical

production systems (CPPS) based on cyber–physical system (CPS) to achieve a more decentralized way of functioning and interacting. Mourtzis et al. [21] explored a cloud-based approach to provide timely maintenance service for the equipment. Wang et al. [35] presented a self-organized multiagent system that incorporates an industrial network, the cloud, and big data to achieve high efficiency. Zhu et al. [36] presented a five-layer web-based structure to support the interaction between the production resources and the application software. Zhong et al. [37] introduced big data analytics onto the shop-floor for increased intelligence. Although these works elaborate the implementing methods for new modes combined with New IT, the synchronous operation and mutual optimization of the physical and virtual spaces are seldom discussed.

In summary, the abovementioned research has rarely paid attention to the following three aspects [6]: (1) building a high-fidelity virtual mirror to describe and mirror the shop-floor thoroughly; (2) keeping the shop-floor and its mirror consistent and synchronized and realizing dual optimization; and (3) converging the data from both physical assets and digital models to generate valuable information for production. In this chapter, these three aspects will be focused on to improve the virtual space and to converge its models, data, and mechanisms with the physical space for the seamless integration and fusion between the physical and virtual spaces.

4.3 CONCEPT OF DIGITAL TWIN SHOP-FLOOR

With the DT technology introduced onto the shop-floor, the concept of DTS is proposed to solve the abovementioned existing insufficiencies. In this section, the concept, interaction mechanism, and operation processes of the DTS are explored.

4.3.1 Concept of Digital Twin Shop-Floor

As shown in Fig. 4.2, the DTS consists of four components: PS, VS, SSS, and SDTD. The PS includes a series of production resources, that is, personnel, equipment, materials, etc., existing objectively in the physical space. The PS organizes the production strictly following the predefined orders from the VS and the services from the SSS. The VS consists of a set of models built in multiple aspects including geometry, physics, behaviors, and rules, to represent the entities in the PS with high fidelity. The VS can operate autonomously and evolve with the PS, providing control orders for the PS and optimization strategies for the SSS. The SSS is an

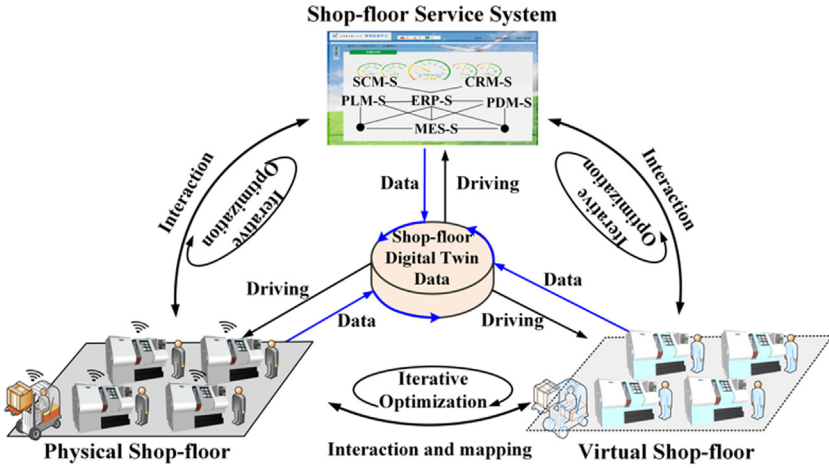


Figure 4.2 Conceptual model of DTS [5].

integrated service platform that encapsulates the functions of an enterprise information system (EIS), computer-aided tools, models and algorithms, etc. into subservices, which are provided for both the PS and the VS toward specific demands. The SDTD includes PS data, VS data, SSS data, and fused data of the three parts, as well as existing domain knowledge for modeling, optimizing, and predicting. The data in the SDTD can eliminate the isolated information islands and converge the data from both the physical and virtual spaces for more comprehensive and consistent information. There are six bidirectional connections among the PS, VS, SSS, and SDTD to make them connect and interact with each other [6].

The interaction processes in the DTS are detailed in Fig. 4.3. These processes are iterative loops for continuous optimization [5].

1. *Interaction between PS and SDTD.* Actual data from the PS (e.g., production schedule, working states of machines, inventories of materials) are injected into the SDTD continuously for data cleaning, transformation, and fusion. The processed data in the SDTD can be obtained by the constructed agent for each entity to make the entity aware of the states of itself and others. If necessary, the behaviors of the entities can be adjusted to better implement cooperation considering a global production target.
2. *Interaction between VS and SDTD.* Simulated data from the VS (e.g., predefined orders, simulated analysis results, predicted states) are transmitted to the SDTD and fused with data from other sources. For the

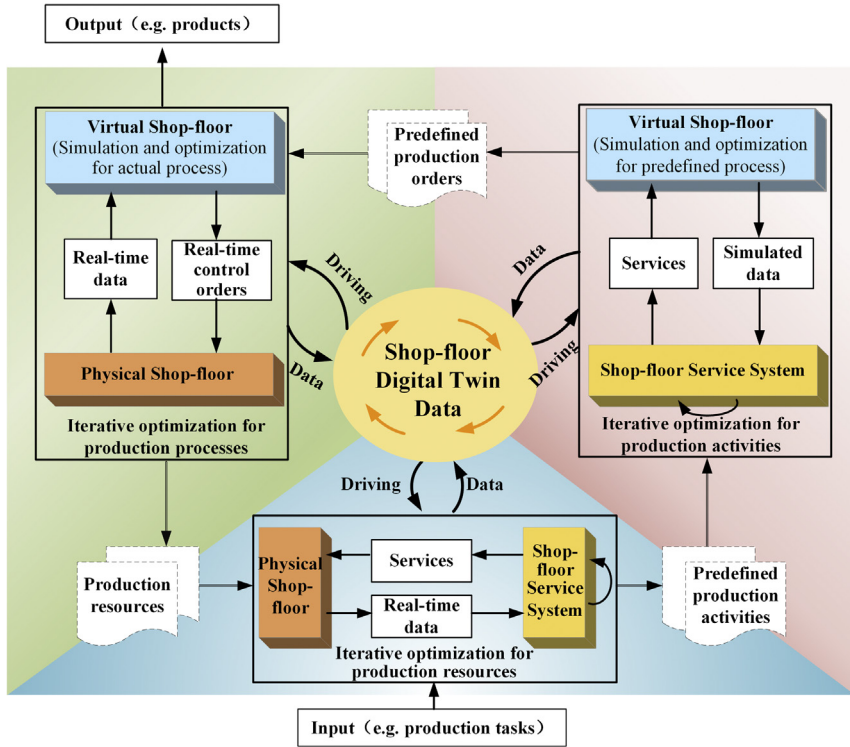


Figure 4.3 Interaction mechanism in DTS [5].

VS, the models and operation mechanisms are built and updated based on the relations, constraints, and rules mined from both the actual and simulated data in the SDTD.

3. *Interaction between SSS and SDTD.* Services data from the SSS (e.g., market data, scheduling for resources, machine maintenance strategies) are transmitted to the SDTD for data processing and fusion. For the SSS, the SDTD provides the corresponding data, algorithms, and models that can be encapsulated into subservices to feed the service composition and the subsequent service process toward the specific demand.
4. *Interaction between PS and SSS.* When the task is transmitted to the DTS, services in the SSS provide the initial production plan to allocate the production resources in the PS and start the production. During the production, real-time data collected from the PS are transmitted to the SSS. Based on the data, the SSS analyzes the current demands of the resources and invokes the corresponding services autonomously,

such as the services for prognostics, maintenance, scheduling, and energy consumption optimization.

5. *Interaction between VS and SSS.* Services for the PS predefine the production activities and can be transmitted to the VS for verification before execution. The VS will feed back the simulated data and optimization advice to the SSS to adjust the predefined production activities until they can fully pass the validation. According to the validated activities, a set of predefined production orders can be generated to guide the actual production. Whereas, for VS, the SSS provides services to ensure the high fidelity of the VS through calibrating the model parameters to sustain its performance with the PS, which supports more accurate validation.
6. *Interaction between PS and VS.* During the actual production, the PS and the VS operate simultaneously. The PS generates real-time data, and transmits them to the VS to update the states of the corresponding models. The VS can adjust the predefined production orders according to the real-time data from the PS, and feeds back the adjusted orders in real-time to trim the actual production process to make it synchronize with the predefined one. Through the above process, the production process can be optimized continuously.

After the production, the products will be output, satisfying the requirements of delivery, cost, quality, etc. Then the production resources will be ready for the next production operation.

4.3.2 Operation Process of Digital Twin Shop-Floor

As shown in Fig. 4.4, the operation process of DTS is illustrated in three stages, including before production, during production, and after production. In this figure, the yellow blocks, blue blocks, and purple blocks represent the PS, VS, and SSS, respectively, and their operations and interactions are fed by data, namely SDTD [6].

Before production, the orders (e.g., delivery, quantity, cost, and quality) are transmitted to the production plan service in the SSS. To support the formulation of the production plan, related data are collected, including sensor data (e.g., material stock, personnel workload, and equipment capacity), simulated data (e.g., prediction of equipment fault, assessment of human physical power, and analysis on material performance), and EIS data (e.g., product lifecycle data, process document and market data). Since these data come from the PS, VS, and SSS, data fusion service is

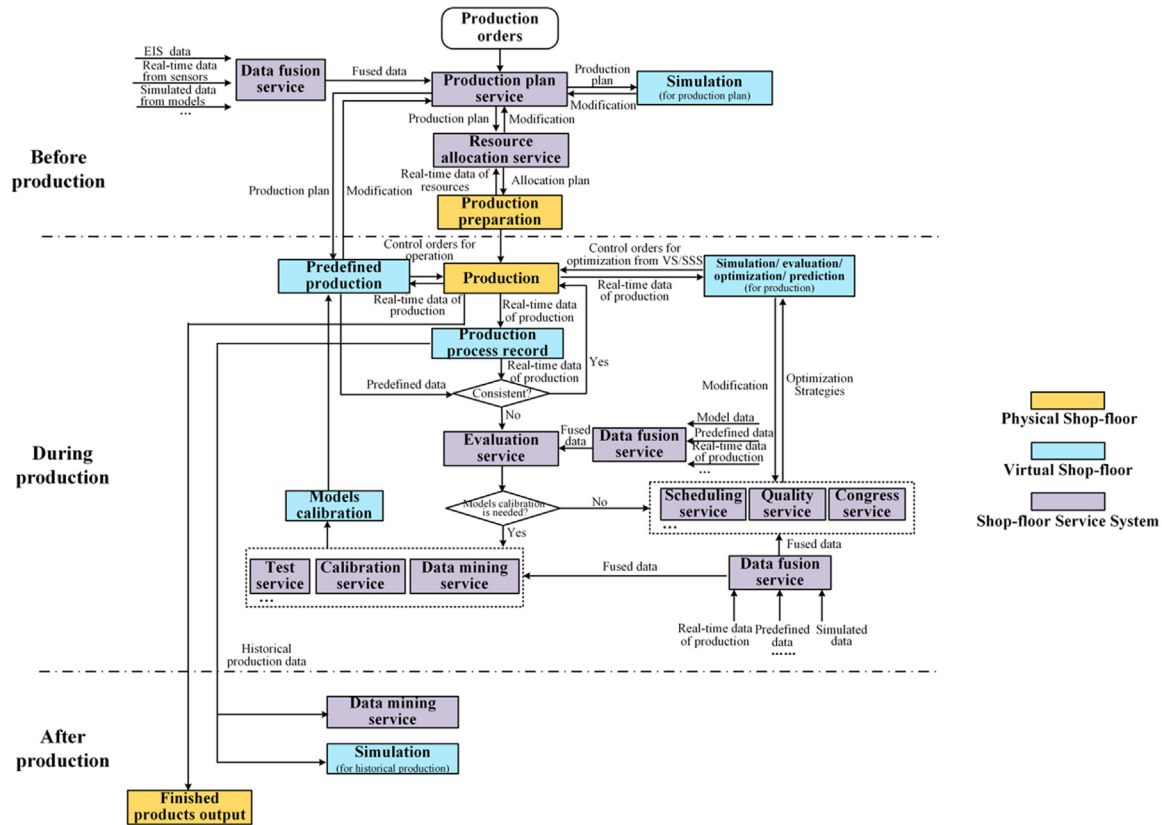


Figure 4.4 Operation process of DTS [6].

provided to fuse them and to generate consistent and valuable information for planning. Driven by the fused data, the production plan service produces a plan, which is then given to the VS for verification. The VS finds potential conflicts in the plan before actual production and feeds back modification strategies to the service. After revision, the plan is transmitted to the resource allocation service, which guides the preparation for production in the PS. If the real-time states of resources change, modification advice can also be given back to the production plan service [6].

During production, the verified plan is transmitted to the VS to predefine the production in virtual space. Then control orders from the VS are given to the PS to start the actual production process and keep it synchronized with the predefined one. On one hand, the real-time data generated by the PS are recorded in the VS, and on the other hand, simulation, evaluation, and optimization, etc. based on models are carried out in real time to generate orders to regulate the production in the PS whenever necessary. During this process, the predefined data from the VS are compared with the real-time data from the PS continuously. If the two parts are not consistent, meaning their difference exceeds the predefined threshold, the evaluation service driven by the fused data will be given. It judges whether the inconsistency is caused by the disturbances in the PS (e.g., equipment failure, material shortage, and emergency order) or by the inaccuracy of models in the VS (e.g., unreasonable setting on boundary and initial condition). If the disturbances exist in the PS, corresponding services, such as scheduling service, quality service, and congress service, will be invoked to eliminate them. These services are given to the VS first for verification, then transformed into control orders to regulate the PS. Otherwise, the inconsistency may be caused by the VS. In this case, services for model calibration are scheduled and implemented on the VS. The production plan verified by the previous models shall be regulated based on the calibrated ones, while the production process will be adjusted accordingly [6].

After production, the finished products are output and transported to the warehouses. At the same time, the history production data are achieved from the records in the models. Based on these data, a data mining service extracts new knowledge for model building and calibration. With the adjustable timeline, the VS can play back the historical situations, which is an effective way to identify the defects in previous productions as well as the corresponding solutions [6].

Thus, DTS involves a constant evolution process along with different rounds of production.

4.4 IMPLEMENTATION OF DIGITAL TWIN SHOP-FLOOR

Based on the concept of the DTS, the implementation method for each component of the DTS is discussed, including the implementation for the PS, VS, SSS, and SDTD, respectively. The implementation method introduced in this section has been investigated in the authors' previous work [6].

4.4.1 Physical Shop-Floor

As shown in Fig. 4.5 [6], except for the traditional production abilities, the PS in DTS is also able to realize interconnection and interaction. It means that from the vertical perspective, the production resources can convey data to the virtual space and receive control orders from it. While horizontally, they can sense other resources and regulate the behaviors of themselves or others.

Data of the production resources should be retrieved in real-time. Considering the heterogeneity, different sensors are applied to data acquisition. For the personnel, portable devices can be used to collect physical power, motion, and work progress. Wired or wireless sensors can be deployed on equipment to collect the states such as machine speed, tool wear, and axis vibration. For equipment with a high degree of automation, these data can be read from embedded modules of its own. RFID

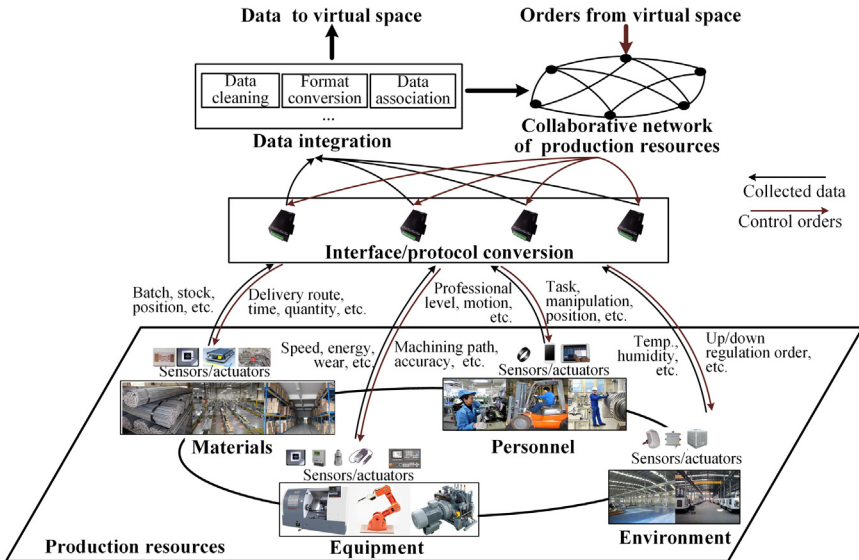


Figure 4.5 Interconnection and interaction in PS.

can be used for lifecycle tracking of materials, and environmental sensors are applied to detecting real-time changes.

However, these data are usually transmitted with different interfaces (e.g., RS232, CAN, and ZigBee) and communication protocols (e.g., Profibus, TCP/IP, and Modbus), which makes it difficult to implement unified data access to the virtual space. Hence, it is necessary to deploy customized access modules, through which the data from different sources can be transformed into a uniformed interface and protocol. In addition, since the accessed data always have different formats, types, and information models, data integration including cleaning, format conversion, association, etc. should be carried out.

Based on the integrated data, a collaborative network that describes the production resources and their relations is built. In the network, the resources are represented by nodes that have the abilities to sense, compute, and interact, while their relations are expressed by edges. According to the orders from the virtual space, a node asks others to cooperate with it and also responds to requests propagated by others, which gives the PS stronger adaptability, flexibility, and robustness.

Orders from the collaborative network are transmitted to the customized access modules for interface and protocol conversion to adapt to different communication modes of actuators in the PS. Finally, the orders can be executed to control and coordinate the production.

4.4.2 Virtual Shop-Floor

As a digital mirror, to simulate the physical counterpart with high fidelity, the VS is built according to four aspects, that is, geometry, physics, behavior, and rules. As shown in Fig. 4.6 [6], the modeling process of a computer numeric control (CNC) machine is taken as an example. First, three-dimension geometry models are built to describe the shapes, sizes, positions, and assembly relations of the machine components. The common tools for these models include SolidWorks, 3D MAX, AutoCAD, CATIA, etc. Secondly, physical properties (e.g., cutting force, torque, and wear) are given to the geometry models to form the physics models, which can analyze the physical phenomena, such as deformation, cracking, and corrosion. The finite element method (FEM) can be used for simulation at this level. Then behavior models are built to describe the machine responsive mechanisms under driving factors such as NC programs and disturbing factors such as manual interferences. Finite state

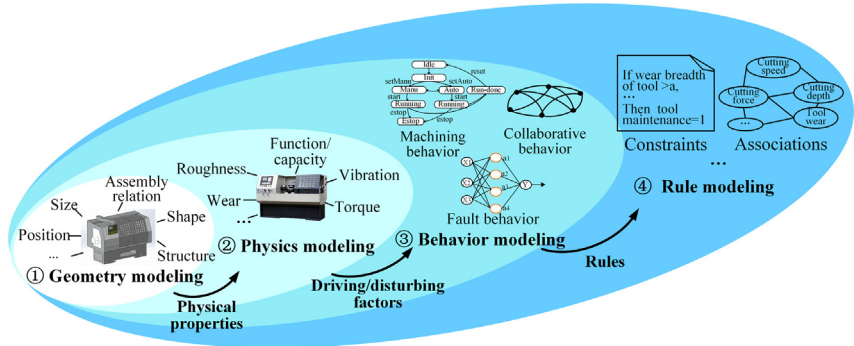


Figure 4.6 Four steps of modeling for CNC machine in VS.

machines, neural networks, complex networks, etc. can be applied to describing the responsive processes. Finally, rules of associations, constraints, and deductions are modeled to describe the domain knowledge and make the above three kinds of models capable of evaluating, reasoning, and predicting. To build the rule models, data mining algorithms, such as Apriori, support vector machine, and K -means, can be used.

After modeling, the four models are fused in both function and structure to form a complete virtual CNC machine. For other entities in the PS, the modeling processes are similar.

To ensure the accuracy of the models, verification, validation, and accreditation (VV&A) is adopted to test the transformations from models to program codes, compare the inputs and outputs of the models with entities, and estimate the models' sensitivity. In addition, with virtual reality (VR) and augmented reality (AR), the VS presents vivid three-dimension images and overlaps the virtual models on physical entities, thus providing an immersive environment for users.

As shown in Fig. 4.7 [6], model operation and model evolution are parallel processes. During operation, the models run synchronously with the physical counterparts. Calibration strategies are generated through comparing the models with entities to support model evolution, which means a higher fidelity to the PS. At the same time, the evolved models can support more accurate estimation, verification, optimization, and prediction for the operation process.

First, the production plan from the SSS is transmitted to the VS for verification. Based on the verified plan, the predefined virtual production process will form and drive the actual production in the PS. As the entities start to work, real-time data generated by them are recorded and compared

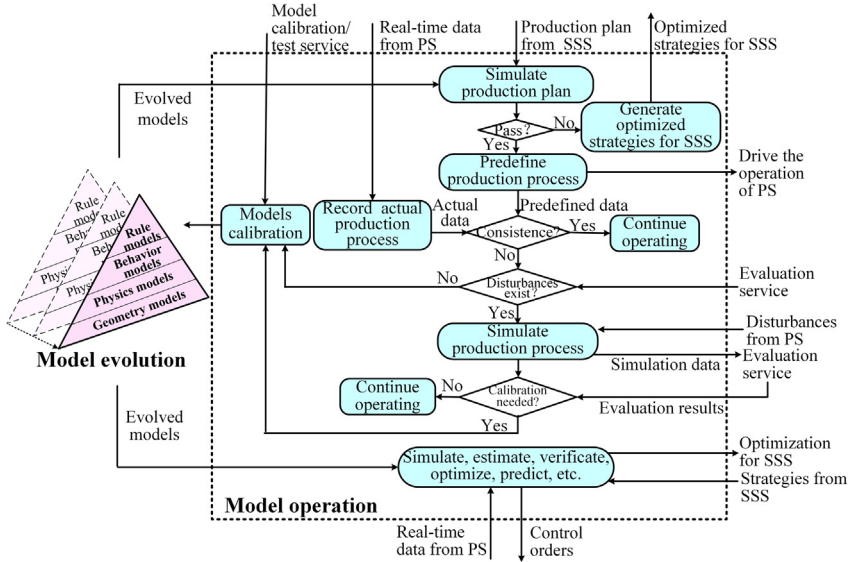


Figure 4.7 Operation and evolution of models in VS.

with data from the predefined process to evaluate the consistency. If the two kinds of data are aligned, accuracy of the models can be confirmed; otherwise evaluation service will be scheduled to estimate the reason for inconsistency. In this process, if disturbances do not exist in the PS, it can be considered that the inconsistency is caused by defects of the models, and that calibrations are needed. If disturbances indeed exist in the PS, they will be captured and provided for the models. Simulation needs to be carried out again with consideration of the disturbances that were unknown to the VS previously. If the simulation results still cannot reach an agreement with the actual states, calibrations are needed; otherwise the models are considered accurate. With the models evolving to approach the entities continuously, the simulations for the PS and the SSS will be more accurate.

4.4.3 Shop-Floor Service System

The SSS provides various services to support the management and control of the PS as well as the operation and evolution of the VS. As shown in Fig. 4.8 [6], resources such as data, models, and algorithms are encapsulated into the subservices and then selected to form composite services for meeting demands from the PS or the VS. The operation mechanism is described as follows.

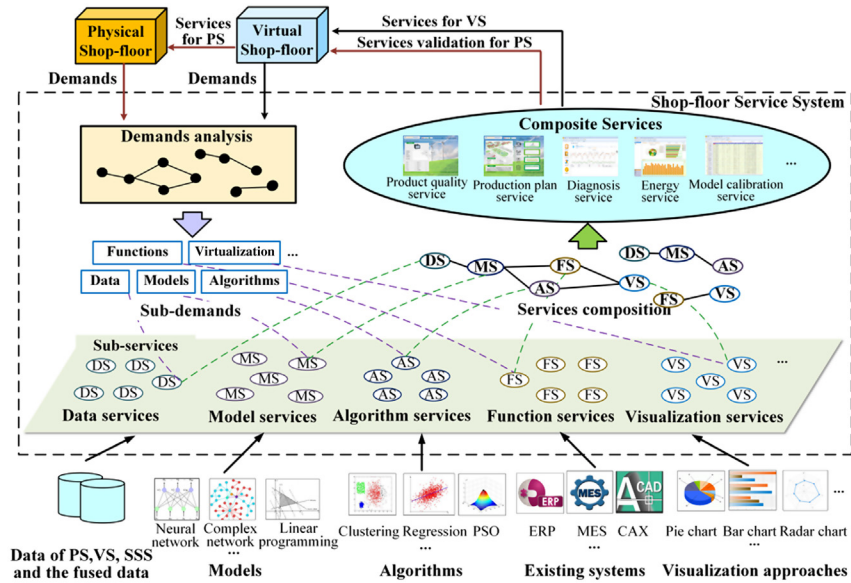


Figure 4.8 Operation mechanism of SSS.

On one side, subservices in the SSS are transformed from resources including data, models, algorithms, existing systems, visualization approaches, etc. To realize the transformations from the resources to the subservices, processes such as service description, virtualization, and encapsulation are needed. On the other side, demands from the PS mainly include production planning and scheduling, quality testing, congress control, etc., which require solving the existing problems quickly and preventing possible faults during production. Demands from the VS mainly require model calibration and test, data mining, etc., to support the model operation and evolution. These demands can be decomposed into subdemands that usually focus on problems like what data should be used, which model is most suitable, and which algorithm provides the best solution. According to the subdemands, the SSS selects suitable subservices from the candidates, and then combines them under certain rules to form the composite services for the PS or the VS under the constraints on time, cost, quality, etc. The services are monitored at runtime and will be recomposed if working improperly. For the VS, the composite services are transmitted to the models directly. For the PS, they are conveyed to the VS first for verification, and then to the PS for execution.

4.4.4 Shop-Floor Digital Twin Data

SDTD mainly consists of PS data, VS data, SSS data, and fused data of the three parts, as well as knowledge for modeling, data processing, and standards. The PS data mainly include production resource data, production process data, environment data, etc. The PS data are directly generated by entities in the physical space without further processing and considered as physical data. The VS data refer to model parameters and data of simulation, evaluation, optimization, and prediction, while the SSS data mainly involve data of various services. The latter two types of data are generated from the virtual space and defined as virtual data. The fused data are the integration and fusion of the physical and virtual data through data comparison, association, combination, and clustering. For example, for a tool, wear data from the physical space can be combined with both the simulated data about stress, deformation, and strength, and service data about maintenance records from the virtual space to form the fused data. It presents a tool correlating both the physical and virtual information, providing a more consistent, accurate, and comprehensive representation compared with the data from a single aspect.

The construction of SDTD is described as shown in Fig. 4.9 [6]. First, data from PS, VS, and SSS with various formats, types, structures, and encapsulations are transformed into a unified form. Second, the data are cleaned to remove the “dirty” data (e.g., errors, duplicate data, invalid data) and replenish the missing data through clustering, regression, and filtering. The related data can be associated and combined based on certain rules. Third, to achieve consistent and comprehensive interpretation, the physical data and the virtual data can be converged to form the fused data through data fusion algorithms, such as Kalman filter, neural network, and Bayesian inference.

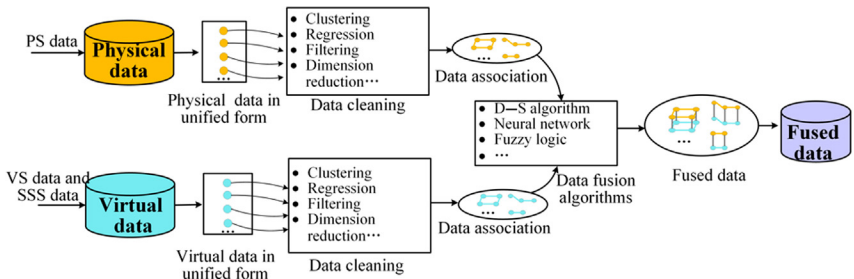


Figure 4.9 Data fusion of SDTD.

SDTD is optimized iteratively through the interaction between different kinds of data. On one side, the historical data is updated and expanded with real-time data joining, while the real-time data can be tested and corrected according to the knowledge accumulated in the historical record. On the other side, the physical data can be evaluated and simulated using the virtual data, while the latter can be compared with the former to confirm accuracy.

4.5 CHARACTERISTICS OF DIGITAL TWIN SHOP-FLOOR

The characteristics of the DTS mainly include the cyber–physical fusion, data driving, integration, and fusion of data from the entire elements, processes and businesses, and iterative optimization [5].

4.5.1 Cyber–Physical Fusion

Cyber–physical fusion can be reflected in three aspects.

1. *Bidirectional mapping between PS and VS.* With the technologies of VR, AR, modeling, and simulation, the VS can provide a highly realistic description for the PS and records the operation process of PS. Whereas, the PS strictly follows the orders from the virtual space and tries to faithfully reproduce the production predefined by the VS. Entities in the PS and models in the VS are in one-to-one correspondence. They operate simultaneously and evolve together.
2. *Real-time interaction between PS and VS.* During the operation, when the PS starts the production, the VS also simulates the production in the virtual space synchronously. On one side, data from the PS can be perceived in real-time and transmitted to the VS for model updating and simulation. On the other side, the VS feeds back control orders to the PS in real-time based on the simulation results to optimize the production. Through the real-time interaction, the PS and the VS can grasp the dynamic changes of each other and make in-time responses.
3. *Fusion of data from physical and virtual spaces.* Data from the physical space (e.g., PS) and the virtual space (e.g., VS and SSS) can be transformed into a unified form and fused seamlessly. It makes the data for production more comprehensive as both the physical and virtual aspects are considered.

4.5.2 Data Driven

The operations of the PS, VS, or SSS are driven by the SDTD as well as real-time data from the other two parts.

1. *Operation of PS.*Data from the SDTD make each production resource in the PS aware of the states of itself thoroughly as well as others, which can drive an automatic selfadjustment considering collaboration. The SSS provides various services to the PS to keep the resources in good condition for production. Control orders fed back from the VS regulate the PS in real-time to keep up with the predefined production.
2. *Operation of VS.*Driven by knowledge mined from the data of the SDTD, models and model operation mechanisms in the VS can be constructed. The services from the SSS can calibrate parameters of the models to sustain their performance with the corresponding entities. In the production, the VS is driven by the real-time data from the PS for model updating, simulation, and prediction.
3. *Operation of SSS.*Data from the PS make the services in the SSS capture the current demands of the production resources. According to the demands, service encapsulation can be fed by the data from the SDTD to form the data subservice, algorithm subservice, and model subservice, which are then combined to form an initial composite service. The initial service will be validated on the VS and revised based on the simulated data until it becomes optimal.

4.5.3 Fusion of Data From All of the Elements, Processes, and Businesses

In the DTS, data from all of the elements, processes, and businesses can be integrated and fused.

1. *Data integration and fusion for elements.*In the DTS, data of elements (e.g., persons, equipment, materials, and environment) can be fully connected to the virtual space, then integrated and fused, which supports the data sharing and keeps the elements interactive with one another.
2. *Data integration and fusion for processes.*From raw materials to finished products, data of various processes (e.g., turning, milling, and grinding) can be efficiently integrated and fused to further support the cooperative process.

3. *Data integration and fusion for businesses.* Data from different businesses, such as cost management, product quality management, and energy consumption optimization, can be integrated and fused, which eliminates information islands and makes these businesses keep interactive to pursue compatible goals of optimization.

4.5.4 Iterative Optimization

In the DTS, PS, VS, and SSS are optimized iteratively through continuous interactions.

1. *Iterative optimization between PS and SSS.* During production, real-time data collected from the PS are continuously transmitted to the SSS to update and adjust the services to adapt to the current production. Whereas, the services in the SSS are provided for the PS to optimize the management of the production resources and keep them in good condition.
2. *Iterative optimization between VS and SSS.* Services for the PS can be validated in the VS before execution. The VS will feed back strategies to modify and optimize the services until they can fully pass the validation. Whereas, the SSS provides services to ensure the high fidelity of the VS through parameter calibration to sustain its performance with the PS, which makes the VS optimized continuously during its running to provide more accurate simulation results.
3. *Iterative optimization between PS and VS.* The PS constantly generates real-time data, and transmits them to the VS to update the status of the corresponding models. At the same time, the VS can feed back real-time control orders to trim and optimize the actual operations in the PS iteratively.

4.6 KEY TECHNOLOGIES FOR DIGITAL TWIN SHOP-FLOOR

As shown in Fig. 4.10, the key technologies for implementing DTS are summarized as follows [5].

1. *Interconnection and interaction in PS.* Related technologies include the perception and access, communication protocol analysis, data encapsulation and publication, multiagent technology, etc.
2. *Modeling, operation, and verification of VS.* Technologies including high-fidelity modeling, model integration and model verification, etc. need to be addressed.

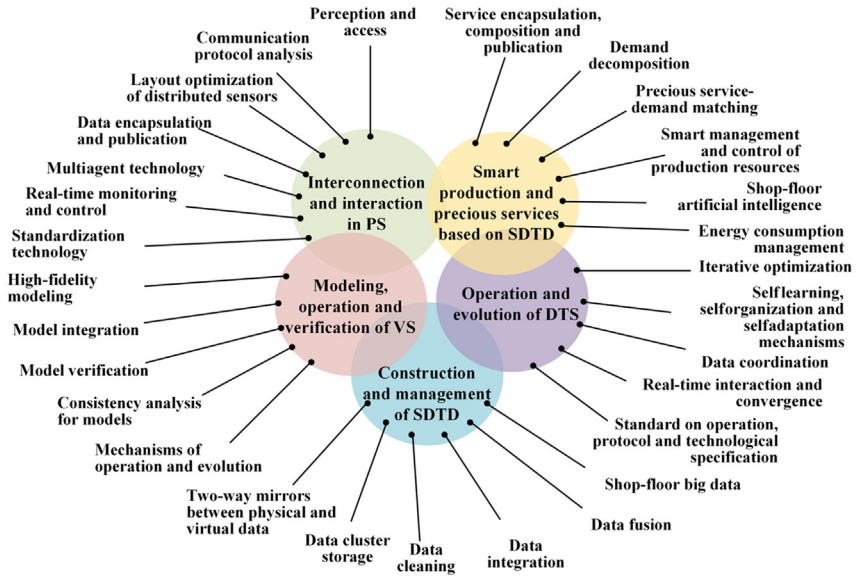


Figure 4.10 Key technologies for DTS [5].

3. *Construction and management of SDTD.* Technologies involve data cleaning, data integration, data fusion, two-way mirrors between physical and virtual data, and shop-floor big data [38].
4. *Operation and evolution of DTS.* Technologies including iterative optimization, selflearning, selforganization and selfadaptation mechanisms, standardization, etc. need to be explored.
5. *Smart production and precious services based on SDTD.* Services encapsulation, composition and publication, demand decomposition, precious service-demand matching, energy consumption management [39], etc. should be studied.

4.7 CHALLENGES FOR DIGITAL TWIN SHOP-FLOOR

Challenges exist when it comes to full realization of the DTS. First, it is necessary to keep an adequate two-way connection between the physical and virtual spaces to support real-time interaction. This mainly addresses challenges in terms of the technologies of sensors, communication, database and data processing, etc. Second, due to the variability, uncertainty, and fuzziness of the physical space, building models in the virtual space to mirror entities with high fidelity is a fundamental issue. In addition, when

inconsistency between models and entities appears, it is difficult to identify and utilize them wisely. Third, as the continuous physical space and the discrete virtual space are in different scales, it is a challenge to figure out how to transcend the divergence to realize the seamless integration of the two sides. With the virtual space evolving with the physical one along the lifecycle, data from entities, models, and systems are generated continuously. In this situation, how to integrate and converge the increasing data is worth exploring. Security is another focus that ensures the normal operation of the physical and virtual spaces against malicious attacks. To guarantee the interests of the enterprises, the equilibrium between the costs and benefits of the DTS should be considered [6].

4.8 SUMMARY

As New IT is being developed and applied continuously, developing DTS to start a new paradigm of shop-floor becomes imperative. To support the seamless integration and further fusion on the shop-floor, DTS provides evolved models with high fidelity, continuous interactions between physical and virtual spaces, and fused data convergence of those two spaces [6]. This chapter provides an insight into DTS and a guideline for future work.

The main contributions are concluded as follows:

- The concept and operation process of DTS are explored.
- The interaction mechanism and implementing methods for the four components of DTS are illustrated.
- The characteristics, key technologies, and challenges ahead are discussed.

Future work will focus on the following aspects [6]:

- Smart interconnection and interaction in the PS.
- Adequate two-way connection between physical and virtual spaces.
- High-fidelity models for the VS.
- Service management and precise service-demand matching.
- Applications of DTS in smart manufacturing.

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CHAPTER 5

Equipment Energy Consumption Management in Digital Twin Shop-Floor

As equipment energy consumption (EC) accounts for a large proportion of total consumption in manufacturing, it is imperative to realize efficient equipment energy consumption management (EECM) to reduce EC and improve energy efficiency. Recently, with the developments of digital twin (DT), fused data and high-fidelity models can be enriched greatly and a physical–virtual fusion environment can be built in the shop-floor, which provides new opportunities to the existing EECM systems, to improve performances of EC monitoring, analysis, and optimization. In this context, the chapter introduces a framework of EECM in the digital twin shop-floor (DTS) first, then explores implementation methods for the framework, and finally discusses the potential advantages.

5.1 INTRODUCTION

Manufacturing is a sector with a large amount of EC, which leads to serious environment problems, such as toxic smog, global warming, and acid rain pollution. Energy consumed by the equipment on the shop-floor occupies the largest proportion of EC in manufacturing [1]. However, as this part of EC can vary dramatically in different conditions during production, there is great potential to save energy. Therefore, research on EECM is abundant and has been devoted to reducing EC and improving energy efficiency for the equipment through EC monitoring, analysis, and optimization [2]. The related research, which has been studied in the authors' previous work [2], is introduced as follows.

In EC monitoring, different sensory devices are attached to the equipment to measure the EC (e.g., the consumption of electricity, water, and gas) and EC-related parameters (e.g., speed, vibration, load) for the sake of further meaningful analysis. For example, Lu et al. [3] built a distributed real-time energy monitoring platform for the shop-floor based on a

wireless sensor network (WSN), and the platform was evaluated. Verma et al. [4] presented a real-time remote monitoring method using the MTConnect standard protocol to collect acoustic data and vibration of an air compressor. Tristo et al. [5] built a power consumption monitoring system for the main subsystems of a machine tool and the collected data are transmitted via Wi-Fi. Tao et al. [6] proposed a five-layer framework for resources intelligent perception and access in manufacturing based on the Internet of Things (IoT), and described key technologies for various resources. Kara et al. [7] provided a middleware infrastructure for integrating different sensors to gather real-time EC data from each stage of the production cycle.

EC analysis is conducted based on the collected data from EC monitoring, to provide a deep insight into the equipment EC. For example, May et al. [8] proposed a seven-step method to define energy-related key performance indicators (e-KPIs) in production and identified weaknesses of energy efficiency based on these indicators. Lindberg et al. [9] presented a set of KPIs for energy, raw material, and operation, and improved the KPIs through changing related process signals or combinations of process signals. He et al. [10] explored relationships between numerical control codes and EC of different components in machine tools to help designers choose energy-efficiency programs. Tao et al. [11] assessed EC and related emission from different perspectives based on the bill of material (BOM). Liu et al. [12] proposed an EC prediction method for a machine tool, considering the start-up period, idle period, and cutting period.

In EC optimization, EC-related parameters of machining, scheduling, tools, etc. are optimized to minimize the consumed energy. For examples, He et al. [13] discussed methods for machine tool selection and operation sequence optimization to reduce the EC generated by idle equipment. Wang et al. [14] elaborated a two-stage EC optimization method, that is, the process stage and system stage. The method mainly optimized key operational parameters for a single machine in the first stage, and adjusted a sequencing plan and schedule for different machines in the second stage. Mourtzis et al. [15] fused measurements from machines to generate meaningful information, which could estimate the EC in new cases and thus optimize machining parameters. Tapoglou et al. [16] calculated optimal cutting parameters based on function blocks, with consideration of available tools and characteristics of the selected machine.

Some researchers have tried to integrate the above functions into information systems for further applications. For examples, Vijayaraghavan and Dornfeld [17] developed a software architecture for energy monitoring and analysis based on MTConnect and complex event processing technique. Karnouskos et al. [18] proposed that simple stand-alone approaches for EC evaluation and optimization should be integrated in a more holistic system supported by information and communication technology-based infrastructure. Soplop et al. [19] explored a method for extending the scope of manufacturing execution systems (MES) to improve energy efficiency.

Although the above works enable EC monitoring, analysis, and optimization, the main limitation is that they mainly focus on the EC data and the related parameters measured from real equipment in the physical space, but neglect the data from the virtual space. Recently, with the developments of DT and DTS, the data from both the physical and virtual spaces have become available and the real-time interactions can also be built, which makes it possible to further improve and optimize the existing EECM [2]. Therefore, this chapter explores the framework and implementation method of EECM under the DTS environment, and discusses the potential advantages of the DT-based EECM compared with the traditional EECM.

5.2 FRAMEWORK OF EECM IN DIGITAL TWIN SHOP-FLOOR

In this section, a machine tool is taken as an example to illustrate the framework of EECM in DTS. As shown in Fig. 5.1 [2], the framework describes the EECM compositions, including the physical machine tool, virtual machine tool, EECM services, and DT data, that belong to the physical shop-floor (PS), virtual shop-floor (VS), shop-floor service system (SSS), and shop-floor digital twin data (SDTD) in DTS, respectively. The four compositions interact with each other based on the mechanism of DTS. When data from the physical and virtual machine tools are consistent, the EECM services can be invoked to conduct EC monitoring, analysis, and optimization based on data from the two machines. To the contrary, if they are inconsistent, the services will find the cause of the difference and make an effort to eliminate it, which is a dual optimization for both the physical machine and the virtual model. DT data from multiple sources are involved and fused in the database, such as data from the

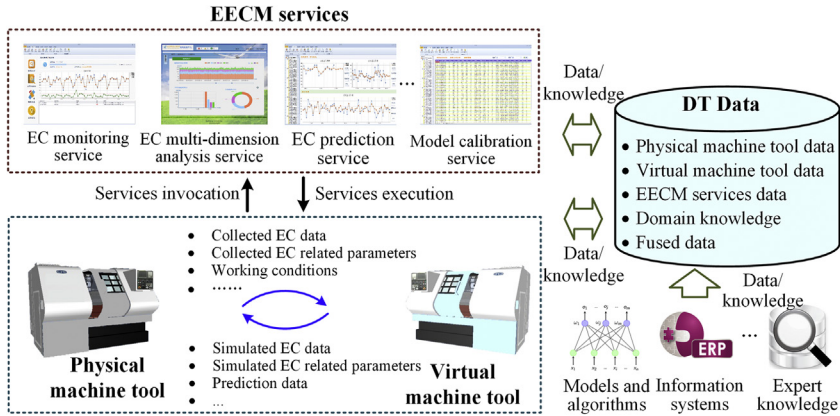


Figure 5.1 The framework of EECM for a machine tool.

physical and virtual machine tools, services, as well as information systems and expert knowledge databases in the external environment, feeding the framework with valuable information and knowledge.

5.3 IMPLEMENTATION OF EECM IN DIGITAL TWIN SHOP-FLOOR

According to the framework in Fig. 5.1, the implementation of EECM in DTS is explored from four aspects, including the implementation methods for the physical machine tool, virtual machine tool, EECM services, and data.

5.3.1 Physical Machine Tool

The physical machine tool is the target of EECM. Different sensors are mounted on it to collect the operation states and working conditions. In the framework of EECM, the EC data and related parameters are particularly critical. The former refers to the consumption of electricity, lubricant, compressed air, etc., while the latter mainly includes the following parameters [2]:

- *Machining parameters:* spindle speed, feed rate, cutting depth, etc.
- *Workpiece parameters:* Strength, hardness, plasticity, etc.
- *Tool parameters:* Angle, teeth number, material, wear, corrosion, cracking, etc.
- *Scheduling parameters:* Scheduling sequence, idle time, working time, etc.
- *Machine specifications:* Machining capacity, function, motor type, etc.

To obtain the EC data and related parameters for the physical machine tool, IoT technology is employed for the real-time perception and

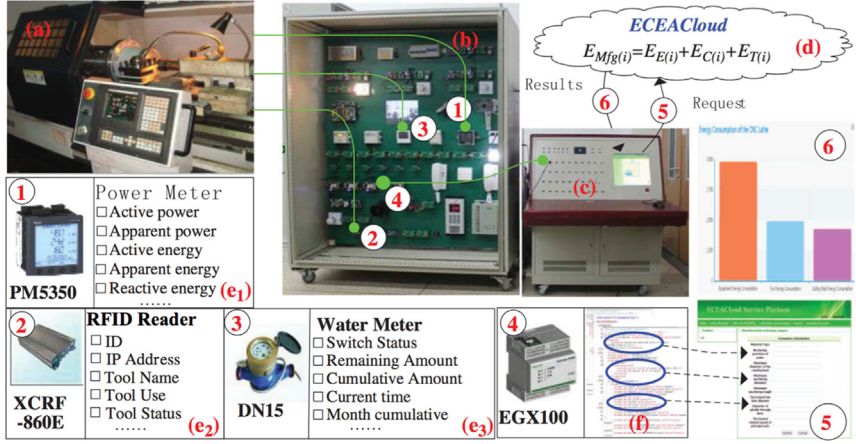


Figure 5.2 Data collection experimental bench for the physical machine tool [20].

collection, and an IoT-based intelligent perception experimental bench has been developed as in Fig. 5.2 [20]. It integrates over 30 intelligent IoT devices (e.g., smart electronic meters, sensors, and communication devices) and 40 data interfaces (e.g., serial port and USB interface). The collected data are sent to the virtual space using wired or wireless transmission equipment in the experimental bench [20].

For example, if one wants to obtain and analyze the EC of the physical machine tool (Fig. 5.2A), one can connect it with a smart power meter (i.e., PM5350 in Fig. 5.2(1)), RFID reader (Fig. 5.2(2)), and smart water meter (Fig. 5.2(3)) in the experimental bench. Then the information about power consumption (Fig. 5.2E1), tool (Fig. 5.2E2), and cutting fluid (Fig. 5.2E3) can be collected. The collected data are documented in a standardized format (Fig. 5.2F) and transmitted via the Ethernet gateway (i.e., EGX 100 in Fig. 5.2(4)) to the software system (Fig. 5.2C). The documented data will be recorded automatically as part of the input of the EECM services (Fig. 5.2(5)) and support the analysis results (Fig. 5.2(6)) [20].

5.3.2 Virtual Machine Tool

In the EECM, the functions of the virtual machine tool are described as follows [2]:

- *Recording:* Data from the physical side can be recorded using the virtual machine tool. If necessary, any situation can be played back.
- *Simulation and prediction:* Under the same working conditions as the physical machine tool, the virtual one can generate an amount of

simulated data and predicted data, including not only the simulated/predicted EC data and EC-related parameters, but also further analysis results.

- *Validation:* As the virtual machine tool can be adjusted with little cost, the machine tool operations can be validated in the cyber environment first, and then performed on the physical one, which reduces the risk involved in actual production processes.

The virtual machine tool can be constructed through building and integrating geometry, physics, behavior, and rule models. As in Fig. 5.3, the geometry model represents the shapes, sizes, and assembly relations of components in the machine, including the main body, chuck, axis, tool, etc. The physical properties, such as the functions (e.g., turning, milling, and grinding), capacities (e.g., workload, working stroke and accuracy), and key parameters (e.g., speed, force, torque) are described by the physics model. The behavior model is constructed to simulate the EC in different stages (e.g., machining stage, idle stage, and overload stage) and levels (e.g., part level, subsystem level, and entire machine level), as well as simulate the related parameters (e.g., cutting force). The simulated data can be compared with the actual ones for further analysis. Finally, the rule model concludes the relations among the EC-related parameters to support the EC evaluation and prediction.

5.3.3 EECM Services

In the EECM, the specific functions are realized by their corresponding services. There are two kinds of EECM services. The first mainly works for the physical machine tool for EC monitoring, analysis, and optimization. The second serves for the virtual machine tool, for example, model test and calibration services, ensuring the virtual mirror always keeps up with the physical entity [2].

EECM services are supported by various subservices, including data services, model services, algorithm services, and visualization services. The initial data, models, algorithms, and visualization methods are stored in the SDTD, which integrates the data from the PS, VS, SSS, and the domain knowledge. They are encapsulated as services in a unified form to show their functions and properties. For example, the data services describe the data type, value, source, and evaluation information. The model and algorithm services mainly describe their functions, quality of service (QoS), inputs, outputs, and parameter settings. The visualization

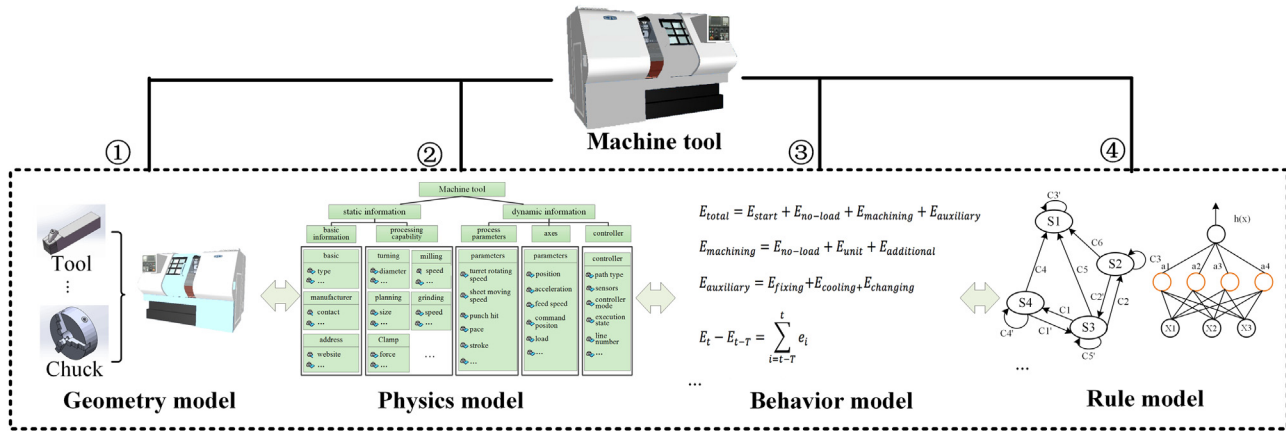


Figure 5.3 Modeling for the virtual machine tool.

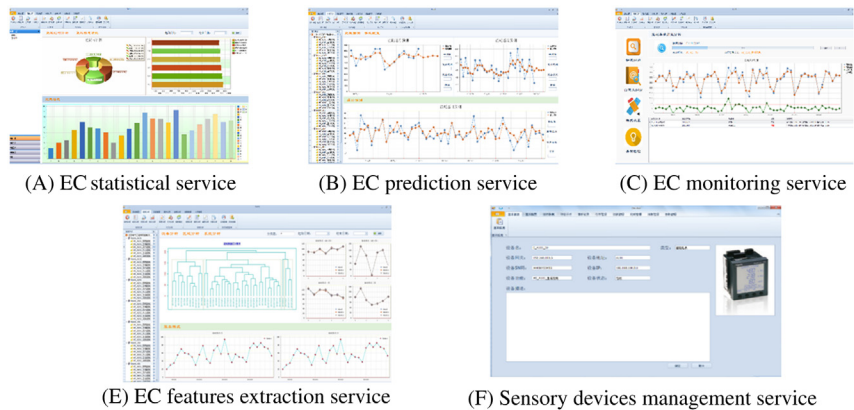


Figure 5.4 EECM services.

services provide the functions, QoS, inputs, and outputs. According to user demands, these subservices are invoked and combined to form a complete EECM service, which can be presented in a software-based representation with various visual forms [2].

Some EECM services are shown in Fig. 5.4. For examples, in Fig. 5.4 (A), the EC statistical service is fed by EC data in different time scales (e.g., a day, a week, or a month), which are analyzed using statistical algorithms. The statistical results are presented with pie charts and bar charts to illustrate the trend of the EC. In Fig. 5.4(C), the monitoring service is supported by the actual EC data and corresponding simulated EC data in the same period. These data can be compared to detect the abnormality in the EC, and then the abnormal deviation is exhibited with line charts.

5.3.4 Digital Twin Data

There are mainly five types of DT data, including the physical machine tool data, virtual machine tool data, EECM services data, domain knowledge, and fused data.

The physical machine tool data consist of equipment operation states and working conditions. Among them, the EC data and EC-related parameters are particularly important for the EECM, including the consumption of electricity, lubricant, and compressed air, and machining parameters such speed, force, and depth. The virtual machine tool data mainly include the model parameters and model operation data, such as the geometric parameters of the component model, simulated machining parameters, and simulated EC data. The EECM services data contain

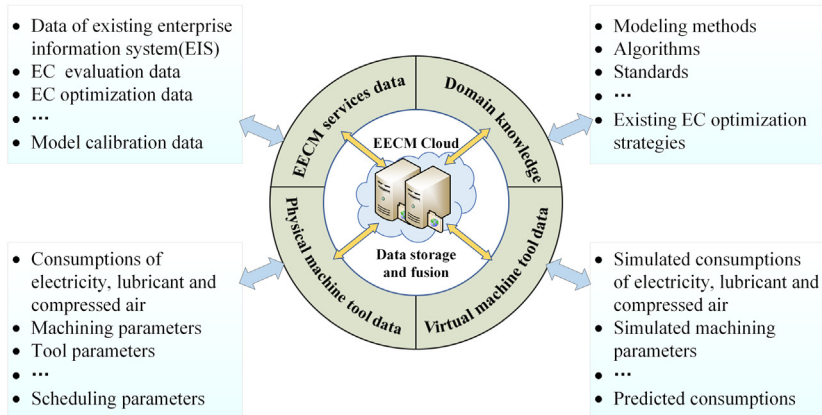


Figure 5.5 Data fusion in cloud environment.

service description data, operation data, and analysis result data, such as the service properties, optimized machining parameters, and optimized scheduling parameters. The domain knowledge can be mined from the accumulated data or obtained from the existing information systems and databases, including modeling methods, algorithms, and EC evaluation standards [2].

As shown in Fig. 5.5, the above four types of data can be stored uniformly and fused in the cloud environment, which is characterized by powerful computing ability. In this process, the data are first cleaned to fill missing values, reduce noise, and eliminate duplicate values, as well as inconsistent and abnormal values. Then the cleaned data can be compared, associated, and combined to obtain fused data through data fusion algorithms. For example, the features of EC (e.g., trend, cycle, and frequency) from different data sources can be extracted through algorithms, such as the principal component analysis (PCA), wavelet analysis, and Fourier transformation, and then fused using a neural network to present a more comprehensive description of the EC.

5.4 POTENTIAL ADVANTAGES OF EECM IN DIGITAL TWIN SHOP-FLOOR

Compared with the traditional mode, as the virtual machine tool is introduced and data from both the physical and virtual machine tools are available, functions of the EECM will become more comprehensive and accurate [2]. Potential advantages of the EECM, which have been studied

in the authors' previous work [2], are introduced in this section, including advantages on EC monitoring, analysis, and optimization.

5.4.1 Advantages in Energy Consumption Monitoring

EC monitoring can be performed in two ways: real-time to historical monitoring, and physical to virtual monitoring. In this way, both gradual and abrupt changes in the EC can be detected accurately.

Real-time to historical monitoring compares the real-time EC data collected from the physical machine tool with the historical data that have been recorded in the virtual machine tool. Since the EC may increase gradually with time due to the degradation of machine tool components (e.g., wear, corrosion, and misalignment), the monitoring can reflect the increase of the EC, and can thus present the current states of the machine. Based on the proposed EECM framework, as the historical data can be corrected through fusing historical sensory data from the physical machine tool and simulated data from the virtual machine tool, they will be more accurate than the initial historical data, thus supporting more precise monitoring.

Physical to virtual monitoring compares the actual EC from the physical machine tool with the simulated EC from the virtual machine tool. In a normal situation, as working conditions of the physical and virtual machine tools are set as the same, data from the two parts are always consistent. However, inconsistency may appear when some sudden disturbances from the physical world (e.g., power grid fluctuation, power leakage, and short circuit) occur. In this situation, data from the virtual machine tool will be taken as the reference to detect the abrupt change of the EC in the physical machine tool. As the virtual machine tool evolves simultaneously with the physical one, the reference data are also updated with time, which makes the monitoring of EC changes more accurate, compared with the traditional EC monitoring method using static references.

5.4.2 Advantages in Energy Consumption Analysis

In the proposed EECM framework, as data from both the physical and virtual machine tools are available, the EC analysis can be fed with sufficient data sources, which enables high-quality analysis. Its advantages in this aspect are introduced mainly in multilevel and multistage analysis, behavior analysis, and statistical analysis.

Multilevel and multistage analysis. Considering the costs of sensory devices, it is impractical to deploy many sensors on the physical machine tool to collect all the EC data at multiple levels (e.g., part level, subsystem level, and entire machine level) and multiple stages (e.g., machining stage, idle stage, and overload stage). Hence, soft sensing has been introduced by some researchers [21]. In the EECM, large quantities of models and data from both the physical and virtual spaces can be obtained, so that clearer relations between the EC data and related parameters can be captured. Therefore, it becomes achievable to build more accurate soft sensing models to calculate the EC at multiple levels and stages, thus reducing high costs from installation and maintenance of the physical sensory devices.

Behavior analysis. The machine tool behavior (e.g., normal, overload, failure) can be analyzed through fusing features from both the physical and virtual machine tools. From the physical aspect, the features can be extracted from the EC data and related parameters in the time domain or frequency domain. From the virtual aspect, features can be extracted from further analysis and prediction. As the features can be obtained from both the physical and virtual machine tools, the behavior analysis can be conducted in a more comprehensive way.

Statistical analysis. It is often necessary to obtain the EC analysis results with statistical significance. As the virtual machine tool operates in the cyber environment with no costs or limits, large amounts of accurate simulated data can be obtained to satisfy the data quantity requirements.

5.4.3 Advantages in Energy Consumption Optimization

In EC optimization, based on the proposed EECM framework, parameter optimization, scheduling optimization, and machine tool upgrading are introduced.

Parameter optimization is a complex optimization problem that not only aims at minimizing the machine tool EC, but also considers a set of constraints, such as the product quality, productivity, delivery time, etc. Searching for the optimal solution to this problem is always difficult with mathematical modeling. However, as the virtual machine tool provides a digital mirror in high fidelity with the physical one, the optimization can be performed in the cyber environment through parameter testing and modifying. In addition, as the virtual machine tool records the operations of the physical machine tool, the historical parameter settings under the same condition can be used as references.

Scheduling optimization. During production, task completion time of the physical machine tool can be accurately predicted by the virtual counterpart. Based on the predicted time, scheduling of the physical machine in the subsequent time can be adjusted in advance, thus shortening the machine idle time to reduce the EC generated in the idle period.

Machine tool upgrading. In the proposed EECM, interactions between the designer and the physical machine tool can be enabled during the machine upgrading process. First, based on the real-time data collected from the physical side, the virtual machine tool that can analyze and reflect the actual EC performance of the physical one can be provided through the Internet to the designer, who will be encouraged to generate new design ideas to pursue lower EC. Then the ideas can be fed back to the virtual machine tool to reflect the expected performances from the designer. Deviation between the actual and expected performances can be analyzed through comparison, and the corresponding upgrading strategies can be generated and validated by the virtual machine tool. Finally, the validated strategies will be performed on the physical machine tool to realize the upgrade.

5.5 SUMMARY

The proposed EECM is an effective way to reduce the EC and improve energy efficiency of the equipment. With the developments of DT and DTS, the data and models are enriched greatly, which provides possibilities to improve and optimize the EECM on the shop-floor. In this context, this chapter has explored the application of EECM in DTS and discusses the framework, implementation methods, and potential advantages in detail. Future works will concentrate on the following aspects [2]:

- further perfection of the models,
- multisource data fusion and management,
- EECM system establishment in DTS.

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CHAPTER 6

Cyber—Physical Fusion in Digital Twin Shop-Floor

As an efficient way to realize seamless interaction, integration, and fusion between the physical space and the virtual space for manufacturing, the concept of digital twin shop-floor (DTS) was proposed in Chapter 4, Digital Twin Shop-Floor. In this chapter, cyber—physical fusion in the DTS is studied from four aspects, including fusion of physical elements in physical shop-floor (PS), model fusion in virtual shop-floor (VS), data fusion in shop-floor digital twin data (SDTD), and fusion of services in shop-floor service systems (SSS). Related criteria and key technologies for these fusions are also discussed.

6.1 INTRODUCTION

To adapt to the manufacturing needs and trends such as socialization, personalization, servitization, intelligence, and greenization, various countries have proposed different national-level manufacturing development strategies as mentioned in Chapter 4, Digital Twin Shop-Floor. One of the common goals of these strategies is to realize cyber—physical fusion, and consequently to achieve smart manufacturing. On the shop-floor, which is the basic “playground” of manufacturing activities, digital twin shop-floor (DTS) has been proposed. For the DTS, cyber—physical fusion is not only the goal, but also one of the core challenges in practice [1]. To realize this fusion, it is essential to implement fusion in the major components of the DTS, respectively. These four processes include fusion of heterogeneous physical elements on the physical shop-floor (PS), fusion of multidimensional models on the virtual shop-floor (VS), cyber—physical data fusion on the shop-floor digital twin data (SDTD), and services fusion on the shop-floor service system (SSS).

Some existing works have laid foundations for the above fusions, including connection and interconnection on the shop-floor [2,3], digital/virtual shop-floor modeling [4,5], shop-floor data and information integration [6,7], shop-floor optimal operations and precision management [8,9],

etc. Based on this, many valuable theories and technologies have been developed to solve the problem of fusion at the shop-floor level from different perspectives. However, to realize the fusion completely, there are still some limitations and deficiencies, which are outlined as follows.

1. Connection and Interconnection on the Shop-Floor

Previous researchers have mainly focused on theoretical, technical, and methodological studies for a single physical element (e.g., a person, a machine, a product) on the shop-floor, such as works on data collection, data transmission and processing, state monitoring and health management, etc. Commercially available devices have also been developed to implement the related functions. Based on this, the smart operations for a single element can be realized to some extent. However, there has been insufficient attention paid to the interconnection between the heterogeneous physical elements (e.g., machine-machine, man-machine, machine-environment, human-machine-environment) on the shop-floor. Particularly, the connection and interconnection of all the heterogeneous physical elements, including persons, machines, materials, and environment are lacking. As a result, theories and universal devices for full interconnection and fusion of the heterogeneous physical elements on the shop-floor would need to be pursued.

2. Digital/Virtual Shop-Floor Modeling/Simulation

Previous researchers mainly focused on the construction and simulation of geometry models or system models for the physical elements on the shop-floor, while models that can depict the actual behaviors, rules, and constraints are relatively rare; as a result, a complete mapping from physical elements to virtual models cannot be achieved. In addition, with respect to the shop-floor simulation, previous research mainly relies on the existing data-driven models, and lacks sufficient explorations into the precise relations among shop-floor behaviors, rules, and constraints, as well as their integration methods. Little attention is paid to multidimension fused models that can describe the shop-floor accurately and thoroughly from various aspects such as geometry, physics, behaviors, and rules.

3. Shop-Floor Data/Information Integration

Based on information systems or platforms such as manufacturing execution systems (MES), enterprise resource planning (ERP), and computer-aided

process planning (CAPP), the information and data involved in the shop-floor can be integrated and shared. Although the integration scope covers the existing data from information systems deployed on the shop-floor and part of the collected real-time data from production, data from the upstream business (e.g., market demand and product design), and downstream business (e.g., product maintenance and disposal) are seldom considered. In addition, simulated data from the virtual models and fused data from both the physical elements and virtual models have attracted little attention. In summary, seamless integration and fusion for data from all of the elements, processes, and businesses on the shop-floor are insufficient.

4. Shop-Floor Optimal Operations and Precision Management

The traditional method for optimal operations and precision management is mainly based on the process of problem analysis—model construction—algorithm design—optimization analysis and control. With Internet of Things (IoT) and big data applied to the shop-floor, researchers around the world have proposed a new idea, which transforms the traditional process to data association mining—dynamic evolution—simulation/prediction—intelligent control based on shop-floor big data. Both methods rely on models and data from the shop-floor. However, owing to the lack of multidimension fused models and data, the current shop-floor still has various deficiencies in operations and management, such as poor consistency between the physical and virtual spaces, low intelligence, and low accuracy. Therefore, to solve these problems, it is urgent to develop reliable services and service fusion that are driven by data and models.

All the deficiencies mentioned above can be summed up as one scientific problem, that is, cyber—physical fusion on the shop-floor. To solve the problem, digital twin (DT) technology is introduced to the shop-floor and the concept of DTS is proposed. In the DTS, through constructing fully interconnected PS, high-fidelity VS, fully integrated and fused SDTD, and highly precise SSS, it becomes possible to realize the cyber—physical fusion on the shop-floor, thus providing a new solution for enterprises to carry out optimal operations and precision management [1].

6.2 REFERENCE ARCHITECTURE FOR DIGITAL TWIN SHOP-FLOOR

To provide references for enterprises to realize the DTS in the future, a five-layer reference architecture for the DTS is designed, as shown in

Fig. 6.1 [1]. These layers include the physical layer, model layer, data layer, service layer, and application layer. Each layer is introduced as follows [1]:

- 1. The physical layer mainly refers to entities such as persons, machines, and materials, and a set of production activities and processes in the PS. It is mainly responsible for production in the physical space, and has capabilities such as data collection and transmission, interconnection, interaction, and cooperation, etc.
- 2. The model layer mainly includes various virtual models combined with real-time data and knowledge in the VS. It produces multidimensional analysis, ideas for optimization, and predictive states to support decision making in the physical space.
- 3. The data layer mainly refers to the data in SDTD. It can provide data to drive the operations of PS, VS, and SSS, and to implement data lifecycle management, such as data processing, integration, and fusion.
- 4. The service layer mainly provides the shop-floor with production services including smart production scheduling, collaborative process planning, product quality management, production process management and control, equipment health management, and energy efficiency optimization analysis.

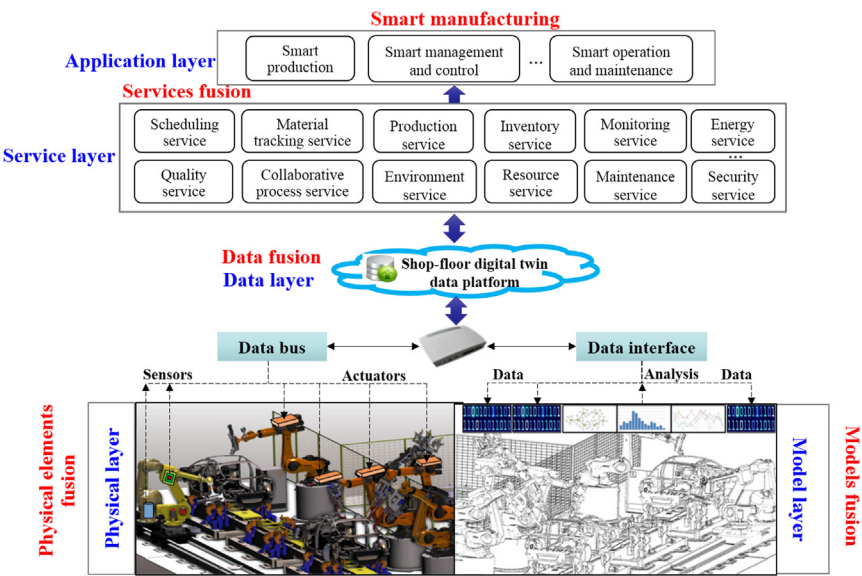


Figure 6.1 Reference architecture for DTs.

5. The application layer mainly refers to practical applications toward specific products during manufacturing, such as smart production, smart management and control, smart operation and maintenance, etc.

In consideration of Fig. 6.1 and the previous analysis of the four components of DTS [10] (PS, VS, SDTD, SSS), the problem of DTS cyber–physical fusion is divided into four key issues, namely, fusion of heterogeneous physical elements in PS (physical elements fusion), multidimensional models fusion in VS (models fusion), cyber–physical data fusion in SDTD (data fusion) and fusion of services in SSS (services fusion), each of which is discussed as follows.

6.3 PHYSICAL ELEMENTS FUSION

To realize smart connection and interconnection of the physical elements on the shop-floor, researchers around the world have made substantial studies, such as data collection and monitoring based on radio frequency identification (RFID) [11], wireless sensor networks (WSN) [12], and intelligent instruments [2]; shop-floor interconnection based on manufacturing service bus [13], Profinet [14], OPC UA [15], AutomationML [16], and IoT [17]; and devices for manufacturing resource perception and access based on Industrial Internet hub solutions [18]. However, in these works, there are still deficiencies such as insufficient interconnection and interaction among the physical elements and incomplete collection of data. Hence, this section puts forward the scientific problem of “person–machine–material–environment interaction and fusion” (i.e., physical elements fusion).

As shown in Fig. 6.2 [1], the physical elements fusion can be achieved from four levels, including the smart connection and interconnection, efficient data communication and integration, real-time control and interaction, and smart cooperation and convergence. Realizing the four-level fusion needs to achieve massive data collection for the heterogeneous physical elements, reliable real-time data transmission to the information layer, multisource multimodal heterogeneous data integration, and precise control of behavior coordination in the hybrid dynamic environment.

The goal of physical elements fusion is to provide real-time data support for realizing models fusion, data fusion, and services fusion in the DTS, completing physical elements smart connection and interconnection, and integrating multisource heterogeneous data. Based on this, physical elements fusion can further realize the following goals.

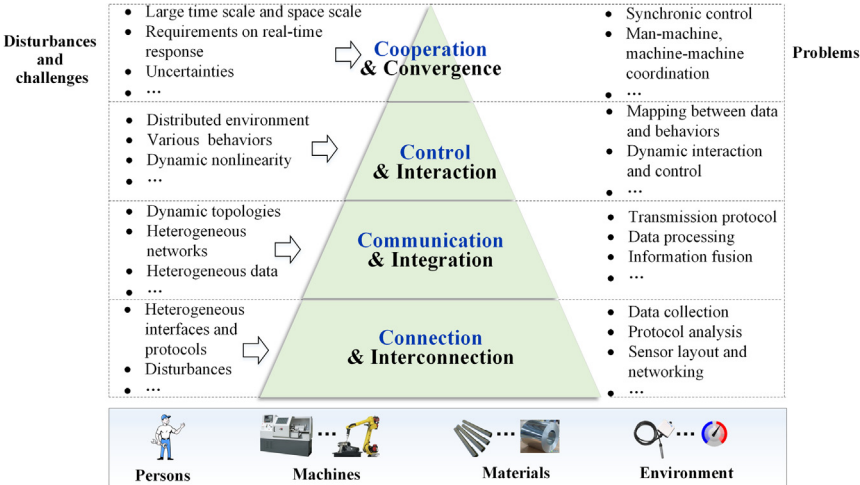


Figure 6.2 Physical elements fusion in PS.

1. *Man—machine interconnection and collaboration*: Accuracy of machines and flexibility of humans should be combined, so that the machine can adjust its work plan quickly to adapt to dynamic and varied production status, and respond to orders from persons through sound, gesture, or software tools, etc.
2. *Machine—machine interconnection and collaboration*: Manufacturing machines in the PS should operate collaboratively, which gives the machines good adaptability, required redundancy, and high robustness.
3. *Machine—material interconnection and collaboration*: Based on RFID and other sensory technologies, smart perception and location for materials, work-in-process (WIP) and products, accurate resource-tracing and smart storage should be implemented.

As shown in Fig. 6.2, realization of physical elements fusion mainly involves the following theories and technologies [1].

1. Man—Machine—Material—Environment Smart Connection and Interconnection

This includes comprehensive perception methods for all the elements, processes, and businesses in the hybrid dynamic PS, such as plug-and-play protocol analysis methods based on selfadaptive matching, optimized layout, and dynamic networking mechanisms for multisensor groups; measurement theories in multitemporal and spatial scales; and universal devices for smart interconnection of the heterogeneous physical elements [18].

2. Man—Machine—Material—Environment Smart Communication and Computing

This includes data transmission protocols of high speed, high stability, and low latency; real-time reliable link scheduling technologies; multiscale heterogeneous data fusion and service encapsulation methods; and dimension reducing processing and distributed storage methods for massive information.

3. Man—Machine—Material—Environment Smart Control and Interaction

This includes semantic mapping rules between multimodal data and object behaviors, key process modeling and control, data-driven selfadaptive control methods, and dynamic interactive real-time control methods.

4. Man—Machine—Material—Environment Smart Cooperation and Convergence

This includes distributed collaboration control theories, virtual reality (VR) and augmented reality (AR) technologies, shop-floor operation pattern recognition and knowledge mining, and SDTD-driven man—machine, machine—machine, and machine—material collaborative control methods.

6.4 MODELS FUSION

In shop-floor modeling, researchers mainly focus on the construction of geometry models, such as the modeling and simulation of shop-floor layout [19], assembly processes [5], machining processes [20], and shop-floor logistics [21]. Based on the geometry models, some other researchers have further explored the modeling of physical factors (e.g., machines' physical properties, process parameters, production capacity), and then simulate the changing of physical parameters [22] and the process plan evaluation [23]. Although the modeling and simulation of both the geometric and physical factors have been discussed, the description and simulation of actual behaviors, rules, and constraints of the shop-floor are seldom studied. In particular, the VS that can comprehensively fuse different models to completely mirror the PS is lacking.

As shown in Fig. 6.3 [1], to realize the complete mapping between the PS and the VS, it is necessary to build models of the shop-floor from

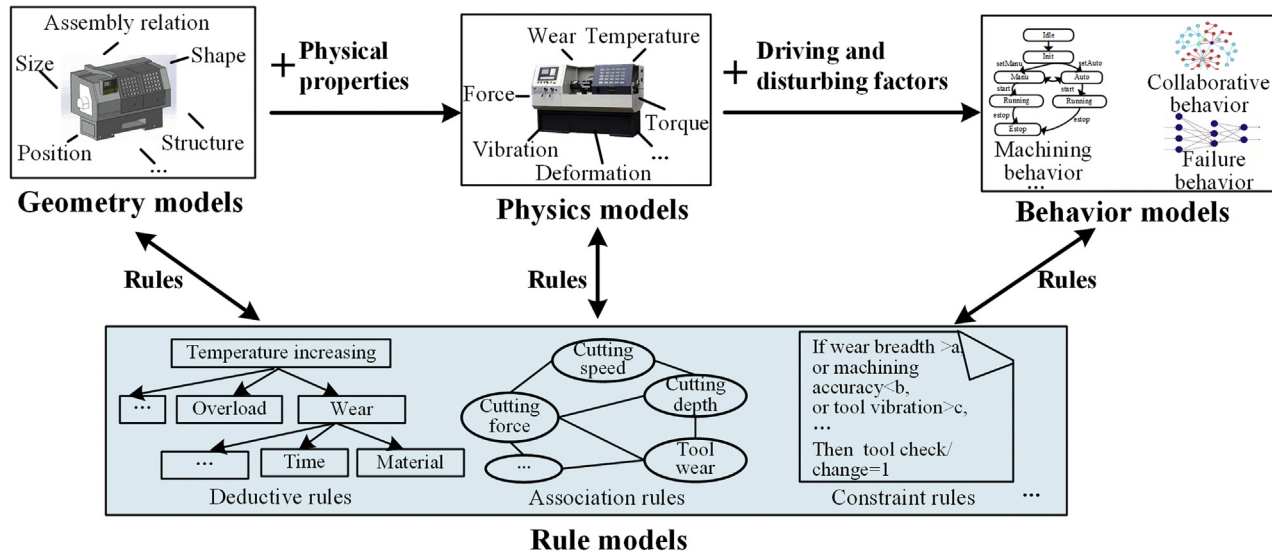


Figure 6.3 Multidimension models fusion in VS.

multiple dimensions including geometry, physics, behaviors, and rules. Meanwhile, the established models should be verified to ensure correctness and effectiveness. Based on this, the models of all dimensions can be correlated, combined, and integrated to form a complete VS model with high fidelity in the virtual space. Hence, this section puts forward the scientific problem of geometry—physics—behaviors—rules based multidimension models fusion (i.e., models fusion). Technologies in models fusion mainly involve multidimension model construction, evaluation and verification, correlation and mapping, and fusion and consistency analysis, which are specified as follows [1].

1. Construction of the Multidimension Models

To realize the multidimension models fusion for the VS, the models must be constructed first. It means that the multidimension models must be built to describe the PS from geometric shapes, physical properties, behavior responses and rules, etc.

2. Evaluation and Verification of the Multidimension Models

After the models are constructed, their effectiveness and correctness must be ensured. It is necessary to verify the constructed multidimension models based on verification, validation, and accreditation (VV&A), and to verify the accuracy of the models' inputs and outputs, simulation confidence, and sensitivity.

3. Correlation and Mapping Mechanism of the Multidimension Models

To realize the fusion of the multidimension models, it is necessary to explore and analyze the correlation and mapping mechanism among the models in each dimension under the premise that the previously constructed models are correct. The geometry models and physics models describe the heterogeneous physical elements on the shop-floor. Based on these models, the behavior models can be built by adding driving and disturbing factors, so that the corresponding elements have the behaviors, response mechanisms, and abilities to perform complex actions. The rule models characterize the rules reflected in geometric parameters, physical properties, and behaviors of the elements, so that the VS is provided with the capability of evaluation, evolution, and inference. Through establishing the correlation and mapping mechanism among the models in

different dimensions, the models can be integrated and fused in both structure and function to form a comprehensive mirror for the PS, with a unified three-dimensional representation.

4. Theory and Method of the Multidimension Models Consistency

To maintain consistency and synchronism of the models in the VS with the entities in the PS, it is necessary to ensure consistency between the models and the corresponding entities on one side, and consistency between the models in different dimensions for the same entity on the other.

6.5 DATA FUSION

Shop-floor manufacturing data have the same features as big data [8,24,25] such as massive volume, multiple sources, heterogeneity, multiple scales, and multiple dimensions. Through establishing relationships between the production process and operational decision making on the shop-floor with big data, the operation rules of the shop-floor can be analyzed, which contributes to improving production efficiency [25,26] and product quality [27], reducing energy consumption [28], and keeping machines in healthy condition [29]. Although relevant studies have achieved integration and fusion of part data on the shop-floor, the integration and fusion of data from all of the elements, processes, and businesses under the cyber–physical interaction environment have not been fully realized. For example, when a mobile phone casing is manufactured, it is required to consider not only the design data, such as style, color, user evaluation, and material, but also the real-time processing data in the PS such as processing temperature, pressure, and surface roughness. Meanwhile, the order data such as processing quality, delivery time, and production tasks also cannot be neglected. Therefore, this section puts forward the scientific problem of fusion of cyber–physical data from entire elements, processes, and businesses (i.e., data fusion).

Data fusion refers to the generation, modeling, cleaning, correlation, clustering, mining, iteration, evolution, and fusion of data from all of the elements, processes, and businesses based on the physical elements fusion, models fusion, and running consistency principles of the shop-floor. Through data fusion, various dynamic evolution processes, evolution rules, and statistical characteristics of the shop-floor can be effectively

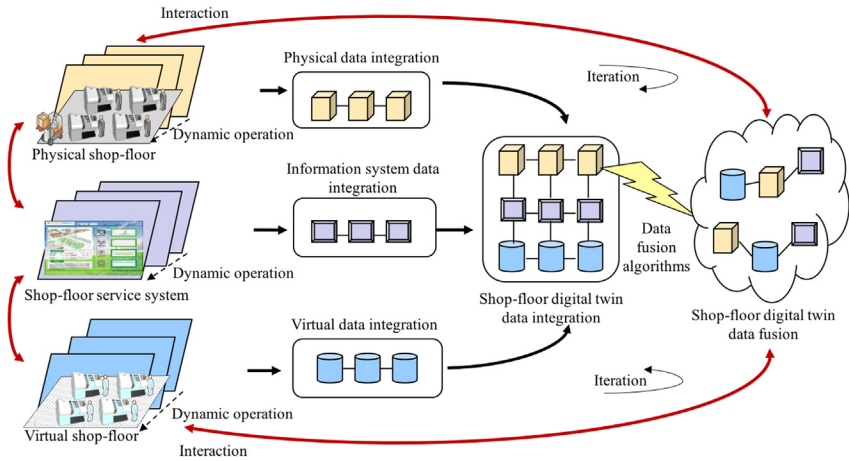


Figure 6.4 Data fusion in SDTD.

described and reflected. As shown in Fig. 6.4 [1], data fusion mainly involves the following criteria and technologies.

1. Data Generation, Modeling, and Cleaning

This includes real-time generation mechanisms of data in SDTD, such as complicated data classification and modeling, empty value estimation and feature reduction for incomplete information, data cleaning, and verification of data accuracy, integrity, and consistency.

2. Data Correlation, Clustering, and Mining

This includes correlation, such as comparison and integration of the real-time data and the historical data in SDTD; mapping between the data and operations; clustering and fusion of the data from different levels, periods, and spaces; data tracing; feature extraction; and behavior pattern recognition.

3. Data Iteration, Evolution, and Fusion

This includes interactive iteration and optimization of the data from PS, VS, and SSS, such as dynamic growth rules of the data, dynamic evolution of data correlation, data evolution rules, and statistical characteristics analysis.

6.6 SERVICES FUSION

The SDTD that fuses the physical and virtual data not only can reflect the states of the PS and the VS, but also drive their operations. The actual demands during the operation of the DTS, such as energy consumption management, precise control, and predictive maintenance, involve a series of complex and dynamic smart decision-making problems. The complexity and dynamism are due to the dynamic and uncertain environment, uncertain states of physical elements, dynamic changes of production tasks, and diverse objectives and constraints. Services can be used to build an important bridge between the manufacturing physical space and the virtual space, which helps to realize smart interconnections and operations [30,31]. The ultimate goal of the physical elements fusion, models fusion, and data fusion in the DTS is to provide various services required in the production based on the SDTD in combination with the existing information systems (e.g., MES). The services herein include production scheduling, product quality management, collaborative process planning [32], production process control, equipment health management, and energy efficiency optimization analysis [33]. Accordingly, smart production and management on the shop-floor can be achieved through fusion and collaboration of the services through dynamic service invocation, scheduling, and combination. The above process is defined as the scientific problem of data-driven services fusion (i.e., services fusion). The shop-floor service fusion and application mainly involve the following criteria and technologies [1].

1. Data-Driven Service Generation

This includes conversion methods from data to services; multilevel mapping mechanisms among physical elements, virtual models, data, and services; digital description and encapsulation for services; and services aggregation.

2. Service Smart Management and Optimization

This includes construction of shop-floor service models, comprehensive assessment of quality of service (QoS), service online monitoring, supply–demand matching mechanisms [34,35], as well as service optimization configurations.

3. Service Fusion and Application

This includes demand analysis for shop-floor complex coupling application, dynamic supply—demand matching mechanisms, service dynamic scheduling, service collaboration and fusion based on service combinations [36,37], and reliability assessments and improvement strategies for service collaboration.

6.7 SUMMARY

Cyber—physical fusion is a common challenge for advanced manufacturing strategies, and also a key scientific problem that needs to be solved for the deployment of the DTS. At present, the DTS has the following shortcomings in production and operation processes: (1) data collection on the shop-floor is incomplete; (2) virtual/digital models are difficult to establish and simulation analysis lacks systematism; (3) bidirectional dynamic mapping and fusion for the physical and virtual data are insufficient; and (4) dynamic and flexible combinations of services toward specific demands are lacking.

Based on research on the concept, composition, operating mechanisms, characteristics, and key technologies of the DTS, this chapter designs a reference framework for the DTS and further proposes the scientific problem, that is, the cyber—physical fusion in the DTS. With respect to the composition of the DTS, cyber—physical fusion is divided into four different perspectives, including the physical elements fusion in PS, models fusion in VS, data fusion in SDTD, and services fusion in SSS. The corresponding criteria and key technologies for these fusions are also discussed. This chapter aims at providing a theoretical and technical reference for researchers to carry out studies on the cyber—physical fusion for the DTS.

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CHAPTER 7

Digital Twin-Driven Prognostics and Health Management

Prognostics and health management (PHM) is critical to improve reliability and decrease downtime cost for complex equipment working in harsh environments. With the development of cyber–physical system (CPS), realizing cyber–physical fusion to optimize traditional PHM is an important breakthrough. In this chapter, digital twin (DT), an emerging technology for cyber–physical fusion, is introduced to improve the efficiency and reliability of PHM. A five-dimension DT is constructed to emulate physical equipment thoroughly with high fidelity. Based on the constructed DT, a PHM method consisting of the framework, procedure, and coevolution mechanism that makes efficient use of digital mirroring, continuous interactions, and fused data in DT is presented in detail. The proposed method is applied to the yaw system and the gearbox of a wind turbine (WT).

7.1 INTRODUCTION

Complex equipment, such as aircrafts, ships, and wind turbines, usually work over decades in harsh environments that are full of randomness and uncertainty [1]. Hence, degradation and failure of the equipment are inevitable during the operation, which leads to downtimes, increased maintenance costs, and postponement of tasks. To address these problems, prognostics and health management (PHM) is introduced as an effective way to support the reliable operation of complex equipment [27].

PHM is developed based on maintenance methodologies and diagnostic techniques [2]. The earliest maintenance practice is breakdown maintenance, which replaces or repairs failed components when the equipment appears to have obvious malfunctions. As this type of maintenance is unplanned, it always upsets the initial plan and affects the completion time of the task. To improve this situation, scheduled maintenance or planned maintenance is introduced. It can be preplanned according to the experts' recommendations and periodically performed to keep the

equipment in good condition through cleaning, maintaining, repairing, calibrating, and inspecting. But as this maintenance is carried out based on predefined intervals, it lacks flexibility and cannot deal with unexpected failures. More recently, condition-based maintenance can analyze the equipment conditions in real-time and acts only when the maintenance is actually needed. It can detect and eliminate both the expected and unexpected faults efficiently in time, and thus decreases the maintenance cost. PHM has evolved from condition-based maintenance, and now has become a more integrated and mature approach that enables health management related to the lifecycle of the equipment [2,3].

Generally, PHM can be classified into three stages, as shown in Fig. 7.1, including observation, analysis, and decision [4]. In the observation stage, real-time data from the complex equipment need to be collected and processed. The real-time data are then compared with expected values in the analysis stage to monitor the current health conditions of the equipment. When the equipment has degraded, prognosis and diagnosis are performed to isolate and identify the fault cause, and predict the remaining useful life (RUL). According to the analysis results, while considering constraints on different aspects, the decision creates maintenance strategies that are presented to persons through the human–machine interface and conducted to ensure the equipment can complete the task.

Related works on these three stages are reviewed as follows:

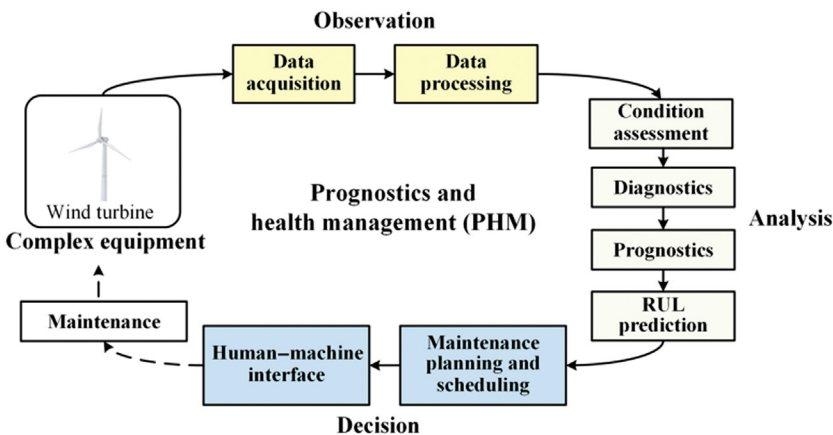


Figure 7.1 The process of PHM for complex equipment.

1. In the observation, sensory devices such as current sensors, torque transducers, and force sensors [5–7] are mounted on the equipment for ubiquitous data collection. Based on the collected data, the conditions of the equipment can be estimated through signals in the time domain [8], frequency domain [9], and time–frequency domain [10].
2. In the analysis, there are three main methods: data-driven, model-based, and hybrid methods [2]. The data-driven method trains the diagnostic or prognostic models based on features directly extracted from the collected data and neglects the complex internal relations among various variables [11]. The model-based method aims at building precise mathematical models to present the physical processes based on the fundamental understanding of the equipment [12]. The hybrid method is the combination of the above two methods and its function can be enhanced through learning from the two sides [13].
3. In the decision, according to the analysis results, maintenance plans are generated to decide the maintenance time, tools and technicians, etc., considering constraints such as the maintenance cost, spare part inventory, and equipment reliability [14,15]. In addition, human–machine interfaces can provide valuable information in various virtual forms [2].

The above works explore the methods in different stages for the PHM. In recent years, more researchers have made an effort to improve the PHM with the new generation information technologies (New IT). Deutsch and He used a deep learning-based approach to predict RUL of rotating components with big data on vibration signals, which enabled automatic feature extraction and RUL prediction without human intervention [16]. Gao et al. discussed the cloud-enabled prognosis in manufacturing, where data across the manufacturing hierarchy are integrated in the cloud environment to improve efficiency [17]. Combining with a cloud platform and the Internet of Things (IoT), a remote monitoring system for a motor was built, providing on-demand monitoring services and allowing quick and accurate decision making [18]. Zaluski et al. developed a data mining-based prognostic model for C-18 aircraft driven by a variety of data, such as the flight data from sensors, built-in test equipment data, and maintenance data [19]. Nabati and Thoben studied the characteristics of data collected from offshore wind turbines to decide whether these data were suitable to be used in big data analytics for maintenance of the turbines [20]. An IoT-embedded cloud architecture was built by Lee to detect and analyze manufacturing equipment status considering overall manufacturing processes [21].

Existing works have contributed much to PHM. However, there is a common deficiency for most of them as they are primarily driven by the equipment in its physical space, with little connection to its virtual model, which affects the efficiency and reliability of PHM. Currently, with the development of cyber–physical system (CPS), attaching importance to the virtual space and implementing the seamless cyber–physical fusion to improve PHM are inevitable trends. Therefore, some researchers have tried to introduce digital mirrors of the equipment from the virtual space to work with the real entities to better support PHM. For example, Lee et al. proposed to build a digital twin model that managed hundreds of similar physical equipment with integrated data from multiple sources, to assess the equipment condition and identify the failure mode through simulation [22,23]. Liu and Xu investigated the concept of a cyber–physical machine tool (CPMT), which integrates the machine tool with its cyber twin that has the ability to transform the data into valuable information for PHM [24]. Zinnikus et al. presented a framework for the fault analysis of a cyber–physical production system, which compares the observations of the physical equipment with the behaviors of the models for anomaly detection and fault diagnosis [25]. Penna et al. proposed a visualization tool for a cyber–physical maintenance environment, which relates the physical elements with virtual ones and makes remote access possible via a web browser [26].

Although these works have attempted to introduce the digital models of complex equipment into PHM, to implement PHM driven by both physical and virtual spaces, some common problems still exist. They include (1) how to build the models to describe the complex equipment thoroughly and accurately, (2) how to establish the interaction mechanism between the equipment and its digital mirror model to make them evolve jointly and support PHM together, and (3) how to fuse the data from both physical and virtual spaces and generate valuable information for PHM. However, few studies have been focused on these problems [27].

As DT has the potential to realize cyber–physical fusion, this chapter proposes a DT-driven PHM method. First, the DT for the complex equipment is built, including the physical equipment (PE), virtual equipment (VE), services (Ss), connection (CN), and DT data (DD). Second, the DT-driven PHM method, including the framework, the three-stage procedure, and the coevolution mechanism is explored. The framework defines the inputs, outputs, and roles of the DT in the proposed method; then the procedure is designed to deal with two types of faults (the

gradual fault and abrupt fault) with different workflows; and the coevolution mechanism illustrates the evolution process for different parts in the DT during PHM. Finally, to verify the efficiency and advantages of the proposed DT modeling and DT-driven PHM methods, the yaw system and the gearbox of a wind turbine (WT) are taken as examples in the case study. Based on the DT, the above three common problems existing in the current work of PHM can be solved as follows: (1) According to the construction of the DT, a five-dimension DT for the equipment can be built, which describes the real equipment in multiple aspects and provides access to the equipment even out of physical proximity. (2) Utilizing the interaction mechanism in the DT, the difference between the physical equipment and the digital mirror can be used to detect the disturbances from the external environment, potential faults in the equipment, and defects in the virtual models. It is a coupled optimization for both the equipment and its digital mirror to better serve PHM. (3) Since the DT generates data from the equipment, the digital mirror, and the fused data, data for PHM activities (e.g., condition assessment, feature extraction, and prognostics) are enriched greatly, which can provide more comprehensive and accurate information [27].

7.2 DIGITAL TWIN FOR COMPLEX EQUIPMENT

7.2.1 Five-Dimension Digital Twin for Complex Equipment

According to the proposed five-dimension definition of the DT in Chapter 3, Five-Dimension Digital Twin Modeling and Its Key Technologies, a DT concept model for complex equipment is built as shown in Fig. 7.2 [27].

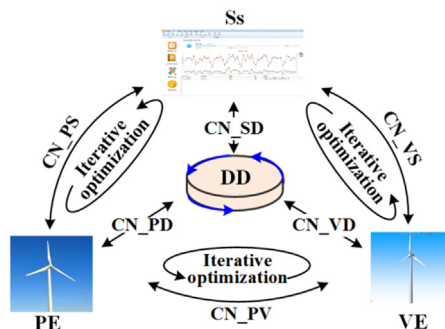


Figure 7.2 Five-dimension DT concept model for complex equipment [27].

The DT includes five components, that is, PE, VE, Ss, DD, and CN. PE refers to the physical equipment, operating in the real environment and completing prescribed tasks. VE represents the virtual equipment, which is the high-fidelity digital counterpart of the PE. It is composed of a set of models in multiple dimensions to describe the PE thoroughly. Ss stands for the services for both PE and VE. It optimizes the operations of the PE, and ensures the high fidelity of the VE through calibrating the VE parameters during its operation. DD is the DT data, including data from PE, VE, Ss, and domain knowledge, as well as the fused data. As the DD includes data from both physical and virtual aspects as well as their fusion, it can provide more comprehensive and accurate information. CN is the connection that ties different parts of the DT together [27].

7.2.2 Modeling for Each Dimension of Digital Twin

Then, specific models of the five-dimension DT are built according to the DT modeling methods explored in Chapter 3, Five-Dimension Digital Twin Modeling and Its Key Technologies. To illustrate the models, a WT is taken as an example of complex equipment. For each dimension, some typical models, which have been studied in authors' previous work [27], are shown in Fig. 7.3 and illustrated as follows.

1. *PE (i.e., physical WT)*. The physical WT includes mainly the subsystems and the mounted sensors. The subsystems consist of the blade, generator, gearbox, and yaw system for transforming wind energy into mechanical and electrical energy. Sensors are deployed to collect the generator temperature, gearbox vibration, and power output.
2. *VE (i.e., virtual WT)*. According to expression (2) in Chapter 3, Five-Dimension Digital Twin Modeling and Its Key Technologies, the virtual WT consists of a geometry model, physics model, behavior model, and rule model, expressed as G_v , P_v , B_v , and R_v , respectively. In G_v , the components (e.g., the blade, gearbox, and shaft) are constructed in three dimensions, then assembled to form the geometrical WT. In P_v , the blade deformation, gear tooth stress, bearing temperature, etc. can be simulated using the finite element method (FEM) to reflect the physical properties. B_v includes the power generation, yawing, pitching, untwisting, etc. For example, the power generation can be described by the curve of the wind speed and power output. The yawing can be expressed by the integral relation between the yaw angle (φ_y) and the yaw rate (ω_y), and the yaw rate (ω_y) is denoted by a

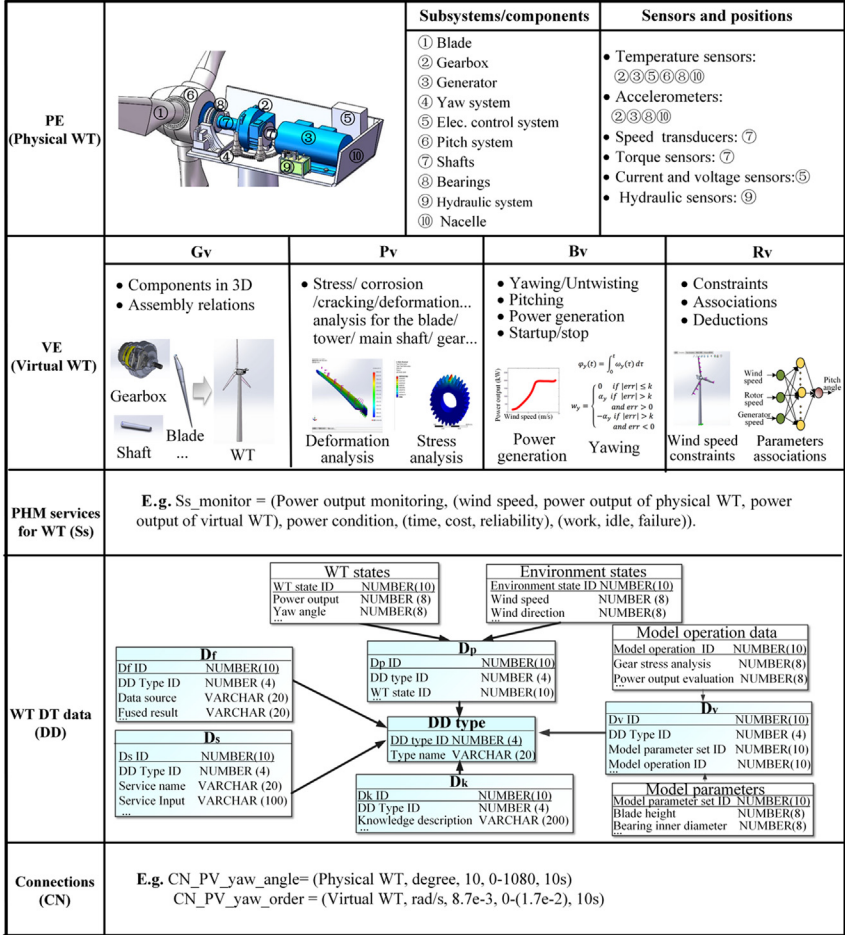


Figure 7.3 Modeling for DT components for a WT [27].

piecewise function of the yaw error (err) [28]. In R_v , the constraints for wind speed can be simulated through force analysis, and the associations of WT parameters can be mined using a neural network.

3. Ss . In Ss , the encapsulation of power output monitoring service is modeled as an example. According to expression (3) in Chapter 3, Five-Dimension Digital Twin Modeling and Its Key Technologies, it can be represented as $Ss_monitor = [Power\ output\ monitoring, (wind\ speed, power\ output\ of\ physical\ WT, power\ output\ of\ virtual\ WT), power\ condition, (time, cost, reliability), (working, idle, failure)]$. Its inputs include the wind speed, power output of the physical WT, and

power output of the virtual WT, while the outputs are the evaluated power output condition. The service can be scheduled considering its quality (i.e., time, cost, and reliability) and state (i.e., working, idle, or failure).

4. *DD*. According to expression (4) in Chapter 3, Five-Dimension Digital Twin Modeling and Its Key Technologies, *DD* is modeled with five components, where D_p is the data from the physical WT, D_v is the data from the virtual WT, D_s is the data from the *Ss*, D_k represents the domain knowledge, and D_f denotes the fused data of D_p , D_v , D_s , and D_k .
5. *CN*. The connection between the physical WT and virtual WT (i.e., *CN_PV*) is taken as an example. According to expressions (5) and (6) in Chapter 3, Five-Dimension Digital Twin Modeling and Its Key Technologies, data from the physical WT (i.e., yaw angle) is expressed as *CN_PV_yaw_angle* = (physical WT, degree, 10, 0–1080, 10 s), and the order from the virtual WT (e.g., yaw rate) is denoted as *CN_PV_yaw_order* = (virtual WT, rad/s, $7.7e-3$, 0–($1.7e-2$), 10 s).

7.3 DIGITAL TWIN-DRIVEN PHM METHOD

The constructed DT is then applied to create the DT-driven PHM method. In this method, faults are classified into two types: gradual fault and abrupt fault [27]. The former can be predicted as caused by the gradual degradation of the components, such as wear, aging, and fatigue. The latter is unpredictable and can happen suddenly due to random disturbances unknown to VE, such as human interference, environmental change, and incorrect installation. To deal with the two types of faults, the DT-driven PHM method, including the framework, procedure, and coevolution mechanism are studied as follows.

7.3.1 Framework

As shown in Fig. 7.4, the framework introduces the inputs and outputs, as well as the roles of DT components (PE, VE, *Ss*, *DD*, *CN*) in the proposed method.

7.3.1.1 Inputs

The inputs are a series of data, including mainly the data collected from the equipment in the remote field; the modeling and calibrating methods for the DT; and the models, algorithms, rules, and standards for PHM.

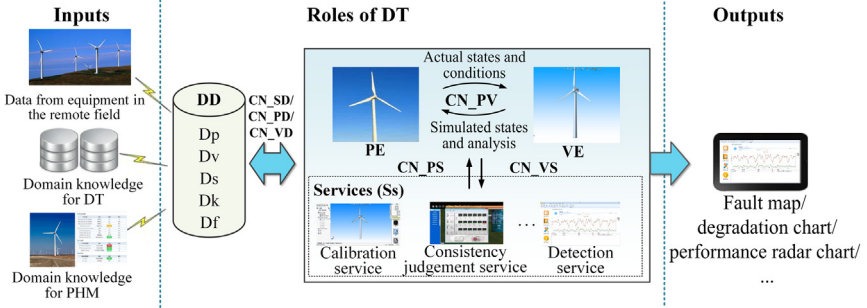


Figure 7.4 The framework of DT-driven PHM method.

These data are stored in DD. Based on the inputs, the DT can be constructed.

7.3.1.2 Roles of DT

As the DT is constructed, the roles of different DT components are presented as follows [27]:

1. PE is the target of PHM. Malfunctioning of any part in the PE may cause faults. Data of PE can reflect its current states. It mainly includes the operation states and the working conditions, denoted as $X = x_i | 1 \leq i \leq n$ and $C = \{c_j | 1 \leq j \leq m\}$, respectively, where x_i is the i th actual state parameter (e.g., speed, force, torque), c_j is the j th condition (e.g., environmental parameter, load, control order), while n and m are the numbers of parameters.
2. C and X are loaded to VE, which makes the VE operate under the same working conditions with the PE. On one hand, the VE operates synchronously with the PE and generates simulated operation states, denoted as $Y = y_i | 1 \leq i \leq n$, where y_i is the mapping to x_i , representing the i th simulated state parameter, and n is the parameters number. On the other hand, since the VE can perform further analysis (e.g., stress analysis, deformation analysis, and state prediction), data from the VE are more abundant than those from the PE. Hence, other data from the VE are represented as $Z = z_k | 1 \leq k \leq l$, where z_k denotes the k th analysis parameter and l is the parameters number.

For the gradual fault, since the degradations are accumulated in the PE gradually and recorded by the VE at the same time, data in X and Y will be similar under the premise that models in the VE are of high fidelity to the PE. For the abrupt fault, as the random disturbances can

happen suddenly and are usually unknown to the VE, data in X and Y will be different.

3. Ss mainly includes the model calibration service, consistency judgment service, degradation detection service, inconsistency cause judgment service, fault cause identification, and prediction service and maintenance service. Actually, these services correspond to the functions in different steps of the procedure in [Section 7.3.2](#). The suitable service can be invoked to implement the corresponding function on demand.
4. DD fuses data from the PE, VE, Ss, and the input data. It makes the information from DD more comprehensive and accurate. On one hand, DD is used for DT modeling and calibrating; on the other hand, features can be extracted from DD, and fused for degradation detection, failure identification, and prediction, for the PHM.
5. CN_PV enables the interaction and comparison between the PE and VE, which is critical for fault identification in different fault types. As CN_PS connects the PE and Ss, and CN_VS connects the VE and Ss, services can be invoked and executed timely once the demands from the PE and VE are obtained. Due to CN_PD, CN_VD and CN_SD, PE, VE and Ss can store data in DD and obtain the processed data from DD in real-time.

7.3.1.3 Outputs

The outputs are the analysis results from DT-driven PHM method, which can be displayed in multiple forms, including the degradation chart, performance radar chart, fault map, risk chart, etc. They are outputs for the engineers to obtain the health status of the equipment as well as maintenance strategies.

7.3.2 Procedure

As shown in [Fig. 7.5](#), the procedure of the DT-driven PHM method is classified into three stages, that is, observation (steps 1–3), analysis (steps 4–6), and decision (step 7). These steps, which have been studied in the authors' work [\[27\]](#), are described as follows.

Step 1: DT Modeling and calibration. Establish DT for the complex equipment according to [Section 7.2](#). During the operation in the following steps, if the VE deviates from the PE due to defects in modeling, perform calibration according to [Section 7.3.2.1](#).

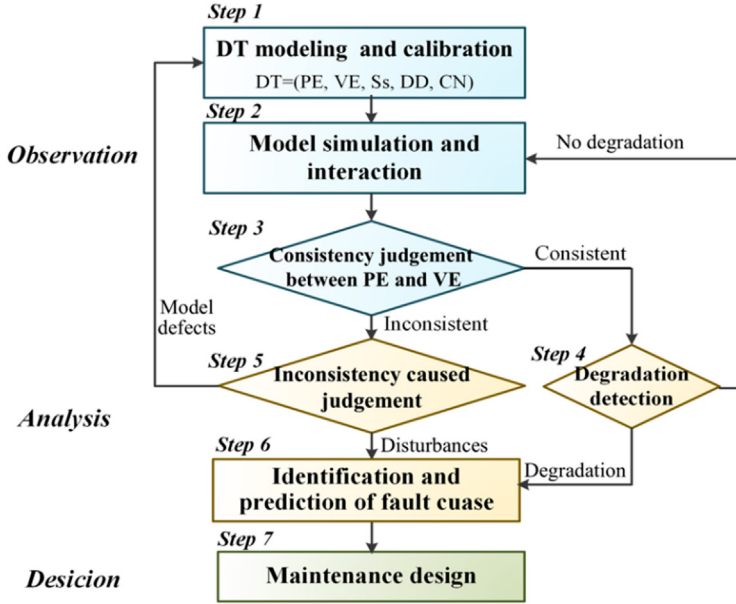


Figure 7.5 The procedure of DT-driven PHM method [27].

Step 2: Model simulation and interaction. Load X and C to the VE for model initialization and simulation. Keep PE and VE interacting to make both of them reach the last states.

Step 3: Consistency judgment. Let $\mathbf{p}_a = (x_1 \dots x_i \dots x_t)$ and $\mathbf{p}_e = (\gamma_1 \dots \gamma_i \dots \gamma_t)$ denote the states of PE and VE at the current moment (i.e., t_n), where $x_i \in X$, $\gamma_i \in Y$ and $t \leq n$. If $\|\mathbf{p}_a - \mathbf{p}_e\| \leq T_p$, where T_p is a predefined threshold, PE and VE are considered consistent, meaning within the tolerable differences caused by inevitable fluctuations (e.g., changes in the actual environment). Then go to step 4. Otherwise, they are inconsistent and go to step 5.

Step 4: Degradation detection. During operation, performance degradation will inevitably appear on the PE. As the PE and VE are consistent, the VE will also have the same performance. Hence, the gradual degradation can be reflected on indicators expressed by the parameters from both PE and VE, including X , Y , and Z . If some indicators exceed their upper or lower thresholds, go to step 6. Otherwise, go to step 2.

Step 5: Inconsistency caused judgment. Judge the inconsistency cause according to Section 7.3.2.2. If it is caused by the defects of modeling, go back to step 1 for calibration. If it is caused by the abrupt fault of

the PE, go to step 6. In addition, if the inconsistency is caused by both aspects, go to step 1 for calibration first.

Step 6: Identification and prediction of fault cause. According to the faults from step 4 (i.e., the gradual fault) and step 5 (i.e., the abrupt fault), identify and predict the cause of the fault according to [Section 7.3.2.3](#). Then go to step 7.

Step 7: Maintenance design. According to the analysis results, select the maintenance strategies among a set of standby alternatives. Perform the selected strategies on the VE first to check if any conflict exists. If so, modify the strategies to eliminate the conflicts; otherwise, carry them out on the PE.

7.3.2.1 Model Calibration

There are two possible reasons in the modeling that could lead to the deviation between PE and VE: (1) CN_PV fails to keep the normal communication between PE and VE, and (2) VE is inaccurate and fails to generate the precise representation for PE.

To solve these problems, the calibration process is presented in [Fig. 7.6](#). First, examine CN_PV through the cyclic redundancy check (CRC) and the delay check. In the CRC, test the integrity and accuracy of the transmitted data. In the delay check, test whether the communication is in real-time. Second, calibrate the models in VE through

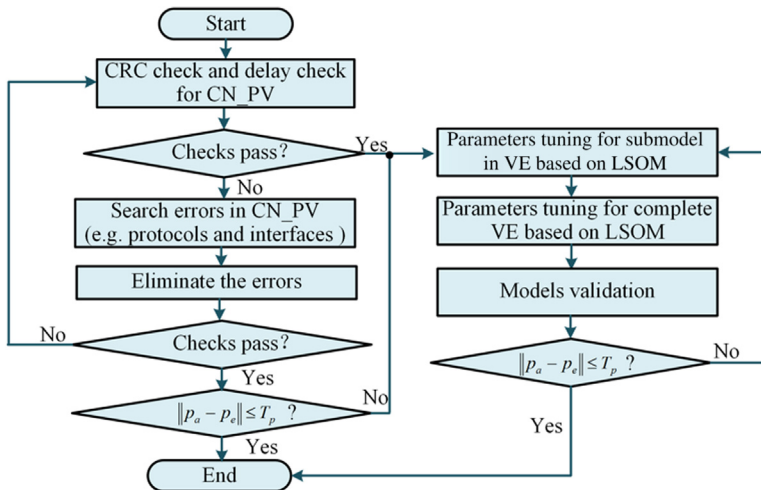


Figure 7.6 Calibration for CN_PV and VE.

parameters tuning for both the submodel and the complete VE based on the least-squares optimization method (LSOM) in [29].

7.3.2.2 Inconsistency Caused Judgment

Assume that at t_n , the PE and the VE are judged as inconsistent (i.e., $\|\mathbf{p}_e - \mathbf{p}_a\| > T_p$). To identify the inconsistency cause, take the time series of historical states of the PE under the same condition (i.e., C) as the reference. Represent the reference as $\mathbf{P}_{hs} = (\mathbf{x}_{hs1} \dots \mathbf{x}_{hsi} \dots \mathbf{x}_{hst})$, where t is the parameter number, $\mathbf{x}_{hsi}$ is the time series for the i th parameter, denoted as $\mathbf{x}_{hsi} = (x_{hsi}(1) \dots x_{hsi}(q) \dots x_{hsi}(s))^T$, and s is the element number. Accordingly, for the i th parameter of $\mathbf{p}_a$ (i.e., x_i) and $\mathbf{p}_e$ (i.e., y_i), the time series from the current (i.e., t_n) to a future moment (i.e., t_{n+s-1}) are denoted as $\mathbf{x}_{si} = (x_{si}(1) \dots x_{si}(q) \dots x_{si}(s))^T$ and $\mathbf{y}_{si} = (y_{si}(1) \dots y_{si}(q) \dots y_{si}(s))^T$, respectively. Denote $\mathbf{P}_{as} = (\mathbf{x}_{s1} \dots \mathbf{x}_{si} \dots \mathbf{x}_{st})$ and $\mathbf{P}_{es} = (\mathbf{y}_{s1} \dots \mathbf{y}_{si} \dots \mathbf{y}_{st})$ as the time series for $\mathbf{p}_a$ and $\mathbf{p}_e$. $\mathbf{P}_{hs}$, $\mathbf{P}_{as}$, and $\mathbf{P}_{es}$ are the multivariate time series. Measure the similarity of $\mathbf{P}_{as}$ and $\mathbf{P}_{hs}$, as well as $\mathbf{P}_{es}$ and $\mathbf{P}_{hs}$ according to the correlation based dynamic time warping (CBDTW) described in Ref. [30], which considers the correlation between the variables and provides an accurate measurement. Based on the measurement result, judge the inconsistency cause. The process is shown in Fig. 7.7.

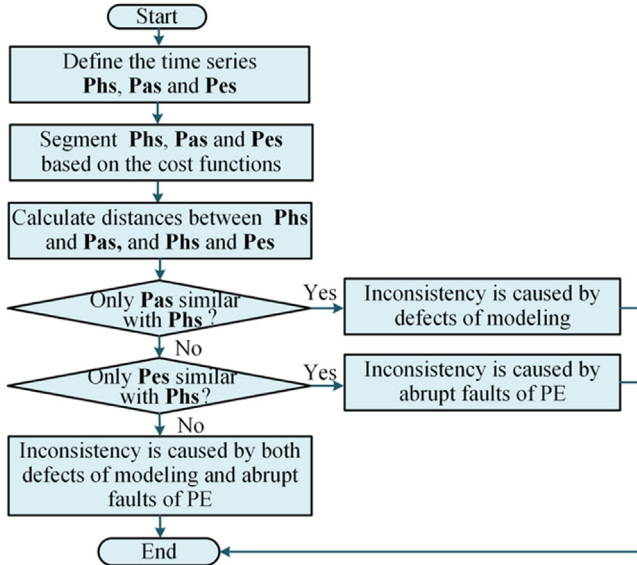


Figure 7.7 Similarity measurement.

7.3.2.3 Identification and Prediction of Fault Cause

1. *Gradual fault*: From step 4, before the gradual fault occurs completely, localize the fault cause in advance. It includes two phases, as shown in Fig. 7.8. In the training phase, extract the forewarning features for different fault causes from the historical data of PE and VE, including X, Y, and Z. Then train the prediction models, which fuse the features from both the PE and the VE, and relate them to the fault causes through methods such as the neural network. If the trained models can pass the accuracy test, they will be considered as good models and used for the prediction. In the prediction phase, input the forewarning features extracted from the real-time data of X, Y, and Z into the models and predict the cause.
2. *Abrupt fault*: From step 5, localize the components related to the abrupt fault first based on the deviation between PE and VE. Then check the components one by one through comparing their parameters from the VE with those from the PE, to finally identify the source of disturbances.

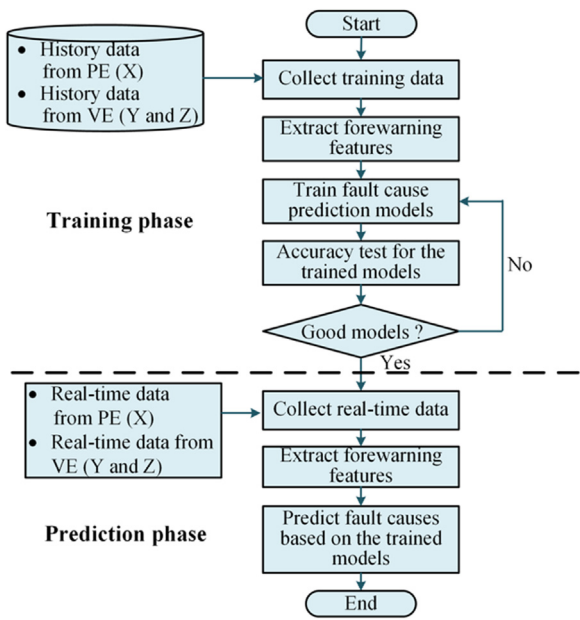


Figure 7.8 Model training and fault cause prediction.

7.3.3 Coevolution Mechanism

During DT-driven PHM, the PE, VE, and Ss evolve with each other continuously, as shown in Fig. 7.9. For PE_i , VE_i , and Ss_i , i represents the i th interaction among the three parts.

The PE and the VE always make an effort to keep consistent with each other. Assume that PE_i and VE_i are consistent in the beginning. To distinguish the current “consistency” with others, it is labeled as “*consistency_i*.” However, since the states of PE_i (denoted as p_{ai}) are continuous and variable, the states of VE_i (denoted as p_{ei}) may appear to have deviated due to defects and limitations in the models. In this situation, model calibration service will be performed to calibrate the VE_i to keep up with the PE_i . After that, the new consistency (i.e., *consistency_{i+1}*) is reached. From *consistency_i* to *consistency_{i+1}*, the VE evolves to better understand the PE, which makes the VE provide more accurate data and models for the PHM. As to the other aspect, if random disturbances from the external environment (always unknown to VE_i) exist on PE_i , the deviation between p_{ai} and p_{ei} will also appear. The disturbances will be captured by the inconsistency cause judgment service and eliminated by the maintenance service to achieve new consistency. By this process, the disturbance sources are reduced gradually and the PE is more likely to operate as the expected process provided by the VE, which makes the faults of the PE more predictable.

With the accumulation of data, the functions of Ss are also updated and improved continuously, which makes Ss_i evolve to Ss_{i+1} . For example, since the fault cause prediction models in Section 7.3.2.3 can be

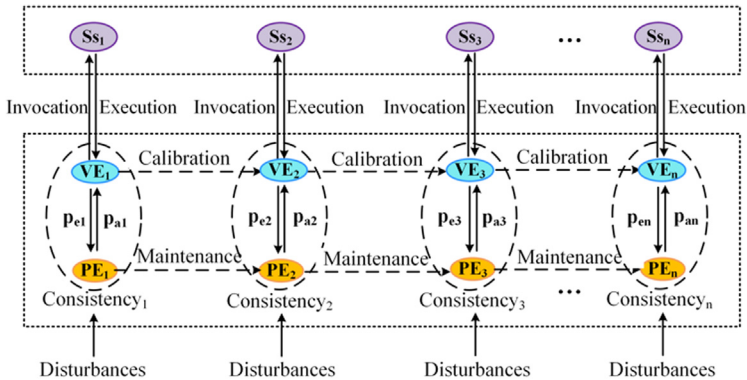


Figure 7.9 Coevolution mechanism of DT-driven PHM.

optimized online with more data obtained for training, the fault cause identification and prediction service based on the trained models is also refined accordingly.

7.4 CASE STUDY

7.4.1 Problem Description

A WT is complex equipment that transforms wind energy to mechanical and electrical energy. Since the WT usually works over decades and its working conditions are characterized by stochastic loading, rapidly varying torque, and a dusty environment, most components may fail and the failure rates are shown in Table 7.1 [31]. Hence, PHM is crucial to keep the WT operating reliably. However, some problems still exist as follows. Because of the harsh environment, it is not practical to access the WT all the time for prognosis and maintenance. The features extracted from the physical aspects are mainly used for fault cause identification or prediction, but features from the virtual side are ignored. Since signals from the physical aspect (e.g., the vibration signals) are easily interfered with by other sources, the diagnostic results can be affected [27]. To address the above issues, the DT-driven PHM is applied in the case study.

The gearbox and the yaw system are among the top three components that are prone to malfunctioning. The gearbox is used to transform the low speed drive from the wind power into high speed for the generator, and its failures are mainly caused by the gradual degradation of gears in it. The yaw system can align the nacelle with the wind direction to make the WT obtain the maximum wind energy. Failures of the yaw system

Table 7.1 Fault rates of WT components

Components	Failure rates	Components	Failure rates
Electrical control	13%	Blades	5%
Gearbox	12%	Brake system	5%
Yaw system	8%	Control system	2%
Entire turbine	7%	Entire nacelle	1%
Generator	5%	Shaft/bearing	1%
Hydraulic system	5%	Others	30%
Grid	5%		

Note: The bold characters outstand the gearbox and yaw system which are among the top three components that are prone to malfunctioning. The two components are used to verify the proposed method in the following content, which can better illustrate the efficiency of the proposed method to some extent.

are mainly caused by the wind vane with mechanical structures, which are prone to damage by disturbances from the external environment (e.g., the abrupt drop of temperature). In this section, firstly, the DT of a WT is constructed according to Section 7.2, and the DT-driven PHM method in Section 7.3 is then applied to the yaw system and gearbox to deal with the abrupt fault and gradual fault, respectively.

7.4.2 Digital Twin-Driven PHM for Yaw System

The yaw system is part ④ of the physical WT in Fig. 7.3. In this case study, the fault of the yaw system is caused by the sudden disturbance from the environment and it leads to yaw error. Related symbols are listed in Table 7.2.

According to steps 1 and 2 in Section 7.3.2, with the constructed DT of the WT, the simulation and interaction are started. According to step 3, $\mathbf{p}_a$ and $\mathbf{p}_e$ are denoted as $\mathbf{p}_a = P_{ao}$ and $\mathbf{p}_e = P_{eo}$, respectively. P_{ao} is calculated according to the output voltage and current collected from the voltage and current sensors deployed in part ⑤ (i.e., electronic control system) of the physical WT in Fig. 7.3. P_{eo} is simulated by the virtual WT based on the relation existing between W_s and P_{eo} [32]. The deviation is calculated according to Eq. (7.1). As $\|\mathbf{p}_e - \mathbf{p}_a\| \geq T_p$, $\mathbf{p}_a$ and $\mathbf{p}_e$ are considered inconsistent.

$$\|\mathbf{p}_e - \mathbf{p}_a\| = \frac{|P_{ao} - P_{eo}|}{|P_{ao}|} \quad (7.1)$$

To identify the inconsistency cause, according to step 5, the historical power output under the same W_s is selected as the reference and denoted as P_{ho} . Time series of P_{ho} , P_{ao} , and P_{eo} in 600 s are taken out for comparison in Fig. 7.10, where the sampling interval is 10 s. Hence, $\mathbf{P}_{hs}$, $\mathbf{P}_{as}$, and $\mathbf{P}_{es}$ are denoted as $\mathbf{P}_{hs} = (P_{ho}(1) \dots P_{ho}(60))^T$, $\mathbf{P}_{as} = (P_{ao}(1) \dots P_{ao}(60))^T$, and $\mathbf{P}_{es} = (P_{eo}(1) \dots P_{eo}(60))^T$, respectively. $\mathbf{P}_{hs}$, $\mathbf{P}_{as}$, and $\mathbf{P}_{es}$ are univariate time series. Through distance calculation, it is judged that $\mathbf{P}_{es}$ is similar

Table 7.2 A list of symbols of yaw system

Symbols of yaw system in physical WT		Symbols of yaw system in virtual WT	
A_{dy}	Yaw angle	A_{ey}	Yaw angle
P_{ao}	Power output	P_{eo}	Power output
W_d	Wind direction		
W_s	Wind speed		

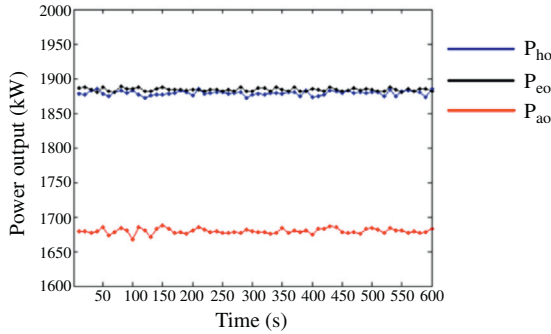


Figure 7.10 Time series of P_{ao} , P_{eo} , and P_{ho} .

to P_{hs} , and P_{as} deviates from P_{hs} . Hence, the deviation is caused by the physical WT.

According to step 6, to identify whether the deviation is caused by the yaw system, the following comparison is performed. For both physical and virtual WTs, when A_{ay} and A_{ey} change in two directions (θ or $-\theta$), the trends of P_{ao} and P_{eo} are different. For the physical WT, P_{ao} increases obviously in one direction (i.e., θ) and reduces in another (i.e., $-\theta$). Meanwhile, for the virtual WT, P_{eo} reduces in both directions. For the physical WT, when A_{ay} changes in the appropriate direction, P_{ao} has the ability to reach the normal level. It illustrates that other components (e.g., blade, gearbox, and generator) work normally and the misalignment of the nacelle (i.e., part ⑩ in PE in Fig. 7.3) caused by the yaw system is the bottleneck for the power output. For the virtual WT, since the nacelle has faced the wind direction, the change of A_{ey} only brings a decrease in P_{eo} . Hence, the yaw system of the physical WT can be identified as the fault cause.

Through further checking of parameters collected from components of the yaw system, the deviation of the wind vane is finally identified as the fault cause, which is interpreted in Fig. 7.11. The wind vane measures the angle β between the zero scale and the wind direction. In normal condition, as in Fig. 7.11A, the zero scale aligns to the central axis of the nacelle. Hence, β is equal to α , the angle between the central axis and the wind direction. The yaw system controls the nacelle to turn until β becomes 0 degrees, which makes the nacelle face the wind direction exactly. However, in Fig. 7.11B, since the zero scale deviates from the central axis due to disturbances from the external environment, α and β

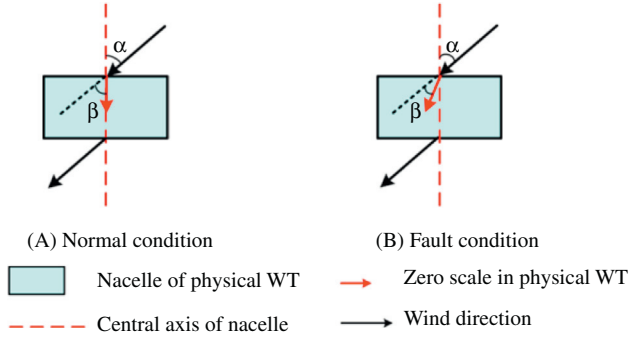


Figure 7.11 The fault of wind vane. (A) Normal condition. (B) Fault condition.

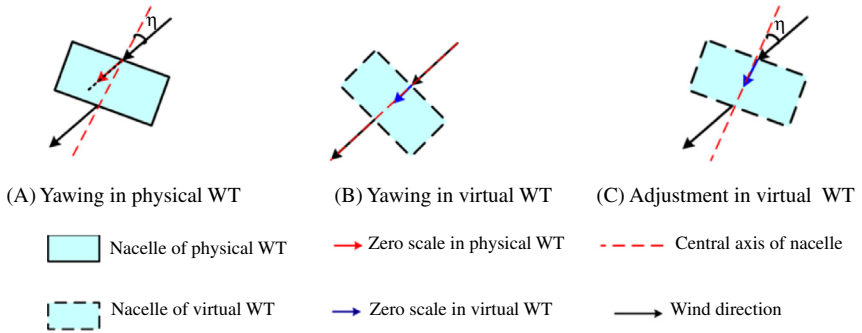


Figure 7.12 Simulation for the maintenance strategy. (A) Yawing in physical WT. (B) Yawing in virtual WT. (C) Adjustment in virtual WT.

are different. Hence, the direction of the nacelle, generated based on β , will become inaccurate and lead to yaw error.

According to step 7, simulation based on the virtual WT is performed to design the maintenance strategy. The yawing of the physical WT is presented in Fig. 7.12A, where the zero scale deviates from the central axis with the angle η and P_{a0} is lower than the normal value. In Fig. 7.12B, as the zero scale aligns to the central axis, P_{e0} can reach the normal level. In Fig. 7.12C, to obtain the angle η , the nacelle direction can be adjusted in the virtual environment until the P_{e0} is equal with P_{a0} . Then η can be obtained. Based on η , the position of the zero scale in physical WT can be calibrated.

It can be seen that the DT-driven PHM method for the yaw system does not need to access the WT first while predicting the fault cause and the maintaining procedure, which improves the efficiency of PHM for the WT.

7.4.3 Digital Twin-Driven PHM for the Gearbox

The gearbox is part ② of the physical WT in Fig. 7.3. In this case study, the fault causes of the gearbox include gear tooth wear, fatigue, and breakage, which are caused by gradual degradation. Related symbols are listed in Table 7.3 [27].

According to step 3, the states of the gearbox in the physical and virtual WTs are denoted as $\mathbf{p}_a = (T_{ai}, T_{ao})$ and $\mathbf{p}_e = (T_{ei}, T_{eo})$, respectively. T_{ai} and T_{ao} are collected from the torque sensors mounted on the input and output shafts (i.e., part ⑦ in the physical WT in Fig. 7.3), while T_{ei} and T_{eo} are simulated by the virtual WT according to the transmission relations among the wind wheel, gearbox, and generator rotor [33]. The deviation is calculated according to Eq. (7.2) [27]. As $\|\mathbf{p}_e - \mathbf{p}_a\| \leq T_p$ is satisfied, the gearboxes of the physical and virtual WTs are considered consistent.

$$\|\mathbf{p}_e - \mathbf{p}_a\| = \frac{\sqrt{(T_{ei} - T_{ai})^2 + (T_{eo} - T_{ao})^2}}{\sqrt{(T_{ai})^2 + (T_{ao})^2}} \quad (7.2)$$

According to step 4, to detect the degradation, the vibration signals of the gearbox in the physical WT are collected from the accelerometers deployed at the low speed stage, medium speed stage, and high speed stage of the gearbox. The energy for every n samples of vibration time series is calculated according to Eq. (7.3) [27], where a_i is the amplitude for the i th sample. Since E_v exceeds the upper threshold, the gearbox is considered to have degraded.

$$E_v = \sum_{i=1}^n |a_i|^2 \quad (7.3)$$

Table 7.3 A list of symbols of gearbox

Symbols of gearbox in physical WT		Symbols of gearbox in virtual WT	
T_{ai}	Input torque	T_{ei}	Input torque
T_{ao}	Output torque	T_{eo}	Output torque
f_m	Meshing frequency	T_{cm}	Maximum contact stress
FC	Frequency center	T_{rm}	Maximum bending stress
VF	Frequency variance	N_m	Number of gear meshing
MSF	Mean square frequency		

According to step 6, to identify the cause (i.e., gear tooth wear, fatigue, or breakage) before the complete failure, forewarning features of the gearboxes in both physical and virtual WT are extracted. From the gearbox in the physical WT, the frequency spectrums of vibrations for the normal condition as well as the three degraded conditions are presented in Fig. 7.13 [27]. The features are mainly reflected on f_m , harmonics of f_m , and their sidebands. In Fig. 7.13A, the amplitudes of f_m and the harmonics are low and the sidebands are not obvious. However, in Fig. 7.13B, as the tooth surface is worn, the amplitudes of the harmonics have risen obviously. In Fig. 7.13C, when tooth fatigue has occurred, the sidebands of f_m and the harmonics broaden, and the amplitudes are observed to be high. In Fig. 7.13D, when the tooth is broken, the frequencies are ample and dispersed, while their amplitudes are higher.

To distinguish the above characteristics, FC , VF , and MSF are calculated according to Eqs. (7.4)–(7.6), where f_k denotes the frequency of the k th spectrum line and X_k represents the amplitude of f_k [34]. Since the

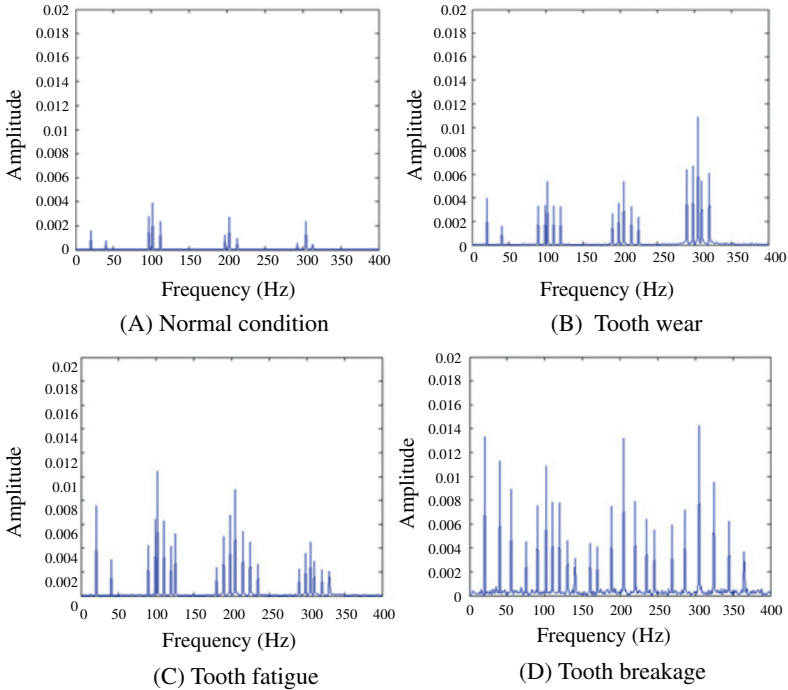


Figure 7.13 Frequency spectrums of vibrations for gearbox in the physical WT. (A) Normal condition. (B) Tooth wear. (C) Tooth fatigue. (D) Tooth breakage.

three indicators can reflect the different positions, energies, and dispersions of the vibration frequencies for different fault causes, they are selected as the forewarning features.

$$FC = \frac{\sum_{k=1}^K f_k X_k}{\sum_{k=1}^K X_k} \quad (7.4)$$

$$VF = \frac{\sum_{k=1}^K (f_k - FC)^2 X_k}{\sum_{k=1}^K X_k} \quad (7.5)$$

$$MSF = \frac{\sum_{k=1}^K f_k^2 X_k}{\sum_{k=1}^K X_k} \quad (7.6)$$

From the gearbox in the virtual WT, as different fault causes can be reflected on the different stress levels (e.g., contact stress and bending stress in Fig. 7.14 [27]) on the tooth, T_{cm} , T_{rm} , and N_m are selected as the forewarning features. T_{cm} and T_{rm} can be obtained by analyzing the forces acting on the tooth surface and root using FEM, and N_m can be obtained through recording the gear meshing number.

Extreme learning machine (ELM) [35], a single-hidden layer feedforward network, characterized by the fast learning speed and good generalization performance, is applied to fuse the features from the gearboxes in both physical and virtual WTs and build the fault cause prediction model. As shown in Fig. 7.15 [27], the constructed network contains 6 input nodes (FC , VF , MSF , T_{cm} , T_{rm} , and N_m), 1 output node (the fault cause), and 10 hidden nodes.

The model is trained based on 90 groups of samples, including 30 for tooth wear, 30 for tooth fatigue, and 30 for tooth breakage. For each sample, it consists of the values of six features and the corresponding fault

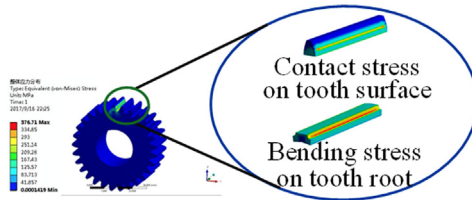


Figure 7.14 Stress analysis of gear for the gearbox in the virtual WT.

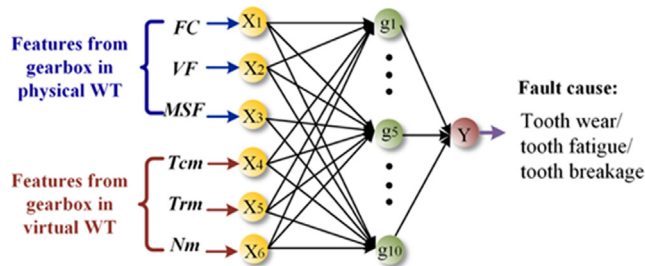


Figure 7.15 Fault cause prediction model driven by DT for the WT.

Table 7.4 Accuracy of the DT-method

Accuracy	Tooth wear	Tooth fatigue	Tooth breakage
Training accuracy	0.93	0.93	1
Testing accuracy	0.93	0.92	1

cause. Then the trained model is tested on 40 samples, including 15 for tooth wear, 13 for tooth fatigue, and 12 for tooth breakage. For each condition, the training or testing accuracy is calculated using the ratio between the number of samples that are classified into the corrective condition and the total number of samples in this condition. The results are shown in Table 7.4, where the proposed method is denoted as the DT-method.

Finally, according to step 7, based on the fault cause, for example, tooth breakage, the maintenance strategy will suggest welding repair. The welding process is first carried out to the damaged gears in the virtual environment for verification. Then the verified maintenance is performed on the gearbox in the physical WT [27].

To further validate the advantages of the DT-method, its performance is compared with the traditional fault cause prediction method for gearboxes, which usually uses vibration signals from the gearbox in the physical WT and is denoted as P-method here. In this case, the P-method is also built using the ELM. The inputs of the P-method are FC, VF, and MSF, which are also used in the DT-method [27]. The training accuracy and testing accuracy for the P-method are shown in Table 7.5. Compared with the P-method, both the training and testing accuracy of the DT-method are improved prominently.

The advantage of the DT-method mainly comes from the construction of the DT. According to the procedure of the DT-method, under

Table 7.5 Accuracy of the P-method

Accuracy	Tooth wear	Tooth fatigue	Tooth breakage
Training accuracy	0.83	0.87	0.90
Testing accuracy	0.80	0.77	0.83

the premise that the gearboxes in the physical and virtual WTs are consistent, the constructed DT can fuse the vibration and stress signals for prognosis. It can depict the gearbox from both the physical and virtual aspects. However, the P-method only focuses on the vibration signals from the physical WT. As the signals are prone to interference from other vibration sources, the accuracy of the P-method can be affected easily [27].

7.5 SUMMARY

In this chapter, the methods of DT modeling and DT-driven PHM are proposed for complex equipment. The aim of the chapter is to give an insight into PHM driven by DT and provide a guideline for future works. The contributions of the chapter are summarized as follows. First, it extends the previous three-dimension definition of the DT to a five-dimension definition for complex equipment, which expresses the properties and characteristics of the DT in a more comprehensive and realistic aspect. Second, the DT-driven PHM method, including the framework, the three-stage procedure, and the coevolution mechanism are presented in detail, which takes advantages of the digital mirror models, physical–virtual interaction, and data fusion in the DT. Third, based on the above works, a case study introduces the DT-driven PHM method in the yaw system and the gearbox of the WT to deal with different fault types. It is verified that the proposed method can predict the fault cause and the maintenance procedures without accessing the WT first, and improve the prediction accuracy of fault cause benefiting from the cyber–physical fusion.

Considering the implementation costs and complexity of the DT, the proposed DT-driven PHM method will be useful for monitoring high-value and major equipment in a plant, and there must be sufficient data for DT modeling. To scale up the PHM application, challenges mainly include [27]:

- Constructing high-fidelity digital mirrors for complex equipment with different properties and behaviors.
- Processing large amount of DT data.

- Balancing costs and benefits of the DT.
Future works will focus on the following aspects:
- Increasing the accuracy of the constructed DT.
- Refining the operation mechanism for the DT-driven PHM.
- Further application of the proposed method to other types of complex equipment.

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PART 3

Digital Twin and New Technologies

CHAPTER 8

Digital Twin and Cloud, Fog, Edge Computing

Digital twin (DT) is the preferred means to achieve the interoperability and integration between the physical and virtual spaces. From a hierarchical perspective, DTs can be divided into unit level, system level, and system of systems (SoS) level. To meet the practical requirements on smart manufacturing, three complementary technologies, that is, edge computing, fog computing, and cloud computing, are introduced to combine with DTs in different levels for function enhancement. In this chapter, the three-level DTs in manufacturing are described first, then the perspectives of unit-level DT, system-level DT, and SoS-level DT respectively supported by edge computing, fog computing, and cloud computing are discussed.

8.1 INTRODUCTION

From the manufacturing development perspective, a fusion process between the physical and virtual spaces of manufacturing is evolving over time. First, before the advent of information technologies, manufacturing only involved the physical space. Specifically, all manufacturing activities were done by humans and/or machines. The lack of information limited manufacturing efficiency and production capacities. In the second stage, the virtual space appeared. A variety of manufacturing-related software products were developed and applied in production. However, due to technological limitations, the advantages and potential applications of virtual space had not been fully exploited. In the third stage, the physical space and the virtual space began to interact with each other. The virtual space has strengthened the management of the manufacturing process, optimizing the logistics, capital flow, and information flow. However, during this stage, the physical and virtual spaces of manufacturing were not synchronized. At present, with the continuous developments of new generation information technologies (New IT), real-time interaction and

further integration between the physical and virtual spaces is becoming an inevitable trend in smart manufacturing [16].

According to the horizontal analysis on various manufacturing strategies and vertical analysis on manufacturing development, the cyber–physical fusion of manufacturing is gaining increasing importance. To this end, digital twin (DT) has become the preferred technology. The DT creates virtual models for physical objects in a digital way to simulate their behaviors, serving as a bridge between the physical and virtual spaces [1]. Through the cyber–physical closed loop, the DT could promote smartness of all manufacturing processes [16].

Smart manufacturing is a value creation process in which multiple subjects participate. Depending upon different magnitudes of the data in smart manufacturing and different time-sensitivities of smart applications, the DTs can be divided into three different levels, that is, the unit level, system level, and system of systems (SoS) level [2]. Each level has a data closed loop, including status perception, data analysis, decision making, and execution. However, as various levels have different roles and infrastructures, how to build various levels of DTs toward smart manufacturing becomes a central issue [16].

Edge computing, fog computing, and cloud computing may offer solutions to the abovementioned issues. Cloud computing enables ubiquitous, convenient, on-demand network access to the shared resources pool [3]. Because of on-demand resource sharing, high computing and storage capabilities, as well as low costs, cloud computing is a perfect technological means for the SoS-level DT. Fog computing shifts the computing, storage, and networking capabilities of cloud to the edge network [4]. By directly processing data on the network (e.g., network routers, various information systems, etc.), fog computing can help implement the system-level DT. Edge computing allows data processing to be performed at closer proximity to the data sources [5]. The edge can be defined as the end points with both data producers and data consumers. For the unit-level DT, edge computing will be very beneficial [16].

In this chapter, edge computing, fog computing, and cloud computing are introduced for implementing the unit-level, system-level, and SoS-level DTs, respectively, toward smart manufacturing. The three-level DTs in manufacturing are described first, then edge computing, fog computing and cloud computing are briefly reviewed. Finally, a framework of the three-level DTs based on these technologies is discussed.

8.2 THREE-LEVEL DIGITAL TWINS IN MANUFACTURING

By building a DT system that integrates data from different objects, activities, and processes, the innovation and efficiency of product design, production planning, and implementation can be enhanced effectively. The DT is an important supporting technology for smart manufacturing. To better understand the DT, a systematic and hierarchical view should be established. For smart manufacturing, both small objects such as a piece of equipment or a production line, and large objects such as a shop-floor or an entire factory, can be considered as DTs. Therefore, the DTs need to be divided into different levels, including unit level, system level, and SoS level, to adapt to objects in different granularity [6]. The DT in each level is a systematic model that is comprised of a previous level or levels. The system-level DT can be considered as an integration of multiple unit-level DTs that cooperate with each other. The multiple unit-level DTs or multiple system-level DTs constitute the SoS-level DT, that is, a complex system including complicated relations, associations, and combinations. As shown in Fig. 8.1, the unit-level DT mainly serves for a single object (e.g., a piece of equipment), which is the smallest unit participating in manufacturing activities. The optimization of manufacturing can be achieved through the adjustment of the equipment. With respect to the system-level DT, a smart production line comprised of machine tools, robot arms, automated guided vehicles (AGVs), etc. can be considered as a system with specific production capacities. For the unit-level and system-level DTs, the virtual models are ultrahigh-fidelity mappings through digital descriptions of the equipment and production line from the perspectives of geometric shape, function, and operating status. Basic attributes, real-time status, and other data are transmitted to the virtual models to drive simulation and prediction. Meanwhile, parameters of the virtual models are fed back to optimize the physical entities. By the closed-loop interaction process, the physical entities and virtual models evolve simultaneously. For the SoS-level (e.g., a shop-floor), accurate shop-floor management and reliable operations, which are inseparable from services, are especially crucial. As shown in Fig. 8.1, (1) physical entities are a set of objective entities that have specific functions to complete manufacturing tasks according to inputs and outputs; (2) virtual models are the digital images of the physical entities and can completely and truly reflect the lifecycle of the physical counterparts; (3) services integrate various functions such as control, optimization, and prediction to

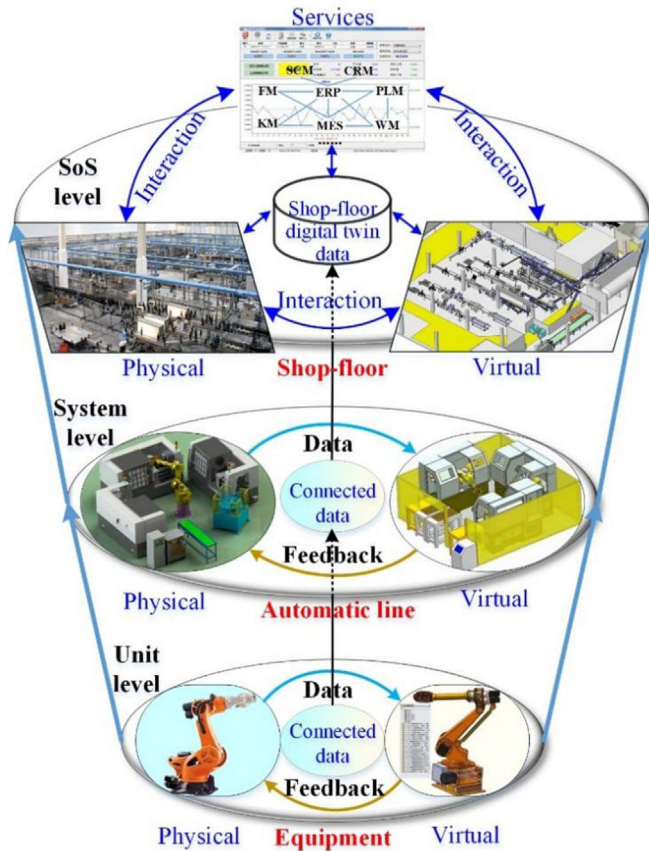


Figure 8.1 Unit-level, system-level, and SoS-level DTs in manufacturing [6].

provide application services according to the requirements; (4) data is the core driver of the DT, including the data from the physical entities, virtual models, services, as well as their fusion data; and (5) connections among them place the above four parts in pairs, ensuring real-time interaction and iterative optimization. Based on the five-dimension structure of the DT, the digital twin shop-floor (DTS) provides a new way to practice smart manufacturing [6].

8.3 FROM CLOUD COMPUTING TO FOG COMPUTING AND EDGE COMPUTING

As complementary technologies, fog computing and edge computing, which provide computing, storage, and networking services between end

devices and traditional cloud computing [4], offer bright prospects for expanding smart manufacturing applications [16].

8.3.1 Cloud Computing

The National Institute of Standards and Technology (NIST) defined cloud computing as “a model for enabling ubiquitous, convenient, on-demand network access to a shared pool of configurable computing resources (e.g., networks, servers, storage, applications, and services) that can be rapidly provisioned and released with minimal management effort or service provider interaction” [3]. Because of the significant benefits, such as ubiquity, convenience, on-demand resources sharing, high computing and storage capabilities, as well as low costs, cloud computing has become attractive to many large companies (e.g., Amazon, Google, Facebook, etc.) that deliver services over the Internet to gain economic and technical benefits [7]. From the user perspective, cloud hides the underlying complexity and heterogeneity through infrastructure-as-a-service (IaaS), platform-as-a-service (PaaS), and software-as-a-service (SaaS) [8]. Users can outsource their tasks to service providers, instead of purchasing expensive devices for small tasks. The rise of cloud computing has changed the way industries and enterprises do business and has created brand new opportunities for them. However, the problems of network unavailability, overfull bandwidth, and delays make cloud computing unable to solve all problems [9].

8.3.2 Fog Computing

The new generation of smart devices, which are capable of processing data locally rather than sending them to the cloud, enable a new distributed computing paradigm, namely fog computing [9]. Fog computing was initially introduced by Cisco, which is one of the world-leading providers of network solutions. Fog computing is considered as an extension of cloud computing to the edge network, providing services (e.g., computing, storage, etc.) closer to near-user edge devices, instead of sending data to the cloud [9–11]. By directly processing data on the network, for example, network routers, the fog computing paradigm can improve efficiency, reduce the amount of data that have to be transferred, and enhance security [8]. Fog computing can be characterized by location awareness, low latency, edge location, real-time interaction, interoperability, support for online interplay with the cloud, etc. [4,12]. These

characteristics make application services deployment more convenient. Therefore, fog computing will play a key role in numerous delay-sensitive applications for smart manufacturing.

8.3.3 Edge Computing

Edge computing is a scattered architecture that is driven by increasing smart terminals, massive networked devices in IoT, and decreasing cost of computer components [13]. Due to its bright application prospects, many companies have devoted themselves to edge computing, such as Intel, ARM, and Huawei, who are main providers of information and communication technology (ICT) equipment [14]. Similar to fog computing, edge computing also allows computation to be performed at the edge of the network, but at the closer proximity to data sources. Therefore, the edge can be defined as the resources more toward the terminals' side, which are not only data consumers, but also data producers [5]. Through data analysis and processing in edge devices, edge computing can achieve sensing, interaction, and control between underlying objects and objects. In manufacturing, edge computing can meet the critical requirements of agile connectivity, real-time optimization, and other applications at the manufacturing site.

8.4 THREE-LEVEL DIGITAL TWINS BASED ON EDGE COMPUTING, FOG COMPUTING, AND CLOUD COMPUTING

In light of the widespread applications of New IT in manufacturing, manufacturers are facing an explosive growth of data [15]. The essence of the DT is to acquire data from the physical entities and environment through sensors, and then compute and analyze them in the virtual space, so as to control the physical entities and environment. As a result, the data closed loop is established, forming the interaction and fusion between the physical space and the virtual space. The various levels of DTs in manufacturing have different requirements for data processing and data circulation, such as latency, bandwidth, security, etc. Edge computing, fog computing, and cloud computing with complementary properties provide new ideas and ways for the implementation of the unit-level, system-level, and SoS-level DTs. From the perspective of efficient use of resources, DTs deployed at the unit level based on edge computing, system level based on fog computing, and SoS level based on cloud computing can achieve asset control, management, optimization, and business optimization [16].

8.4.1 Unit-Level Digital Twin Based on Edge Computing

The basic requirements that need to be met in constructing a unit-level DT include (1) state perception, (2) computing and processing of data, and (3) control of physical entities. As an architecture that extends computing, networking, and storage capabilities from the cloud to the edge, edge computing enables perception, computing, and control for objects through data analysis and processing in edge nodes. As shown in Fig. 8.2, the manufacturing resources (e.g., machine, robot, component, AGV, etc.) with physical devices (e.g., machine body, spindle, tools, etc.) and cyber parts (e.g., embedded system, sensory device, etc.), form the unit-level DT. The cyber parts can monitor and perceive information from the physical devices, and exert control onto the physical devices through an actuator that is capable of receiving control instructions. Due to the perception, data analysis, and control capacities, edge computing can be deployed on the unit-level DT, which can be considered as an edge node. Because of the data circulating on the unit device, edge computing can implement smaller applications that can help provide better real-time responses. The real-time applications provided by edge computing at the unit level include real-time processing and analysis for sensory data, data buffering, high-performance real-time control, actuator monitoring, fault diagnosis, health feature extraction, cycle count accumulation, fault

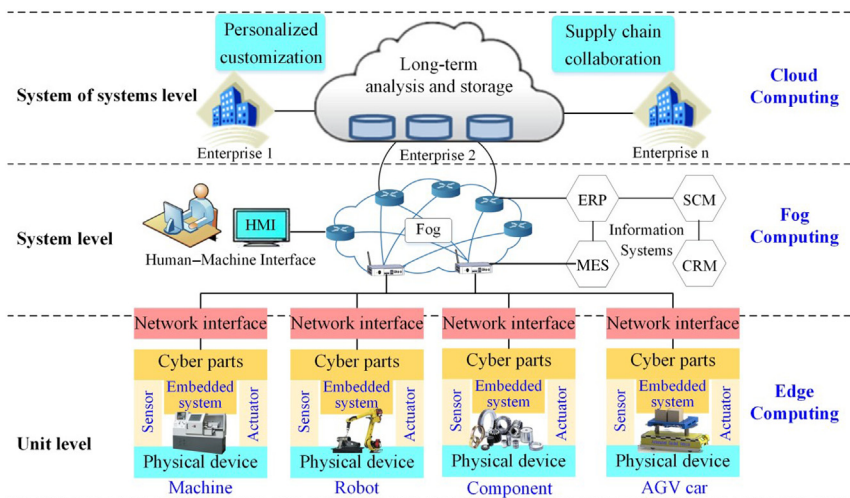


Figure 8.2 Three-level DTs in manufacturing based on edge computing, fog computing, and cloud computing [16].

handling, and safe shutdown. For example, on a machine tool or robot arm, sensors are used to detect whether some malfunctions may pose safety hazards. In this case, delay is not tolerable. If the data is sent to the cloud, the response time may be too long. However, with edge computing, the time delay is reduced, allowing decisions to be made in real-time since the data is not far from the data source. Edge computing architecture does not rely on an Internet connection, which is beneficial to the unit-level DT [16].

8.4.2 System-Level Digital Twin Based on Fog Computing

As shown in Fig. 8.2, multiple unit-level DTs are connected to the network through the network interface and the information management systems, for example, enterprise resource planning (ERP), manufacturing execution system (MES), supply chain management (SCM), and customer relationship management (CRM). The system-level DT integrates a variety of heterogeneous unit-level DTs. Besides, through the human-machine interface (HMI), each unit-level DT can be accessed and controlled to monitor and diagnose states and health conditions of the corresponding system. The system-level DT emphasizes the interconnections and interoperability among its component elements. On this basis, it focuses on the real-time and dynamic collaborative control of different elements, to achieve the coordination and unification of the physical space and the virtual space. In general, the system-level DT is geographically concentrated (mainly within the manufacturing enterprise), which is well suited to the fog computing model. The data of the system-level DT (i.e., enterprise internal data) can be directly dealt with by fog computing to improve efficiency, rather than be spread to and returned back again from the cloud. Taking delays, network traffic, cost, etc. into account, fog computing provides services at the edge of a network, promoting real-time interaction, scalability, and interoperability. The fog computing environment can be constituted by network components (such as routers, proxy servers, base stations, and others). These components can provide different computing, storage, and networking capabilities to support the implementation of smart manufacturing applications. The unit-level DTs use sensors to collect operating data on each device and analyze them. The fog computing environment summarizes the data from all the unit-level DTs, and provides actionable information (such as cooperation with each other to finish tasks) for smart manufacturing. The

system-level DT in fog computing provides applications that have lower real-time requirements but stronger computation requirements, such as model adaptation, model training, spectrum computing, wavelet transform, alarm priority processing, remote monitoring, accident recording, remote maintenance, performance assessment, and asset visibility. For example, in the actual production process, the robot places materials or parts on the conveyor belt or AGV for transportation, and then machine tools transform them into desired products. In this way, the combination of real-time edge computing and fog computing with interoperability and scalability will provide the best performance for the system-level DT [16].

8.4.3 System of Systems—Level Digital Twin Based on Cloud Computing

The collaborative optimization between multiple system-level DTs can be achieved by developing a smart service platform. As a result, multiple system-level DTs constitute a SoS-level DT. For example, multiple production lines or factories collaborate with each other through a smart cloud service platform, achieving the enterprise-wide system integration during the entire product lifecycle. The SoS-level DT involves various participants and resources, which may be geographically dispersed. The data of the SoS-level DT is more abundant and diverse. Therefore, the requirements that need to be met in the SoS-level DT include (1) providing distributed data storage and processing and (2) providing data and smart services for enterprise collaboration. As shown in Fig. 8.2, the cloud computing architecture is conducive to the organization and management of a large number of connected devices, as well as the combination and integration of enterprise's internal and external data. In the cloud computing architecture, a variety of different types of storage devices can work together through the application software, to jointly provide data storage and business access for enterprises. Besides, massive data mining must be supported by the distributed processing and virtualization technologies, which are the typical characteristics of cloud computing. Therefore, cloud computing is the ideal technology for long-term and massive data storage and analysis of the SoS-level DT. In addition, services with the characteristics of interoperability pave the way for collaboration among the system-level DTs. In virtue of the cloud architecture, various unit-level DTs, or system-level DTs can be encapsulated into services, so as to become plug-and-play components and be shared by other participants [17]. In this way, manufacturers not only can conveniently provide their services to

their collaborative stakeholders, but also use various resources in a pay-as-you-go manner [18]. In summary, the cloud computing architecture provides more architectural flexibility, and higher utilization of external data for product innovation and value creation. For example, supply chain application and product customization can benefit from understanding the aggregate data view [16].

8.5 SUMMARY

Regarding the development of manufacturing servitization, socialization, and demand personalization, the promotion and application of smart manufacturing constitute an inevitable trend. Achieving the interaction and integration of the physical space and the virtual space of manufacturing through using state-of-the-art technologies, constitutes an important prerequisite for implementing smart manufacturing. To this end, DT is the preferred technology. The DT can be applied to the entire manufacturing process, the entire supply chain, and the entire product lifecycle. Depending on the scope of resource allocation, the DT can be stratified to the unit level, system level, and SoS level. Because the data generated in different levels have different requirements at time scales, edge computing, fog computing, and cloud computing offer ideal technical solutions for different levels of cyber–physical systems (CPSs) and DTs. The cooperation among edge computing, fog computing, and cloud computing can better meet the requirements of each level of the DT [16].

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CHAPTER 9

Digital Twin and Big Data

Smart manufacturing is becoming the focus of global manufacturing. As intelligence comes from data, manufacturing big data is beneficial to all aspects of manufacturing. It can accurately predict requirements, and quickly identify errors and bottlenecks, thus innovating products and services, and improving manufacturing processes. Achieving the communication and interaction between the physical and virtual spaces of manufacturing is also an important bottleneck in smart manufacturing. Digital twin (DT) paves the way to solve the above bottlenecks. In this chapter, the similarities and differences between big data and DT and how the two concepts can be integrated to promote smart manufacturing are discussed.

9.1 INTRODUCTION

As shown in Fig. 9.1, an increasing number of devices are being connected to the Internet. Gartner predicted that more than 20 billion devices (most from the manufacturing industry) would be connected to the Internet of Things (IoT) by 2020 [1]. As a result, a large volume of various data is being generated, which will be over 40 zettabytes (ZB) by 2020 [2], including structured, semistructured, and unstructured data. With powerful storage and computing capability of cloud computing, big data analysis models and algorithms are running to organize, analyze, and mine these raw data [3,4], and hereby obtain valuable knowledge. Meanwhile, artificial intelligence (AI) with selflearning ability is becoming increasingly more intelligent through data analytics [9].

In manufacturing, big data involves a large volume of structured, semi-structured, and unstructured data generated from the product lifecycle. The increasing digitalization of manufacturing is opening up opportunities for smart manufacturing [5]. The manufacturing data are collected in real-time and automatically using IoT [6]. Through big data analysis based on cloud computing, manufacturers can find the bottlenecks of

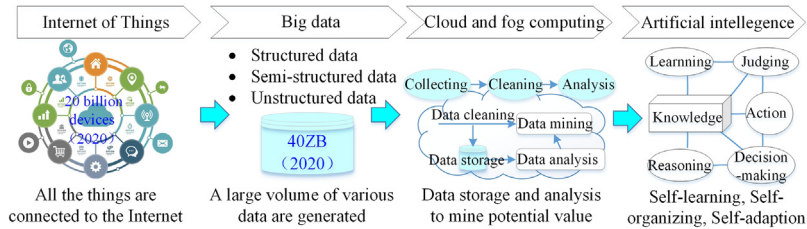


Figure 9.1 New IT and their applications [9].

manufacturing processes, recognize the causes and impacts of problems, and find solutions. Therefore, manufacturing processes are improved to enhance production efficiency, making manufacturing leaner and more competitive. All the valuable information from manufacturing big data is fed back to support product design, manufacturing, and prognostics and health management (PHM). This can help traditional manufacturing transform into smart manufacturing [9].

In addition, the interaction and convergence between the physical space and the virtual space of manufacturing is getting more and more attention. The digital twin (DT) paves the way for cyber–physical fusion. The DT is used to create virtual models for physical objects to digitally simulate their behaviors [7]. The virtual models can understand the state of the physical entities through sensing data, so as to predict, estimate, and analyze the dynamic changes. And the physical objects can respond to the changes according to the optimized scheme obtained from the simulation [5]. Through the cyber–physical closed loop, the DT could achieve the optimization of the entire manufacturing process [8–9].

In conclusion, collecting and analyzing a large volume of manufacturing data to find laws and knowledge has become the key for smart manufacturing; meanwhile, the DT breaks the barriers between the physical space and the virtual space. However, are there any similarities and differences between big data and the DT? What are their respective advantages? Is it possible to structure a bridge to bring big data and the DT together? How can they be merged together? These questions are all worth deeply thinking about and exploring [9]. Therefore, this chapter reviews the concepts of big data and the DT, as well as their applications in manufacturing. On this basis, they are compared with each other from different aspects. In addition, big data and the DT are combined together to promote smart manufacturing.

9.2 BIG DATA

The brief history, concept and characteristics of big data have been explored in the authors' previous work [24], and they are introduced in this section.

9.2.1 Brief History of Big Data

The story of how data became big started many years before the current fervent interest in “Big Data.” 50 years ago, a rapid increase in the volume of data and information appeared, known as the “information explosion” [10]. Since then, papers related to information collection [11], storage [12], and processing [13] have greatly contributed to the boom of big data. The first article in the Association for Computing Machinery (ACM) Digital Library to use the term “Big Data” is “Application controlled demand paging for out of core visualization” [14] in the *Proceedings of the IEEE 8th Conference on Visualization* in 1997. From then on, the concept of big data has gradually become accepted by scientists and researchers. In 1989, Dresner, a Gartner Group analyst, proposed “business intelligence” as “concepts and methods to improve business decision making by using fact-based support systems” [15]. Business intelligence and analytics (BI&A) has emerged as an important area of study for both practitioners and researchers [16] who are dedicated to data analytics problems. In 2011, the paper entitled “Big Data: The next frontier for innovation, competition, and productivity” [17] by McKinsey Global Institute truly brought big data into the public eye. In the same year, *Science* published a special online collection, “Dealing with Data,” which discussed issues of big data in scientific research. The academic milestones [14,17–23] in the brief history of big data since 1997 are illustrated in Fig. 9.2.

9.2.2 Concept of Big Data

The concept of big data has not been precisely defined, as there is a lot of buzz around it [25]. It has been declared as a new kind of economic asset that is strongly competitive in economics, like currency or gold [26]. The creator of the “Day in the Life” photography series even suggested that big data was an intelligent tool to combat poverty, crime, and pollution [26]. A convincing definition of big data focuses not only on the size of data in storage but also on other important attributes of big data, such as data variety and data velocity [27]. For example, the definition from

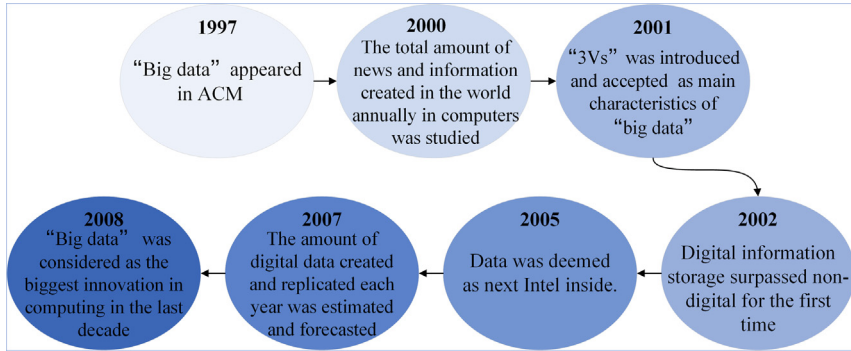


Figure 9.2 Brief history of big data [24].

Wikipedia illustrates that big data is a term for any collection of large and complex data sets that are difficult to analyze using traditional data processing methods. In a more straightforward definition, big data means data that is too big, too fast, or too hard for existing tools to process [25].

9.2.3 Characteristics of Big Data

The properties of big data have been summarized as the five Vs, with the three most famous Vs being volume, variety, and velocity, as introduced by Laney, a Gartner analyst, in 2001 [19].

- *Volume* refers to a large amount of data. The data storage unit in big data has reached the terabyte (TB) and petabyte (PB).
- *Variety* refers to a great number of types of data including weblog, music, video, picture, geographical position, etc.
- *Velocity* refers to the high speed of data processing, which is the most distinct feature from the traditional database.

Nowadays, another two Vs have been added to the model as follows.

- *Variability* refers to the expansion in the range of values of data. Because it can cover the full range of human experience, big data always shows more variance than traditional data sets.
- *Value* refers to the low density and the high overall value of big data. Among the large quantity of data, only a very small part of this information is useful and there is a need to evaluate it. Through data analytics methods, valued information can always be obtained and used in business operations.

According to the five Vs model, both the sheer amount of data and other characteristics of big data can bring challenges to big data

management, where the data set is too large, data values change too fast, and they do not follow the rules of conventional database management and systems [28]. With the need to ensure the real-time or near real-time responses for huge amounts of data in seconds, advanced data analytics methods should be proposed to handle the five Vs of big data.

9.3 LIFECYCLE OF BIG DATA IN MANUFACTURING

The study about “lifecycle of big data in manufacturing” has been explored in the authors’ previous work [31]. Details in the study are introduced in this section. Data is a key enabler for smart manufacturing. However, data is not useful unless it is “translated” into concrete information content and context that can be directly understood by users [29]. Generally, the data needs to pass through multiple steps before concrete information can be obtained from it. The complete journey of data collection, transmission, storage, preprocessing, filtering, analysis, mining, visualization, and application can be referred to as the “data lifecycle” [30]. Manufacturing data is exploited at various points in the data lifecycle. As illustrated in Fig. 9.3, a typical manufacturing data lifecycle consists of data collection, transmission, storage, processing, visualization, and application [31].

9.3.1 Data Sources

In manufacturing, big data refers to the data generated from the product lifecycle, such as design, manufacturing, PHM, etc. [24], which are also featured with high volume, variety, velocity, variability, and value. Manufacturing data usually include the following aspects [31], as shown in Fig. 9.3.

1. Manufacturing site data, including (a) equipment data collected from smart factories using Industrial IoT technologies, with respect to the real-time performance, operating conditions, etc.; (b) material and product data collected from themselves and service systems, such as performance, inventory, context of use, etc.; and (c) environmental data (e.g., temperature, humidity, air quality, etc.).
2. Management data from manufacturing information systems and computer-aided systems, including the data generated from the design scheme, order dispatch, material distribution, production planning, marketing and sales, service management, finance, and so forth.

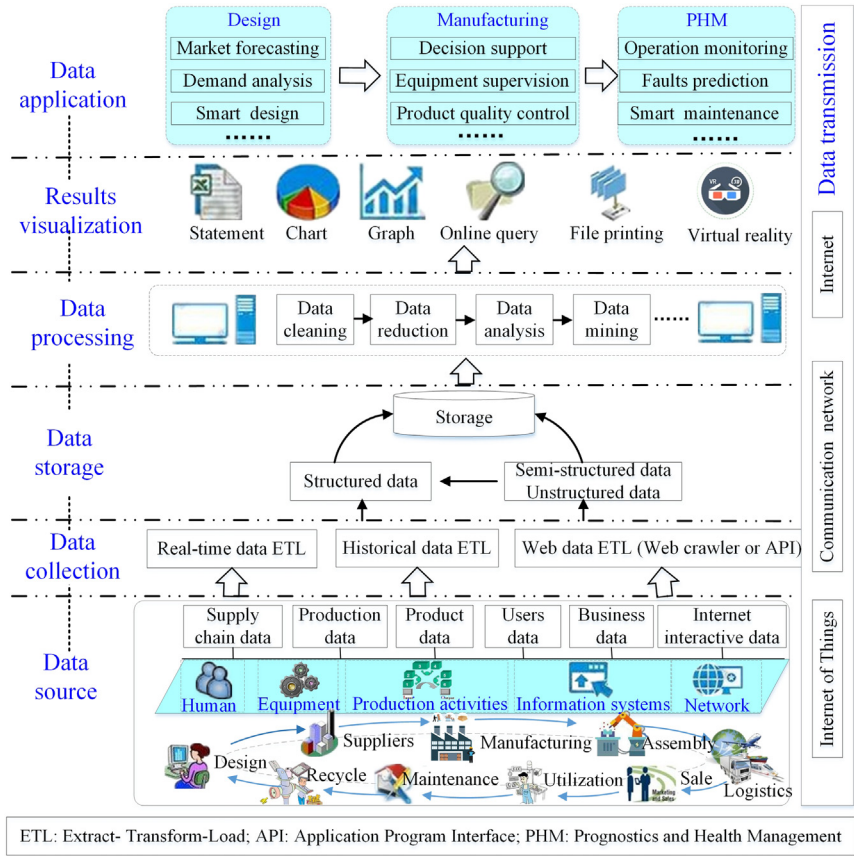


Figure 9.3 Manufacturing data lifecycle [31].

3. Internet data, including (a) user data collected from e-commerce platforms (e.g., Amazon, Walmart, Taobao, etc.) and social networking platforms (e.g., Twitter, Facebook, LinkedIn, YouTube, etc.), such as user comments, preference, and behaviors, etc.; and (b) public data from open websites (e.g., government and public service websites) and some other public net environments.

9.3.2 Data Collection

Data from different sources is collected in a variety of ways. Above all, it is collected by means of the IoT, whereby equipment and product data can be instantly collected through smart sensors, radio frequency identification (RFID), and other sensory devices, making it possible to monitor equipment and product health in real-time [32,33]. For instance, built-in

sensors make it possible to continuously measure, monitor, and report the ongoing operational status of manufacturing equipment and products, such as temperature, pressure, and vibration. RFID enables the automatic identification, tracking, and management of a large number of work-pieces, as well as the materials necessary for production. Moreover, the emerging mobile Internet paves the way for user data collection through smart terminals (e.g., devices like personal computers, phones, laptops, and tablets). Through a software development kit (SDK) or application programming interface (API), for example, basic user data can be collected, including the number of users, user profiles, location, and time. In addition, web crawling [34] is a widely used data acquisition technique for collecting public data based on certain conditions predefined by engineers and AI. Web crawling refers to the technology of deploying “crawlers” (i.e., computer programs) to browse public web pages and collect desirable information. The web crawling technology enables manufacturers to acquire public data in an automatic and efficient manner. Last but not least, management data from manufacturing information systems can be acquired and used at any time through database technologies [31].

9.3.3 Data Storage

The large volume of collected data from manufacturing processes must be securely stored and effectively integrated. Generally speaking, the various types of manufacturing data can be classified into structured (e.g., digit, symbols, tables, etc.), semistructured (e.g., trees, graphs, XML documents, etc.), and unstructured data (e.g., logs, audios, videos, images, etc.) [35]. Traditionally, manufacturing enterprises focused heavily on structured data storage, since it was difficult to directly manage unstructured data within enterprise databases. Object-based storage architecture enables collections of data to be stored and managed as objects, which provides a more flexible solution for integrating semistructured and unstructured data [36]. Also, through cloud computing [37], data storage can be achieved in a highly cost effective, energy efficient, and flexible fashion. In addition, by virtue of cloud services, the distribution and heterogeneity of data are shielded, enabling a highly scalable and sharable mode of data storage [31].

9.3.4 Data Processing

Data processing refers to a series of operations conducted to discover knowledge from a large volume of data. Data must be converted to information and knowledge for manufacturers to make informed and rational

decisions. Above all, data must be carefully preprocessed to remove redundant, misleading, duplicate, and inconsistent information. Specifically, data cleaning involves missing value, format, duplicate, and garbage data cleaning. Data reduction is the process of transforming a massive volume of data into ordered, meaningful, and simplified forms by means of feature or case selection [38]. After data reduction is completed, the cleaned and simplified data is exploited through data analysis and mining to generate new information. The effectiveness of data analysis can be significantly enhanced through a variety of techniques, including machine learning, large-scale computing, and the use of forecasting models. Some advanced data mining methods include clustering, classification, association rules, regression, prediction, and deviation analysis [30]. Through the above data processing efforts, understandable knowledge can be derived from a large quantity of dynamic and ambiguous raw data [31, 39].

9.3.5 Data Visualization

Visualization is intended to clearly convey and communicate information through graphical means, enabling end users to comprehend data in a much more explicit fashion [40]. The most commonly used visualization techniques include statements, charts, diagrams, graphs, and virtual reality [41]. Real-time data can be visualized online via users' smart terminals. Through visualization, the results of data processing are made more accessible, straightforward, and user friendly [31].

9.3.6 Data Transmission

Data is continuously flowing among different information systems, cyber–physical systems, and human operators. Data transmission, therefore, plays a critical role in maintaining communications and interactions among distributed manufacturing systems and resources. The recent advances in IoT, Internet, and communication networks have substantially consolidated the technological foundation of real-time, reliable, and secure transmission of different types of data. As a result, the distributed manufacturing resources can be effectively integrated almost anytime and anywhere [31].

9.3.7 Data Application

Data has entered almost all aspects of daily production and operation in manufacturing enterprises [8]. First, during the design phase, through data

analytics, new insights about customers, competitors, and markets are revealed. Based on the understanding developed through data analytics, designers can accurately and rapidly translate customer voices to product features and quality requirements [42]. As a result, manufacturers will become “closer” to customers, and more agile in terms of coping with the dynamic, changing market. Second, during production, the manufacturing process and equipment are monitored and tracked in real-time. In this way, manufacturers can keep abreast of the changes. Data analytics can lead to informed decisions concerning whether, when, and how to adjust the manufacturing processes and equipment. Additionally, data can facilitate the control and improvement of product quality. Data analytics can provide early warnings of quality defects and rapid diagnosis of root causes, both of which can be rapidly determined. Accordingly, manufacturing systems can be adjusted in a timely manner to control product quality. Lastly, with respect to product usage and PHM, potential product malfunctions can be identified at an early stage [43], which makes precautionary actions possible, such as preventive maintenance, fault prediction, and automatic upgrade. For instance, through the development of prediction models, analysis of historical data can be used to predict the fault occurrence [31,44].

9.4 360° COMPARISON OF DIGITAL TWIN AND BIG DATA IN MANUFACTURING

Both big data and DT have attracted widespread attention, and are considered the key to smart manufacturing. To evaluate the similarities and differences in big data and DT, they are compared from different aspects as exemplified in Table 9.1. The comparison that is discussed in this section has been investigated in the authors’ previous work [9].

9.4.1 Comparison From General Perspective

9.4.1.1 *Similarities Between Big Data and Digital Twin*

First of all, big data originated from the exponential growth of the amount of data resulting from the advance of information technologies. The purpose of DT, meanwhile, is to respond to the desire for interaction and integration between the physical and virtual spaces, which is inseparable from the rapid popularization and application of information technologies [45]. Therefore, although the initial focus of big data and DT are

Table 9.1 Comparison between DT and big data in manufacturing [9]

Items	Big data	Digital twin
Background	The rapid development and wide applications of New IT and the exponential growth of the amount of data	The rapid development of New IT and the desire for cyber–physical fusion
Concept	Including data and processing Focusing on large volume and value	Including physical and virtual spaces and data that ties the two spaces Focusing on virtual–real dual reflection
Functions	Mining behavior features and patterns Insight into trends Data visualization Predicting and analyzing problems Aiding decision making Optimizing and improving processes	Virtual verification Simulation running Ultrahigh-fidelity real-time monitoring Predicting and diagnosing problems Optimizing and improving processes
Applications	Product lifecycle from design to PHM, etc.	Product lifecycle from design to PHM, etc.
Effects	Improving efficiency, customer satisfaction, and management precision Extending life of product and equipment Reducing cost Promoting smart manufacturing	Improving efficiency, customer satisfaction, and management precision Extending life of product and equipment Reducing cost and development cycle Promoting smart manufacturing
Key technologies	IoT Cloud computing Data cleaning Data mining Machine learning, etc.	IoT Virtual reality Augmented reality CPS Simulation, etc.
Data sources	From physical entities, information systems, and Internet in every stage of the product lifecycle	From physical entities, virtual models, and their fusion in every stage of the product lifecycle
Data volume	Large, ranging from petabyte (PB) to exabyte (EB), even zettabyte (ZB)	No specific quantity
Data features	Structured, semistructured and unstructured data	Structured, semistructured and unstructured data
Multisource correlation	Focusing on data attributes and highlighting relationships between features	Focusing on the consistency of multisource data and their evolution and integration
Data acquisition tools	Sensors, RFID, and other sensory devices SDK, API Web crawler, etc.	Sensors, RFID, and other sensory devices Model/data interfaces, etc.
Data processing	Through big data processing tools, algorithm platforms, etc.	No specific methods
Data fusion	Various objects' data fusion in single phases of the product lifecycle	Entire elements, entire process, entire business data fusion in the product lifecycle
Timeliness and accuracy	High	High
Visualization	Table, chart, graphs, and file printing, etc.	Image, video, virtual and augmented reality, etc.
Result verification	Through physical execution processes, or simulation from the third party Relatively slow	Through its own virtual simulation and evolution functions to perform verification before actual execution Relatively advanced

different, their background is roughly the same, namely the advance and wide application of New IT.

All of the functions of big data are from data processing. Big data analysis can identify the behavior features and patterns, provide an insight into the trends, and to help users make decisions. The prediction or optimization abilities of big data are enabled by training data sets or comparing historical results. Whereas, the functions of the DT rely on the simulation and evolution of the virtual models. Via ultrahigh-fidelity virtual models, the DT can simulate the entire operation process independently, according to the actual operating rules of the physical space. Compared with big data, the DT can visually run and verify the manufacturing process in the virtual space. However, although the implementation methods have their own characteristics, both of them share a majority of the same functions. For example, both of them can predict and diagnose the problems, as well as monitor, optimize, and improve the manufacturing process in real-time.

In addition, the same functions lead to similar effects. They both improve production efficiency, customer satisfaction, and precision of management, as well as reduce costs in their own way, thus promoting smart manufacturing. Moreover, they also extend the lifetime of products and equipment and reduce the development cycle.

In conclusion, although big data and DT differ from each other in some details, they are consistent in the overall direction of their background, functions, and effects. Therefore, they have a foundation for cooperating with each other.

9.4.1.2 Differences Between Big Data and Digital Twin

With respect to concept, the contents of big data are relatively simple, emphasizing the large scale and value hidden in the data. To extract value from the massive amount of data, big data uses advanced tools and algorithms that are different from traditional ones [46]. Whereas, the DT is composed of three components, of which data is an integral part. The DT is more concerned with the virtual–real dual reflection. Although both of them involve data, big data is more professional in terms of tools, algorithms, and platforms, and more efficient in data processing than the DT in this regard, as explained in Section 9.4.2.1. Therefore, big data can serve the DT.

In term of applications, both of them are applied in every stage of the product lifecycle from design to PHM. However, compared with the

DT, there are barriers between the various phases of the lifecycle in the application of big data in manufacturing. As designers, manufacturers, service people, etc., may not be from the same company, the data from a certain phase sometimes may only be used in its own phase. Taking the interests and data sharing security into account, it does not fully realize the continuous flow of data in the product lifecycle. In contrast to it, the DT can collect, record, accumulate, and comprehensively process all the data from product design until retirement. It not only can be conducive to design, manufacturing, usage, and PHM of the product, but can also contribute to the next generation of the product. Therefore, the DT can make up for the deficiency of big data to break barriers in the product lifecycle.

In addition, big data and DT share some key technologies, such as IoT. However, big data focuses more on the technologies concerning data (e.g., cloud computing, data cleaning, data mining, machine learning, etc.), while the DT cares more about the technologies concerning cyber–physical fusion, for example, simulation, virtual reality, augmented reality, cyber–physical system (CPS), etc. The combination of their key technologies will be more effective for their application in the product lifecycle.

In conclusion, despite their different emphases, both of them have their own strengths. Moreover, their advantages are complementary to make up for their respective deficiencies. Therefore, they have a bonding point to cooperate with each other.

9.4.2 Comparison From Data Perspective

Data is the common content for both big data and DT. Their respective advantages in terms of data are analyzed as follows.

9.4.2.1 *Advantages of Big Data Over Digital Twin*

First of all, the large scale is inherent in the concept of big data. The volume of big data ranges from PB to EB, even to ZB. This abundant data means rich information and knowledge. In contrast, the DT does not indicate the specific quantity of data, as long as the data is not insufficient as an important part of the DT. However, by comparison, the amount of data that are collected, stored, managed, analyzed, and computed in big data, are certainly more than in the DT. Therefore, big data is more suitable for extracting knowledge from a larger volume of data.

In terms of multisource correlation, big data focuses on data attributes and highlights the relationships between the features. Big data is used to find correlations and knowledge through mining behavior features and patterns. The DT focuses on the consistency of the multisource data, which helps to simulate and deduce the manufacturing process. The inconsistency of the data will lead to conflicts. Therefore, big data is more suitable for greater varieties of data.

Besides, given that the massive data cannot be processed using regular data tools within a reasonable amount of time, big data processing has its own advanced tools, algorithms, and platforms. Whereas, the data processing methods of the DT are not specified. Therefore, the velocity of data processing in big data, which is more professional, is more efficient than in the DT.

In conclusion, big data is more professional and efficient than the DT in terms of volume, value, variety, and velocity of data, which are consistent with the characteristics of big data. This further deepens the role of big data in serving the DT.

9.4.2.2 Advantages of Digital Twin Over Big Data

As the DT is composed of three components, the data sources of the DT are different from those of big data. The data in big data are from physical entities, information systems, and the Internet, which are all generated by activities in the physical space. The data in the DT are not only from physical space, but also from virtual models. In addition, some data are derived from fusion operations, such as synthesis, statistics, association, clustering, evolution, regression, and generalization. In accordance with the different data sources, the data acquisition tools are also different. In addition to the tools they share for data from the physical space (e.g., sensors, RFID, etc.), the DT also needs model data interfaces to collect data from the virtual space, which are not required by big data in manufacturing. Therefore, the DT has more comprehensive data for use.

Furthermore, the data fusion of big data is the fusion of various objects data in a single phase of the product lifecycle, due to the barriers between different phases. Whereas, the data fusion of the DT involves entire elements, processes, and businesses. The DT can achieve data sharing and integration between different phases of the product lifecycle. Therefore, the DT extends the range of application of manufacturing data, and thus avoids duplication and waste.

With respect to how results are presented, big data prefers to use two-dimension and static tools, such as tables, charts, graphs, and file printing. Because of its virtual models, visualization in the DT is more visual, mostly three-dimension and dynamic, using images, videos, and virtual and augmented reality. Furthermore, because the object of big data is just data, the verification of the analysis of results must be through the physical execution process, or through simulation by a third party, which is relatively slow. Whereas, with its own virtual simulation function, the DT can complete the preverification of the results in the virtual space. Therefore, the DT is more advanced and convenient in terms of visualization and result verification.

In general, from the data perspective, big data is more powerful than the DT in respect to data, while the DT is more capable than big data in application. Therefore, it is an ideal option to develop the DT in combination with big data.

9.5 COMPLEMENTARITY BETWEEN BIG DATA AND DIGITAL TWIN

There are many visible and invisible problems in smart manufacturing, which can be reflected by the data. Big data brings more efficient, sharper insight and more intelligence to manufacturing. However, as manufacturing processes become more complex, it is difficult to quickly identify problems arising in the myriad of processes. In the general manufacturing process, designers and manufacturing staff are affiliated with different departments or systems, and work independently. The designers submit product concepts and design schemes to the manufacturing department. The manufacturing department then reflects on how to implement them. This can easily lead to product information loss. Once the design changes, it is difficult to achieve synchronous updates in the manufacturing process. Besides, due to processing and assembly errors and other factors, real manufacturing implementation and production planning may not be completely consistent, hence limiting the manufacturing effectiveness. It is also possible that the manufacturing process cannot meet all the design requirements. Moreover, common product maintenance methods are often passive rather than proactive and are often based on heuristic experience. Many faults can only be diagnosed in the situation where the product is used. Therefore, the questions become: how to feed back the effective information collected from the manufacturing process to the

product design phase? How to realize the correlation and dynamic adjustment between manufacturing planning and implementation? How to promote fault prediction, diagnosis, and maintenance digitally? These are big challenges. The combination of big data analysis and the DT paves the way to solve the abovementioned challenges. The DT can integrate the different phases of the product lifecycle to break data barriers. Meanwhile, big data technologies can quickly process data and provide an insight into manufacturing to help make decisions. Moreover, embracing the concept of manufacturing as a service (MaaS), service-oriented smart manufacturing has received extensive attention. As an effective means, services can enable large-scale smart applications and manufacturing collaboration. Therefore, as shown in Fig. 9.4, the DT, big data, and services form a unity to promote smart manufacturing [9].

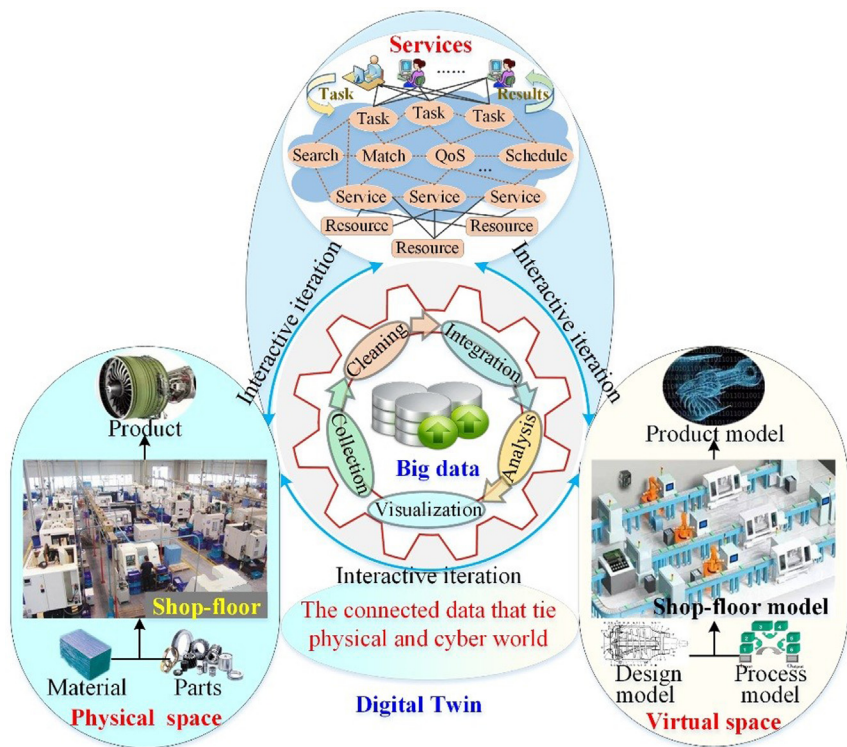


Figure 9.4 Fusion of DT, big data, and services in manufacturing [9].

9.6 FUSION OF DIGITAL TWIN AND BIG DATA IN MANUFACTURING

9.6.1 Product Design Driven by Fusion of Digital Twin and Big Data

In the design phase, product innovation relies on the accurate interpretation of market preferences and customer demands, and the capacity to translate customer voices to product features and quality requirements. Big data enables designers to promote product innovations, and develop more customized products. After the product concept is confirmed based on big data, the function structure and components of the product can be designed in virtue of the DT. In the traditional design pattern, after completing the design, some key components must be first manufactured to allow the manufacturers to assess the design quality and feasibility. However, by taking advantage of the DT and big data, vivid simulation scenarios can be created to predict the finished quality of the designed product, and identify the design defects in the virtual space. The components are manufactured and assembled using virtual models simulating the actual production process. In this case, whether the designed parts can be manufactured, whether the parts can be assembled, whether the parts interfere with each other, and whether the design scheme meets the relevant specifications and functions requirements, can be quickly verified. If the design scheme cannot pass the simulation testing, the product must be redesigned in real-time. When the product is redesigned, the big data technologies will be used to identify problems to improve the design scheme. In the process of product design, the required tools and algorithms are used in the form of services. As a result, the DT in combination with big data shortens the design cycle, thus reducing the cost of time and money [9].

9.6.2 Production Driven by Fusion of Digital Twin and Big Data

In the production stage, the DT with big data can enable optimization of the production planning and real-time adjustment of the production process. Before the production begins, it is necessary to improve the production planning through confirming whether every requisite resource is available. First of all, hypernetwork-based manufacturing resource services of supply–demand matching and scheduling [47] are carried out by the services part in [Fig. 9.4](#), to quickly find available resources. On this basis,

all the processes and the resources for performing production tasks are integrated together for analysis and planning. Combining with big data analysis, the virtual space of the DT simulates and evaluates the different production strategies and planning until a satisfactory production planning is obtained. After acquiring the best production plan, it is delivered to the physical space to guide the actual production. If all the processes in the production conform to the plan, production can be carried out smoothly. Otherwise, faults and errors may occur from time to time. In this situation, real-time data is collected directly from the manufacturing resources to drive the virtual models to simulate the production process. The actual production results and plans are evaluated in the virtual space of the DT. If there are differences, big data analysis is used to find out the reasons and develop solutions, such as adjusting the equipment conditions or revising the plan. In the iterative interaction, the DT with big data ensures that the production can be fully implemented in accordance with the optimal plan. Besides, once the design changes, the production process can be easily updated in the virtual space of the DT, including updating the bill of materials, processes, and assigning new resources. After adjusting the production plans, the entire production process is simulated again to further optimize and maximize utilization of all resources. As a result, using the DT in combination with big data technologies, production planning and implementation can be understood and monitored better to achieve smart production [9].

9.6.3 PHM Driven by Fusion of Digital Twin and Big Data

In the daily operation of a product, through interaction between the physical product and the virtual model, problems can be identified quickly, which improves the efficiency of PHM. First, the virtual model in the virtual space is synchronized with the physical product through real-time data collected by sensors. The working status of the product and healthy conditions of the components are grasped in real-time. In addition, as complementary to the sensors data, the DT also integrates the historical maintenance data and the data derived from big data mining (e.g., the features and patterns of behaviors, operation rules, features of energy consumption, etc.). Through big data analysis of the above data, the DT can continually predict the health conditions and remaining life of the product. The DT with big data can also reveal unknown problems by comparing the actual product responses and expected product responses in a specific scenario. Once the hidden product problems are found, or faults

occur, the maintenance programs are simulated and optimized in the virtual space of the DT to facilitate the actual maintenance. In this way, according to the real-time product status and the historical records, whether the product needs maintenance can be assessed. Similarly, the resources, capabilities, tools, and algorithms are used in the form of services. Therefore, the DT with big data prolongs the product life span and improves PHM efficiency [9].

9.7 SUMMARY

In spite of their differences, DT and big data play complementary roles in the manufacturing industry. Big data can be considered as an important part of the DT. Without big data, most functions of the DT will be castles in the air. Without the DT, the big data analysis and actual manufacturing process will not be in parallel. The DT, cooperating with big data, breaks the barriers between the different phases of the product lifecycle, and shortens the development and verification cycle. Combined with the accurate analysis and prediction capabilities of big data, the DT can implement precise management in many aspects, from product design to verification, and from manufacturing to PHM, etc., resulting in improved production planning and more precise production control. The DT and big data should ideally be used together and complement each other, jointly promoting the development of smart manufacturing [9].

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CHAPTER 10

Digital Twin and Services

Digital twin (DT) provides an effective way for the cyber–physical fusion of manufacturing. Meanwhile, services can make all the business processes and operation procedures in manufacturing realize servitization and standardization, to achieve higher efficiency. Combined with the services, the DT will be easier for usage and will generate more acceptable analysis and evaluation results on product design, manufacturing, usage, prognostics and health management (PHM), and other processes. This chapter specifies and highlights what manufacturing services are, how manufacturing services and the DT converge, and how various components of the DT are used by manufacturers in the form of services.

10.1 INTRODUCTION

As manufacturing evolves toward socialization and servitization, service plays an increasingly more important role in manufacturing [1]. More manufacturers are adopting service logic for their businesses to cope with intense competition and gain more revenue [2]. Services can shield the heterogeneities of resources, which are conducted by different vendors using various standards and communication protocols/interfaces, and enable interaction and integration [3]. With the characteristics of on-demand use, dynamic reconfiguration, and platform independence [4], services endow manufacturing with the advancement of large-scale sharing and collaboration [30].

In view of the concept of everything as a service (XaaS), services can fully release the potential of the digital twin (DT). Through services, each component of the DT, especially virtual models, which are not easy to create rapidly, can be shared and used in a convenient “pay-as-you-go” manner [5]. Moreover, services are an integral part in the working process of the DT, and many actions require the support of third-party services. For example, multisource data fusion requires algorithms, and computing and storage services [30]. Therefore, in this chapter, the DT and services are combined, and DT service generation, management, and application are specified.

10.2 SERVICES IN MANUFACTURING

10.2.1 Concept of Servitization in Manufacturing

The concept of servitization in manufacturing, which has been studied by the authors in their previous work [26], is introduced in this section. The term *servitization* was coined by Vandermerwe et al. in 1988 to describe the phenomenon wherein more companies were adding value to their core products through services driven by customer demand [6]. In fact, servitization was conceptualized as an innovation of organizational capabilities and processes [7]. More companies have already expanded the scope of their services to enhance market competitiveness and gain more revenue [8]. In this context, service-oriented architecture (SoA) [9] provides an effective technique for servitization, and has been applied in various advanced manufacturing modes and systems, such as application service providers (ASPs) [10], manufacturing grids (MGrid) [11,12], product service systems (PSS)/industrial PSS (IPS2) [13,14], and cloud manufacturing (CMfg) [15,5]. For instance, distributed and heterogeneous manufacturing resource services in MGrid can be aggregated for collaborative operations with high efficiency [16]. Moreover, as a cloud-based, service-oriented manufacturing model embracing the concept of manufacturing as a service (MaaS) [17], CMfg can enable sharing of manufacturing resources and capabilities and cooperation in a convenient pay-as-you-go manner through services [18], to maximize the benefits of the manufacturing industry. CMfg has the characteristics of customer centrality, dynamic reconfiguration, on-demand mode, and sharing [19]. Services endow manufacturing enterprises with the advantage of coping with the challenges such as fierce global competition, increasing labor costs, and other kinds of pressures [2,20]. Because services have the characteristics of interoperability and platform independence, they pave the way for large-scale enterprise collaboration, leading to manufacturing socialization [21,22]. A complex manufacturing task can be accomplished collaboratively by many services from different companies [23,24]. In addition, many smart applications such as remote monitoring and diagnosis, predictive maintenance, operational analysis, and product usage (e.g., training, assistance, and interaction) are made possible through services [25]. Services bridge the gap between real production activities and logical collaboration [4], which can promote smart manufacturing applications.

10.2.2 Framework of Service-Oriented Smart Manufacturing

As shown in Fig. 10.1, the abstract framework of service-oriented smart manufacturing (SoSM) consists of two parts, the physical space and the virtual space. The physical space comprises manufacturing resources and human resources. Manufacturing resources include materials, machine tools, machining centers, robots, products, and software resources. Human resources refer to workers on the shop-floor, managers in enterprise, and users in the product's lifecycle. In addition, to interact with the virtual space, the physical space also includes sensor systems and actuator systems. The sensor systems are responsible for collecting various data of manufacturing resources and processes from the physical space. The actuators receive instructions from the virtual space and drive the manufacturing resources to achieve tasks according to smart decisions. All the manufacturing activities are accomplished collaboratively by humans, machines, and other entities in the physical space eventually [26].

The virtual space is composed of two important components. One of them is the virtual mapping of the physical space. Through real-time data acquisition, data integration, and analysis, it can dynamically track and monitor the status and progresses of the physical space, based on the virtual models from physical entities. This component consists of data,

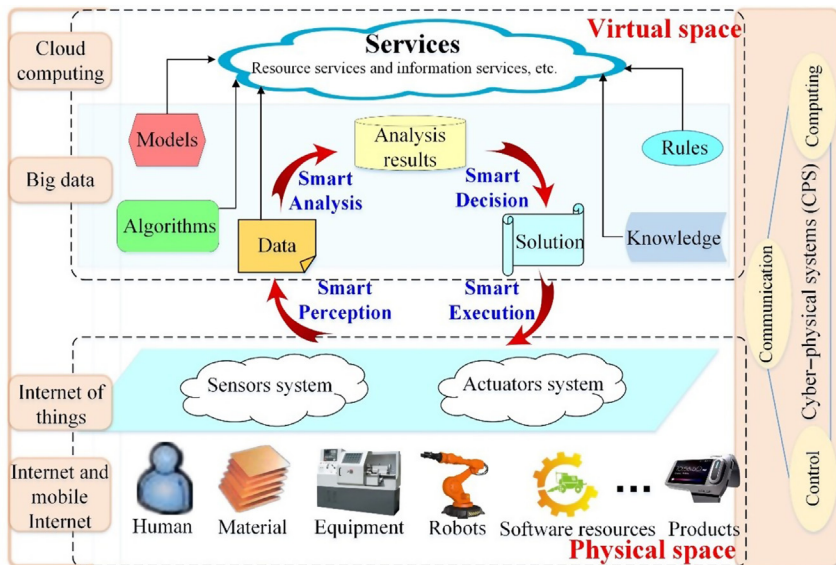


Figure 10.1 An abstract framework of SoSM [26].

models, algorithms, knowledge, and rules. A portion of the data is from physical resources, including status, properties, qualities, and other data from the manufacturing resources. Some data are generated from the interaction between the physical space and the virtual space. Besides the virtual models from physical entities, there are some mathematical models, computing models, and others. The algorithms include data processing algorithms, intelligent optimization algorithms, and data mining algorithms. After data analysis and mining, some knowledge and rules are generated. In addition, some expert knowledge and physical rules are also stored in the virtual space for the users. Another component of the virtual space includes the digital services. These services are encapsulated through a service-oriented approach, such as SoA. Some are resource services, which are encapsulated from the physical manufacturing resources and capabilities based on data from the physical space. Some other services are encapsulated from information elements (i.e., data as services, models as services, algorithms as services, knowledge as services, and rules as services). It should be noted that the physical resources (e.g., material, machines, products) and information elements (e.g., production data and algorithms) can be shared and used remotely by global customers because of service availability [26].

With the comprehensive analysis and processing capabilities for data and the interoperability characteristics of services in the virtual space, SoSM can cope with complex changes in the physical space through effective decision-making processes. In this process, there are four basic functions, that is, smart perception, smart analysis, autonomous decision making, and precise control and execution. First of all, various data from humans, materials, equipment, products, and systems in the physical space are accurately sensed through advanced sensing technologies. Then, the data acquired through perception are identified, processed, and analyzed quickly and accurately in the virtual space, as well as other data operations. According to the results of the analysis, decisions and choices are made autonomously in accordance with the rules and knowledge. Manufacturing resources are invoked in the form of services. Finally, on the basis of the decision making, the physical space is adjusted according to dynamic changes [26].

Humans, materials, equipment, environment, supply chain, and other manufacturing elements are integrated through data, models, algorithms, knowledge, and rules. Internet of Things (IoT), cloud computing, big data, mobile Internet, cyber-physical system (CPS), and other

technologies play important roles in SoSM. With the IoT and Internet technologies, various data in all aspects of manufacturing can be collected, and be converged continuously to the cloud from the terminals. Cloud computing and big data technologies process these original data through sorting, classification, analysis, and other operations to extract the value of the data. Through learning wisdom from the cloud, manufacturing equipment and enterprises can better understand the demands of customers, and these data are encapsulated into services. Through these services and data, coupled with the corresponding software, manufacturing big data are converted into practical information such as yield, quality records, health status of manufacturing elements, and energy efficiency. This information can further help manufacturers better judge and understand the status of manufacturing production. The interconnection and interoperability between the physical space and the virtual space enable SoSM to become a cyber–physical system. Through new generation information technology (New IT), the virtual space in SoSM can be more extensive to connect all things, more intelligent to capture and analyze data, and more proactive to provide services. The physical space can be more efficient to implement manufacturing activities, utilizing the potential of SoSM as a cyber–physical system [26].

10.3 SERVICES IN DIGITAL TWIN

All manufacturing resources and capabilities in the product lifecycle can be virtualized and then encapsulated into services. According to different existence forms and usage patterns, services can be mainly divided into resource services and information resources. The former is developed from hardware resources in manufacturing, such as manufacturing equipment, computing equipment, and materials, while the latter is from software resources, such as professional software, experience models, domain knowledge, and algorithms [27]. In addition, in a specific manufacturing activity, the resources can be extended to certain capabilities, such as in design, simulation, production, and maintenance [27–29].

For the DT, services can be secured from two perspectives. On one hand, services form a crucial component of the five-dimension DT. For example, when a production task is input, the service system can invoke corresponding services to complete the task through service scheduling and combination. On the other hand, the other components of the five-dimension DT (i.e., physical entity, virtual entity, data, and connection)

can also be encapsulated into services. Benefiting from comprehensive servitization, the DT can be uniformly managed in a service platform. When a task is submitted to the platform, the suitable services among all the candidates are invoked and combined to form a complete DT first, and then perform the task collaboratively.

10.4 DIGITAL TWIN SERVICE GENERATION

Models and data are the cores of the DT. However, the creation of virtual models and fusion of data are complex and specialized projects. For users who do not have relevant knowledge, it is difficult to build and use the DT. Therefore, it is imperative that the models can be shared by users and data analysis can be outsourced to third-party professional organizations. Moreover, in the context of manufacturing socialization, the physical resources involved in manufacturing are geographically distributed. With the characteristics of on-demand use, dynamic reconfiguration, and platform independence, services pave way for the problems mentioned above [30]. In the following sections, servitization of physical entities, virtual entities, and data in the DT [33] is mainly discussed.

10.4.1 Physical Entity Servitization

Service encapsulation translates various components of the DT into services with uniform description [31]. As shown in Fig. 10.2 [30], the first and most important step of service encapsulation is to establish the information template, which consists of rich information [32]. For the physical entity, the information includes basic attributes (e.g., name, ID, address), quality of service (QoS) (e.g., time, cost, reliabilities, satisfaction), capacities (e.g., precision, size, process), real-time status (e.g., overload, idle, in maintenance), as well as input and output. The information templates of the physical entity (i.e., PE) can be described as (10.1) [30].

$$PE = \{Basic, QoS, Cap, Status, Input, Output\} \quad (10.1)$$

wherein, *Basic* refers to the basic attributes for identifying the physical entity. *QoS* denotes the evaluation for the performance of the physical entity in conducting the user selection. *Cap* represents the functions of the physical entity. *Status* indicates whether the physical entity is available. Taking an NC machine tool as an example, the production capacity includes the capacities for shape processing (e.g., plane, hole, thread),

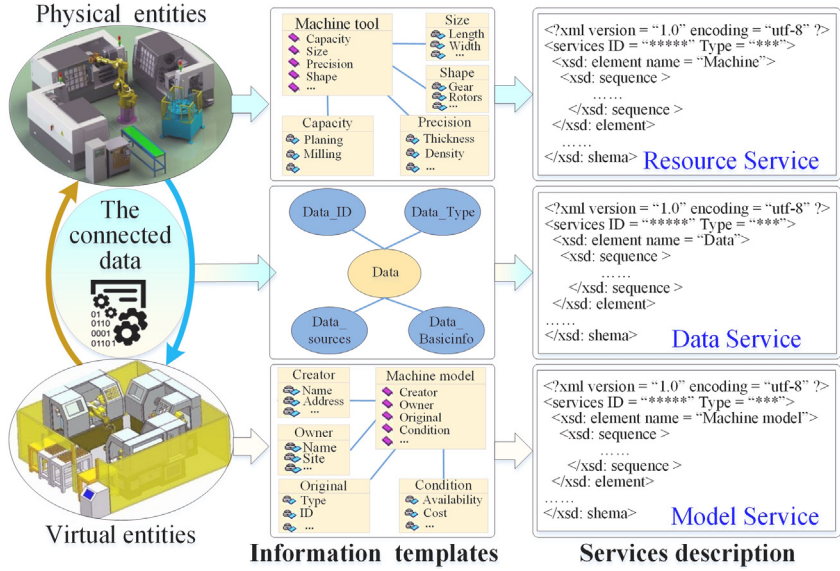


Figure 10.2 DT service encapsulation.

precision (e.g., roughness, deflection), parts processing (e.g., gear, plate), size (e.g., height, width, coordinates). Status can be overloaded, idle, in maintenance, etc. With respect to input and output, the input of the NC machine tool can be materials, such as steel, aluminum alloy, etc. And the output can be a part, or a small product. They can be formulized as (10.2)–(10.5) [30]:

$$\text{Basic} = \{\text{type, name, ID, address, color, delivery_time, ...}\} \quad (10.2)$$

$$\text{QoS} = \{\text{cost, time, reliability, trust, Fun_similarity, ...}\} \quad (10.3)$$

$$\text{Cap} = \{\text{shape_cap, precision_cap, parts_cap, size_cap, ...}\} \quad (10.4)$$

$$\text{Status} = \{\text{load, health, ...}\} \quad (10.5)$$

10.4.2 Virtual Entity Servitization

Similarly, the virtual entity, which is composed of a set of virtual models, can also be described by the information template. Different from the physical entity, the virtual entity can be used by multiple users at the same

time, and can be copied without repetitive creation for the same or equal physical entity. Therefore, when the virtual entity is encapsulated into services that can be shared by users, not only can the creators of the virtual models receive benefits, but the users can also reduce costs and time. The information templates of the virtual entity (i.e., VE) can be described as in (10.6) [30].

$$VE = \left\{ \begin{array}{l} \text{Ori_phy, Creator, Ori_ID, Cur_ID, Owner, QoS,} \\ \text{Online_site, Input, Output, ...} \end{array} \right\} \quad (10.6)$$

wherein, *Ori_phy* denotes the original physical entity corresponding to the virtual entity. *Creator* is the one who builds the virtual entity based on his specialized knowledge. *Ori_ID* is the original identifying number when the virtual entity is first created. *Cur_ID* represents the identifying number of the current copy. *Owner* is the one who possesses the virtual entity. The owner has copyright or may be the creator, and can earn profits through renting out the models or selling copies. Similar to the physical entity, *QoS* denotes the evaluation of the performance of the virtual entity, including cost, reliability, functions, etc. *Online_site* denotes the online address where users can access or download the models. *Input* and *Output* may vary according to the specific models [30].

10.4.3 Data Servitization

In addition, data are very important for smart manufacturing [33]. However, because of the various standards and communication protocols/interfaces, data are often difficult to acquire and understand. Through using the unified templates to describe the data, users can conveniently use the data. In general, data information includes the data provider who owns the data, data sources where the data is collected, data ID used to identify the data, data type (which denotes the kind of the data), and data abstract, which is a brief introduction to data value. Therefore, data (i.e., DD) can be described as in (10.7) [30].

$$DD = \{D_prov, D_source, D_ID, D_type, D_abstract, \dots\} \quad (10.7)$$

The information templates describe most of the information that users want to know and have to know. After the information templates are established, various kinds of physical entities, virtual entities, and data are encapsulated into services based on the service description language, such as XML. As an example, the information template of a punching machine

and a programmable logic controller (PLC) is encapsulated by a service based on XML, as shown in Fig. 10.3 [30]. The information template includes basic information, processing capability, process parameters, and axes information of the punching machine and PLC [30].

10.5 DIGITAL TWIN SERVICE MANAGEMENT

The circulation, transaction, and sharing of DT services depend on efficient management. As shown in Fig. 10.4 [30], after DT service generation, the services are published in the service pool and management platform, where they are managed to be shared by various users. Management for these services covers three stages, that is, the beginning of life (BOL), middle of life (MOL), and end of life (EOL), which are described as follows [18].

The BOL covers the process from the various physical manufacturing resources and capabilities to the virtualized services. It contains the perception and connection of resources, transmission of perception data, data filtering and processing, resource virtualization, service modeling, and digital description. After more services are generated, service aggregation will emerge gradually. As a result all the services are gathered and integrated by various abstract correlations and relations (i.e., the functions, a variety of attributes, characteristics, and invocation histories) along with the continuous operation of the service platform. The MOL is the stage driven by demands from the service generation to be put in use. The operations are service searching and matching, service evaluation, service selection and service composition, service scheduling, and service transaction. The MOL consists of service execution and monitoring, fault tolerance, and logistics service for task implementation. Among these items, the logistics services [34] serve for the specific processes and results of some kinds of services. According to the different requirements of task implementation, the logistics could be the transportation of the results after service execution, as well as the transportation of the materials in the process of service execution. Finally, the services are released and disaggregated after execution in EOL [18].

10.6 DIGITAL TWIN SERVICE APPLICATION

The DT services can be applied in product design, production planning, manufacturing execution, prognostics and health management (PHM),

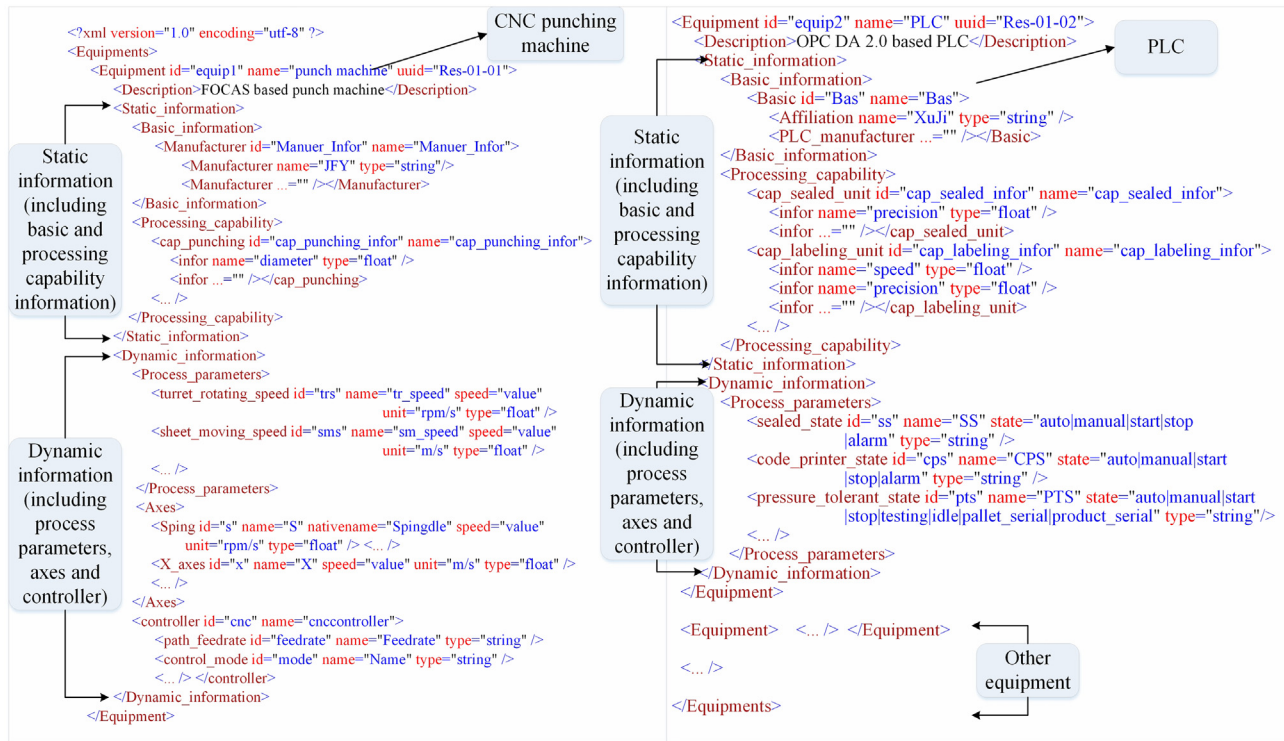


Figure 10.3 Service encapsulation based on XML.

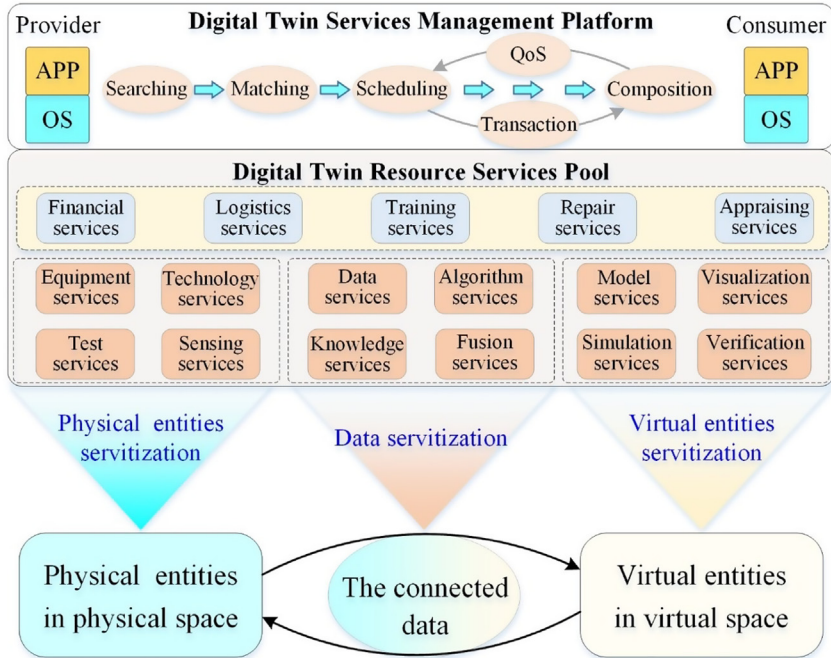


Figure 10.4 Service management framework for DT.

and other domains [35]. In the following sections, some applications that have been studied by the authors in their previous work [30] are introduced.

10.6.1 Digital Twin Service Application in Product Design

In product design, it is the process of back-and-forth interactions among expected space, interpreted space, and physical space [36]. DT-driven design aims to turn the expected product in the designer's mind into the digital representation in the interpreted space based on the existing physical products. To innovate a product, designers have to study plenty of data to acquire valuable knowledge [37]. However, the data concerning the product is one of the most important assets, and is not easy to access. Besides, the designers do not have the professional abilities to process massive data. Service is an answer to these problems. The designers just simply submit their needs to the services management platform. The service managers will match the data services that the designers need with the model services and algorithm services used to process the data. Through invoking and combining these services, the needed results will be returned

to the designers. As a result, the designers can acquire what they want in a “pay-as-you-go” manner [38]. Moreover, after the function structures and components of the product are designed, the design quality and feasibility need to be tested. In virtue of the DT, the designers can quickly and easily forecast product behaviors through verification of the virtual product without waiting until the product prototype is produced [39]. But the virtual verification needs the models of the manufacturing site (e.g., production line or shop-floor), which the designers do not have. After services encapsulation, the model services can be invoked and easily applied in product design, which makes product design more effective to reduce the inconsistencies between expected behaviors and designed behaviors, and greatly shorten design cycles and reduce costs.

10.6.2 Digital Twin Service Application in Production

In general, production is the entire process from the input of raw materials to the output of finished products, which is performed on the shop-floor. In the phase of production planning and production execution, the DT provides an effective method to draw up the plan and optimize the execution process [40]. First, a production task is submitted to the services management platform, and resource services of supply–demand matching and scheduling are carried out to find available resources. Then, based on the real-time status of the physical resources (e.g., machine tools, robot arms, materials), a production plan is drawn up. Digital twin shop-floor (DTS) can simulate the plan in virtual space and identify the potential conflicts even before the actual production [41]. However, building the DTS is a complex and specialized work, especially the models describing physical properties, rules, behaviors, etc. With the help of services, these models do not have to be created by manufacturers themselves. For physical equipment and pervasive rules, their models, which have been established by other manufacturers, can be bought to use in the form of services. Current manufacturers only need to create special models, which are private for themselves. Besides, during the operation of the shop-floor, some services, such as data processing and shop-floor management, would need to be invoked from the service system of the DTS.

10.6.3 Digital Twin Service Application in PHM

The degradation of performance is inevitable during operation of the physical equipment. Equipment malfunction will result in high

maintenance costs and postponement of tasks. PHM is essential to monitor the equipment's condition, and predict and diagnose equipment faults and component lifetimes. In DT-driven PHM, the virtual models of the physical equipment are synchronized with the real states of the equipment. The operation status of the equipment and the healthy conditions of the components are grasped in real-time. A high-fidelity digital mirror for the equipment provides access to the equipment even out of physical proximity. Besides, the interaction of the DT can reduce the disturbances from the external environment, thus improving accuracy. Similar to product design, production planning, and production execution, the models can be accessed through services. Moreover, when failures occur, maintenance services will be invoked to repair or replace the worn components of the equipment.

10.7 SUMMARY

DT has provided a promising opportunity to implement smart manufacturing by integrating the virtual and physical spaces in manufacturing. With services, functions of the DT can be extended and enhanced. In addition, the construction of the DT can be further standardized and simplified. In this chapter, through combining the DT with services, DT service generation, management, and application are discussed, respectively, and methods for servitization of DT components are provided. At present, the DT research is only in its infant stage. More studies need to be carried out to improve and enrich the methods of DT modeling and servitization with consideration of other factors and practical situations.

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CHAPTER 11

Digital Twin and Virtual Reality and Augmented Reality/Mixed Reality

Virtual reality (VR) and augmented reality (AR)/mixed reality (MR) are becoming more mature with the development of related technologies, such as the advanced sensors, computer graphics, real-time registration and tracking, etc. The VR and AR/MR technologies play important roles in the seamless integration and fusion of the physical and virtual worlds, which is crucial for digital twin (DT) implementation. In this chapter, the DT is combined with these technologies. Based on the existing and potential applications of VR and AR/MR in the DT, a general framework is proposed to integrate these technologies with the DT to provide more immersive and interactive services to users. According to the proposed framework, an assembly process is then discussed.

11.1 INTRODUCTION

The history of virtual reality (VR) began in 1962, when a device called the Sensorama Simulator was invented and patented to create a virtual environment that could provide users with multisensory experiences in terms of sound, scent, wind, vibration, etc. [1]. However, the environment enabled by the Sensorama Simulator was not interactive [2]. To solve this problem, Sutherland developed the first head-mounted display (HMD) system several years later, which could generate an interactive view in accordance with users' head position sensing [3]. Krueger also created a VR interactive environment, in which the computer could perceive a participant's actions and respond in real-time with visual and auditory displays [4]. In the 1980s, the Virtual Programming Languages (VPL) Company produced the popular DataGlove and the EyePhone HMD, which were the first commercially available VR devices [2]. In 1992, CAVE automatic virtual environment appeared [2]. Instead of using an HMD, it projected stereoscopic images on the walls of a room, thus

improving users' immersive feelings. In 1995, the VR modeling language was developed by Silicon Graphics, allowing people to create and interact with various virtual models conveniently and rapidly [5]. In the 21st century, VR has been applied to different fields, including entertainment [6], education [7], health care [8], manufacturing [9], etc.

Augmented reality (AR) can be viewed as a variation of VR [5]. In contrast to VR, which creates a completely virtual environment, AR overlays virtual information on the real world to augment the reality rather than replace it. The term *augmented reality* was proposed by Caudell and Mizell in 1990 [10]. In 1993, Rosenberg invented the first functioning AR system, that is, virtual fixtures, which could overlay computer-generated percepts on a real workspace to enhance human performance [11]. Feiner et al. developed a system to generate maintenance and repair instructions for a user wearing an experimental head-mounted, see-through display [12]. The system enables the user to interact with both the real surroundings and virtual information, and can be also considered as one of the early prototypes of AR. In 1998, Raskar et al. introduced the spatially augmented reality (SAR), with which virtual objects could be integrated directly within the physical space [13]. In 1999, ARToolKit was developed by Kato to provide an efficient and rapid method for AR system construction [10]. ARQuake was invented by Thomas et al. to convert the traditional desktop first-person perspective application to an outdoor/indoor mobile AR application for the first time [11]. In 2008, the Wikitude AR travel guide was launched for the Android phone, which added navigation data to real-time streetview [5]. In recent years, with the technical development of related hardware and software, AR has become an effective tool in activities such as planning, guiding, and training in different fields.

Mixed reality (MR) is another form of AR. In 1994, Milgram and Kishino [14] first proposed the concept of the “virtuality continuum” and specified that MR could provide a hybrid and continuous space that extends from a completely real environment to a completely virtual environment. MR is essentially the same as AR, and both of them focus on the fusion of physical and virtual elements. However, AR is a more commonly used term in related research and MR is less mentioned than AR. Therefore, in the remainder of this chapter, AR is used to denote AR/MR for short.

In recent years, VR and AR have been closely coupled and have mutually complemented each other. As they have shown potential for

seamless integration and fusion of the physical and virtual spaces, great importance has been attached to their development. In this chapter, VR and AR are combined with the digital twin (DT) to augment the two spaces of the DT and further enhance the fusion, thus helping the DT to achieve more efficient applications. In the following sections, VR and AR in design, manufacturing, and service are reviewed and then compared from different aspects. Based on the existing and potential applications of these technologies in the DT, a general application framework is proposed and then discussed in terms of an assembly process.

11.2 VR IN DESIGN, MANUFACTURING, AND SERVICE

VR can provide an immersive, interactive, computer-mediated experience in which a person perceives an artificial environment by means of human–computer interaction equipment, such as computers, HMDs, headphones, and data gloves [1,15]. Related technologies of VR include human–computer interaction, computer graphics, haptic feedback, dynamic environment modeling, solid vision, etc. As the VR is characterized by immersion, presence, interactivity, and navigation [16], many researchers have explored VR applications in the design, manufacturing, and service fields.

11.2.1 VR in Design

VR provides a completely virtual environment for designers to develop prototyping, interact with product models, evaluate alternate designs, etc. It can improve visualization of the product, provide intuitive model manipulation, and realize collaborative design. Some related studies on factory layout design and product design are introduced as follows.

In factory layout design, Iqbal and Hashmi [17] discussed how to design and analyze a factory layout in a 3D virtual environment by means of VR and developed a new aisle system to reduce the material handling distance. Menck et al. [18] proposed a collaborative platform based on VR for factory layout design. Users in different places can cooperate, explore, and analyze within the same immersive system in real-time. Gebhardt et al. [19] proposed a holistic VR application, called the Factory Layout Planning Assistant, which not only serves as a platform for realistic virtual walk through simulation, but also provides insights for decision making. Ebner et al. [20] developed a VR-based multitouch table, which allows the users to select necessary items from an object library and place

them on a layout with the touch of a finger. Errors or information can be evaluated and shown on the table in real-time.

In product design, Ryken and Vance [21] combined VR with finite element analysis, sensitivity analysis, collision detection, etc., to create an interactive environment, where designers can modify part shape, evaluate stress, and check for interference for new parts. Bruno et al. [22] used an interprocess communication library to link a VR environment and a numerical simulation environment together. In this way, product behaviors can be simulated and predicted using numerical simulation software and then visualized in the VR environment. Sung et al. [23] proposed a VR-based cable harness design system, which can automatically capture design knowledge embedded within log files, and transmit the knowledge to users during a design task.

11.2.2 VR in Manufacturing

VR can be applied in manufacturing for process planning, simulation, inspection, verification, etc., which can help operators better understand the manufacturing process and improve manufacturability, durability, and profitability of the final product. VR applications in computerized numerical control (CNC), virtual assembly, and robotics are mainly introduced.

In CNC simulation, Luo et al. [24] proposed a hierarchy structure of NC behavior simulation based on VR and developed a prototype system. Lin and Fu [25] presented a virtual modeling method to generate structures of machine tools in response to the functional and structural requirements from users. A virtual machine tool is developed for designing, testing, optimizing, controlling, and machining in a virtual environment. Kao et al. [26] built a virtual controller for the CNC milling machine based on VR software and Microsoft visual basic programming environment. The virtual controller can parse users' NC codes, simulate tool paths, and drive the virtual CNC according to the tool path.

In assembly verification, Seth et al. [27] developed a low-cost VR application that could perform collision detection and behavior simulation in a virtual environment for assembly. The application could run on many types of VR systems (e.g., Desktop, Cave, Power wall, etc.) and operating systems (Linux, Windows, Irix testbed, etc.). In the Virtuelle Werkstatt project, Biermann et al. [28] designed a knowledge-based approach to assemble CAD-based parts in a VR environment, and proposed a multimodal interaction using gesture and speech to conduct

human—computer interaction in an intuitive and natural way. Wang et al. [29] developed a virtual assembly and machining system for a machine tool using Visual C++ and Open Inventor, which could simulate the assembly and machining processes and key parameters. The system was validated on CK6430, a machine tool produced by a Shanghai factory. De Sa and Zachmann [30] presented several interaction paradigms for virtual assembly and maintenance, which enables inexperienced users to work with virtual prototypes efficiently.

In robot path planning, Chen et al. [31] built a haptic-based virtual robot arm driven by PHANToM device and proposed a workspace mapping method based on robot kinematics analysis. The virtual robot arm can be used in both free path planning and constraint-based path planning. Yap et al. [32] developed a VR-based support system for robotic work cell layout planning and programming. In their work, a VR-robotic work cell layout (VR-RoWL) and VR-based robot teaching system (VR-RoT) were constructed to complete the layout planning for robots and to overcome safety issues in programming, respectively. Liu et al. [33] presented a feature mapping algorithm to convert the design features of workpieces to the machining features of the weld seams, which supports motion navigation for welding robots in the virtual environment.

11.2.3 VR in Service

With the aid of VR, services can be simulated and optimized in an immersive virtual environment before actual execution, avoiding potential risks to the operators as well as equipment. Meanwhile, VR can enhance the operators' abilities and thus improve the quality of related activities. Some works on VR in maintenance, logistics, and training are introduced in the following sections.

In maintenance simulation, Lecakes et al. [34] developed a VR system to visualize multiple sensor measurements for the health management of rocket engines. The system can display various forms of data coupling with graphics for operators. Liu and Hao [35] presented a virtual maintenance simulation-based maintainability analysis framework, which includes a virtual maintenance prototyping module, a maintenance simulation module, a maintainability analysis module, and a system management module. Peng et al. [36] built a desktop VR-based approach for complex product maintainability design and verification. The efficiency of the approach was verified on a mobile robot system.

In logistics planning, Reif and Walch [37] considered VR as an innovative way to plan a logistics system. Using a data glove and a treadmill, an immersive workspace for order picking planning can be built, allowing workers to grab goods out of the storage location and walk through the virtual warehouse. Xu [38] assigned the logistics warehousing space through building a space distance calculation model and simulating the layout of the warehouse based on VR technology.

In training, Bhatti et al. [39] presented a desktop computer-based virtual training system, which can provide knowledge on assembly sequence to help operators in procedural learning and skill development. Gutiérrez et al. [40] proposed a multimodal training system that provides different aids and learning strategies according to task demands and user preferences. The system includes four main elements: (1) a task to be learned, (2) a virtual assembly scenario related to the task, (3) evaluation models of users' performance, and (4) a set of learning strategies. Li et al. [41] presented a low-cost desktop VR system for maintenance training, which integrates a disassembly sequence planning model and evolutionary algorithms.

For the above studies, the immersive virtual environment enabled by the VR technology can plan, inspect, simulate, optimize, and validate the activities in design, manufacturing, and service before actual execution. However, as VR provides a completely virtual environment, the users are isolated from the familiar physical environment and do not have a realistic experience, which decreases the fidelity and credibility of the constructed scenes [42]. Building a completely virtual environment also imposes a heavy burden on modeling and computation processes, and presents challenges to real-time performance. Given the aforesaid shortcomings of VR, AR is thus introduced into design, manufacturing, and service.

11.3 AR IN DESIGN, MANUFACTURING, AND SERVICE

AR (including MR) contains a set of innovative techniques (e.g., real-time data acquisition, human–computer interaction, scene capture, real-time tracking and registration, etc.), and can augment the view of the physical world by embedding computer-generated elements or objects [10,42]. It enables people to work in a physical environment that is fed with valuable and abundant virtual information and models. In this way, the high cost of constructing a fully virtual environment can be reduced, and users are allowed to interact with both the physical and virtual objects in an intuitive and natural manner [42]. To date, AR has been widely researched in the design, manufacturing, and service fields.

11.3.1 AR in Design

As AR can provide designers with intuitive visualization, design evaluation information overlaid on the physical environment in an interactive modification mode, and a collaborative platform, it can facilitate the design process. Some works about this technology in design are introduced, mainly including the factory layout design and product design.

In factory layout design, Rauterberg et al. [43] proposed a natural user interfaces (NUI)-based framework to combine real and virtual objects, and then developed a prototype called Build-it, which enables user manipulation (e.g., selection, fixing, positioning, and rotation) during the design process. Siltanen et al. [44] discussed the application of AR in space reservation for layout planning. Yun et al. [45] developed an AR-assisted factory layout planning and analysis method, which overlays 3D models of machines in the physical environment considering spatial constraints and collision detection.

In product design, AR plays a major part in the prototyping process [46]. Ng et al. [47] developed a gesture-based design evaluation system, which can use hand gestures as the main interaction input to manipulate and modify the virtual prototypes in real-time. Shen et al. [48] built a client/server framework to support collaborative product design. In the framework, with the use of augmented virtual information and highlighted features, the designers at different locations can observe and modify the virtual objects in real-time and communicate with remote users. Park [49] built an AR-based reformable mock-up, which enables evaluations of various shapes, colors, textures, and user interfaces for the product in a tangible manner, thus realizing more efficient and realistic product design and evaluation. Barbieri et al. [50] presented a method to perform usability tests in an MR environment for household appliances. Different kinds of prototypes were used to complete the evaluation process, including the virtual, mixed, and physical prototypes. Saleem et al. [51] constructed an MR environment, where a real system runs in conjunction with a simulation environment for frictional parameters design for a servopneumatic system.

11.3.2 AR in Manufacturing

In manufacturing, AR can be applied to superimposing virtual information on real production processes to perform planning, guidance, optimization, etc. Some typical applications of AR in CNC simulation, assembly, and robotics are introduced as follows.

In CNC simulation, Zhang et al. [52] proposed a multiregional computation scheme to realize an AR-assisted in situ CNC simulation system, which includes a real cutter and a virtual workpiece. The system can provide users with a familiar physical environment, thus enhancing the users' inspection of the machining process. Kiswanto and Ariansyah [53] integrated AR with machining simulation of a three-axis CNC machine to assist users in validating an NC program. Weinert et al. [54] developed a milling simulation system for a five-axis CNC machine to recognize and avoid collisions and excessive cutting forces.

In assembly planning, Ong et al. [55] integrated assembly product design and planning (PDP) activities with workplace design and planning (WDP) activities. At the early design stage, WDP information can be fed back to optimize the assembly platform design and product assembly design. Wang et al. [56] presented a method to construct a mixed environment through incorporating real objects into the virtual environment, using laser scanned 3D models with color-based marker tracking. In the mixed environment, engineers are able to conduct hands-on assembly design verification. Wang et al. [57] proposed an AR-aided interactive manual assembly design system, which allows users to simulate manual assembly in a bare-hand interaction manner.

In assembly guidance, Yuan et al. [58] proposed an AR approach for guiding an assembly process. The approach develops a virtual interactive tool to control the AR system and a visual assembly tree structure for information management, so that the assembly operator can easily step through a predefined assembly plan. Zhang et al. [59] applied radio frequency identification (RFID) and AR in assembly guidance to provide just-in-time information rendering and intuitive information navigation. Wang et al. [60] developed a novel human cognition-based interactive augmented reality assembly guidance system (CARAGS), which provides multimodal AR guidance in response to users' needs for quicker and more accurate task execution. Evans et al. [61] evaluated the ability of the Microsoft HoloLens, a wireless AR HMD, in an AR assembly application. The research indicates that although the HoloLens could deliver AR assembly instructions, there still exist some deficiencies, such as inaccurate location tracking for the parts and the assembly station. Rodriguez et al. [62] developed an MR-assisted system for manual operations based on project mapping technology, which can provide instructional guidance directly at the assembly workstation to guide users step by step until the assembly procedure is completed successfully.

In robot path planning, Chong et al. [63] proposed an approach for robot programming using AR, which allows a user to guide a scalable virtual robot in a real environment to move according to the user's required operation paths. Fang et al. [64] proposed an AR-based robot programming system to preview and evaluate the motion of a robot before the execution of a task, considering dynamic constraints. With the system, a user can perform modification and selection using an interactive device, to obtain a smooth and collision-free path. Gianni et al. [65] demonstrated an AR-based framework to evaluate the capability of a robot to plan safe paths in real outdoor scenarios, considering dynamic environmental changes.

11.3.3 AR in Service

Applications of AR in service mainly include maintenance, logistics, and training. Via AR, the real environment can be augmented with text, labels, models, and videos, which will conduce to fewer mistakes, faster speed, and higher quality of the service process.

In maintenance, Wang et al. [66] proposed an AR-assisted equipment maintenance based on infrared markers. The authors applied the method to air filter replacement and gearbox maintenance, and 3D models and annotations could be placed in the real scene to facilitate the maintenance process. Henderson and Feiner [67] designed an AR-based prototype to augment a military mechanic's natural view with text, labels, and animated sequences, thereby supporting quicker task location in maintenance. Suárez-Warden and Mendivil [68] proposed a method of voltage test for electrical diagnosis aided by AR for equipment. Four types of 3D models of the equipment components and corresponding AR markers were designed to trigger different scenes of the voltage test. Zhu et al. [69] proposed a context-aware AR system for routine and ad hoc maintenance activities, during which the system can render useful context information to operators and adapt to dynamic maintenance situations.

In logistics, Ginters and Martin-Gutierrez [70] used AR and RFID for item visualization in a warehouse, which allows users to be immersed in a natural environment and to manage the objects without avoidable mistakes. Kang and Zhang [71] proposed a tracking solution for outdoor logistics based on hybrid radio frequency recognition technology and AR, which facilitates target recognition, goal recognition, content display, and logistics tracking. Reif and Walch [37] explored the AR-based order

picking. Using an HMD, a worker can be fed with important data in his view, such as position and process-oriented information, which could reduce searching time and prevent mistakes.

In training, Westerfield et al. [72] combined the AR graphics with adaptive guidance from an intelligent tutoring system (ITS) to form a training system for manual assembly tasks. Based on the framework, a prototype was built to teach novice users to assemble a computer motherboard. Schwald et al. [73] presented an AR system, mainly including an optical see-through HMD, a tracking system, and a special stand, for training and assisting in maintaining equipment. Webel et al. [74] applied AR to training of assembly and maintenance skills. Based on the needs of trainees, 3D animation and contextual information can be superimposed on the real environment. Through a teleconsultation component, trainers can modify or enhance the training protocol according to the trainees' performances. Träskbäck and Haller [75] tried to apply MR to onsite training for dangerous industrial processes (e.g., an oil refinery). In the research, the user requirements for such an MR-based training tool were discussed mainly to lay the foundation for future works.

Research on AR applications is promising and growing. In the above works, AR enhances the activities in design, manufacturing, and service through embedding the virtual scene generated by the computer into the real scene. The realistic experience of the user can be retained and large amounts of modeling works can be avoided. With AR, the virtual and physical worlds will be merged more tightly to create a new environment, where more immersive scenarios as well as more natural interactions could be enabled.

11.4 COMPARISON BETWEEN VR AND AR

VR and AR have played important roles in the fusion of the physical and virtual worlds. According to references [76–79], similarities and differences between them are compared from different aspects, as depicted in Table 11.1.

Some aspects in Table 11.1 are discussed as follows:

- *Environment*: VR provides a fully virtual environment, which mainly consists of virtual graphics, images, models, etc. In such an artificial space, users can build their own worlds and customize their experiences. AR creates an augmented physical environment through superimposing the virtual elements on the physical ones to display

Table 11.1 Comparison between VR and AR

Items	VR	AR/MR
Environment	Fully virtual environment	Overlaying virtual objects on the physical environment
Image source	Virtual objects (e.g., insights, graphics, images, models)	Combination of physical objects and virtual objects
Perspective	Positions and sizes of virtual objects will change according to the users' perspective in the virtual world	Virtual objects behave based on the users' perspective in the physical world
Presence	In the virtual world with no sense of the physical world	In the physical world, but superimposed with virtual objects
Interaction	Allowing the users to interact with virtual objects for manipulation, operation, control, etc.	Allowing the users to interact with virtual and physical objects at the same time in a natural way
Immersion	Immersing the users into a virtual world, with visual, auditory, haptic, and other feedback	Permitting the users to stay in the familiar physical environment that is supplemented by virtual objects, and to get visual, auditory, haptic, force, and other feedback
Related products	Oculus Rift, HTC Vive, Sony PlayStation VR, Samsung Gear VR, Google Cardboard, etc.	Microsoft HoloLens, Meta 1 & 2 Headsets, ODG headsets, Google Glass, etc.
Applications	Industry, real estate, tourism, construction, film and television, entertainment, health care, education, etc.	Industry, military, advertising, retail, tourism, navigation, entertainment, sports, health care, education, etc.

information that users cannot directly detect with their own senses [80]. It helps to enhance the users' perceptions and understandings of the physical world.

- *Perspective:* VR provides a user-centric perspective. The virtual objects will change their positions and sizes according to the users' perspective in the virtual world [76]. In the AR environment, the virtual objects behave based on the users' perspective in the physical world [76], and align with the physical objects around the users.

- *Interaction:* VR only allows the users to interact with the virtual objects. They can observe, manipulate, operate, and control the virtual world using interaction devices, such as HMDs, data gloves, and handles. But as the users are put into a completely synthetic environment, the interactions could feel different from the familiar ones, which may lead to uncomfortable feelings. With AR, the users can interact with both the physical and virtual objects at the same time. Since AR incorporates fundamental human skills, that is, interacting with the physical objects, it can make the users behave in a more natural way [43].
- *Immersion:* VR can fully immerse the users into a virtual world. It enhances depth perception and the sense of space for users through building a stereoscopic artificial environment, which allows the users to look around, walk around, and even fly around [2]. It also augments the immersion through feeding the users with visual, auditory, haptic, and other feelings. AR permits the users to move freely around the familiar physical environment and the supplementary virtual elements can provide visual, auditory, haptic, force, and other feedback. In this way, more realistic experiences can be obtained by the users.

11.5 DIGITAL TWIN AND VR AND AR

11.5.1 Existing Studies and Applications of VR and AR in Digital Twin

As VR and AR can create an interactive and immersive environment to augment the seamless integration between the physical and virtual worlds, some researchers and organizations (especially some reputable companies) have attempted to apply them in the DT. To date, the existing and potential applications of these technologies in the DT mainly focus on visualization, physical–virtual integration, user experience improvement, remote collaboration, etc., as presented below:

1. *VR in DT.* Rockwell Automation attempted to employ the DT to build a safer and more immersive environment for training, and to leverage VR and AR to enhance visualization for the DT [81]. Siemens has developed realistic virtual substitutes for gas turbines, which are visualized with VR goggles [82]. In the VR environment, sensor data is translated into colors to express the temperature distribution, which facilitates maintenance. Alibaba Cloud focuses on how to use the DT to generate an integrated data model to improve the existing VR systems [83]. Huawei has combined the DT with VR for

virtual assembly [84]. It has constructed DTs for products and assembly stations, with which designers can simulate different assembly scenes wearing VR glasses and gloves. Oracle enables users to visit and work in the DT-based virtual plant via Gear VR [85].

2. *AR in DT.* PTC has combined the DT with AR and VR to guide maintenance for a physical machine [86]. GE applied a Microsoft HoloLens into the DT to superimpose a virtual steam turbine over the actual one, allowing users to look inside the turbine to identify the damaged parts [87]. Peaxy believed that data visualization enabled by AR is essential for the DT to solve real business problems [88]. Microsoft has applied MR into a process DT that encompasses the entire production environment. With MR, holograms of the production line overlaid with real-time data and information were developed, which permits users to work remotely and to collaborate with teams in disparate locations [89]. Besides these companies, other researchers have also carried out related studies. For examples, Kritzler et al. [90] proposed the concept of virtual twin based on AR, which is an interactive virtual representation of the corresponding physical object. Schroeder et al. [91] retrieved data from the DT via web services and displayed the data to users in real-time via an AR system. Peuhkurinen and Mikkonen [92] researched how to use the existing web technologies to develop DT applications in an MR environment. In future, by combining with technologies such as VR, AR, artificial intelligence (AI), and the Internet of Things (IoT), the DT can be created in a cost-efficient way and enable users to work with the digital substitute remotely, eliminating or reducing the need to check the real object on site [93,94].

In view of the above works, it is clear that combining VR and AR with the DT has become an inevitable and beneficial trend. However, a general application framework that can introduce VR and AR into the five-dimension DT and enable these technologies to be coupled with each DT component is still lacking. To solve the problem, such a framework is discussed in the following section.

11.5.2 Application Framework of VR and AR in Digital Twin

Based on the existing and potential applications of VR and AR in the DT, a framework that applies these technologies to the DT is proposed, as shown in Fig. 11.1. In the DT, the physical entities (PEs), virtual

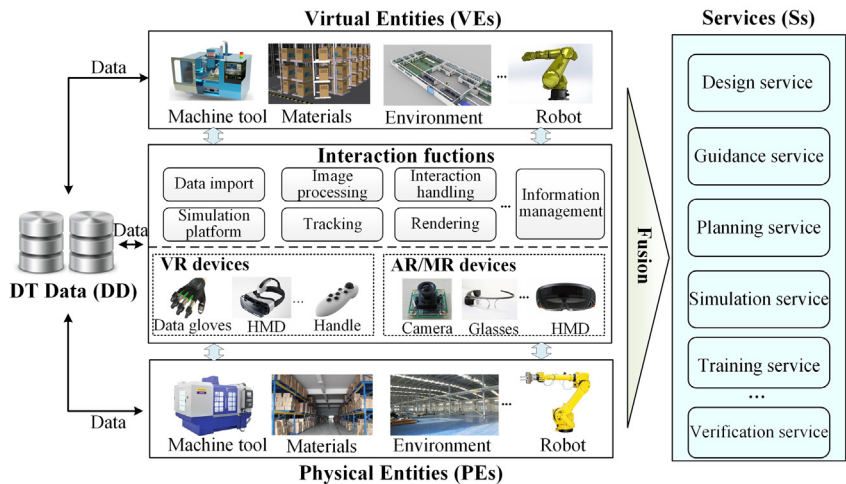


Figure 11.1 Application framework of VR and AR in DT.

entities (VEs), DT data (DD), and services (Ss) are connected with each other. The PEs (e.g., physical machine tool, robot, and materials) and VEs (e.g., virtual machine tool, robot, and materials) are kept synchronous and interactive through advanced sensory devices. The DD can store and fuse data from both the PEs and VEs to provide valuable information for services. Based on this, the introduction of VR and AR can further enhance the interaction and fusion between the physical and virtual spaces, thus feeding users with higher quality services in design, planning, guidance, and training. Due to the different characteristics of VR and AR, their roles in the framework are also different.

1. With VR, the high-fidelity VEs can be presented in a stereoscopic synthetic environment. With VR devices (e.g., microphones, handles, touch screens), sounds, operations, positions, etc., from users in the physical space can be captured and input into the virtual world. The VEs behave autonomously based on the inputs, supported by related functions such as image processing, demand analysis, real-time computing, model invocation, and updating. Meanwhile, insights from the DD (e.g., real-time equipment states, work instructions, and maintenance codes) can also be generated and invoked to demonstrate the corresponding VEs. The VEs and DD will be presented to the users in visual, auditory, haptic, and other modalities through devices such as HMDs, data gloves, and screens. As the VEs and DD can be provided in an interactive and immersive manner, services supported by them

will be more intuitive to improve decision making on layout planning, manufacturing operation, and maintenance.

2. With AR, gestures, operations, and positions of the PEs can be captured using devices such as data gloves, cameras, and RFIDs. Through image processing, tracking, and information rendering, virtual elements including virtual models from the VEs and insights from the DD can be superimposed on top of the corresponding PEs. The combination of PEs and virtual elements can be presented to the users via display devices, such as HMDs, iPads, and projectors. Meanwhile, haptic feedback can also be obtained through data gloves. In this way, a mixed environment where the PEs, VEs, and DD are tightly coupled with each other can be built to enhance the users' perception and understanding of the physical activities in design, operation, and maintenance. Accordingly, corresponding services can be improved using an augmented display that feeds the users with precise virtual models and information as well as a familiar physical working environment synchronously.

Through introducing VR and AR, the DT can be further augmented with regard to physical–virtual fusion. The VEs and DD can be coupled in a completely immersive virtual environment with VR, while the PEs, VEs, and DD can be converged in a mixed environment with AR. Based on this, the services provided to users can be improved to support the design, manufacturing, and maintenance in a more efficient way.

11.6 DIGITAL TWIN-DRIVEN ASSEMBLY COMBINING VR AND AR

11.6.1 Digital Twin-Driven Assembly Mechanism Combining VR and AR

In the DT-driven assembly, PEs (e.g., persons, parts, tools, and machines) can be organized in an orderly manner to complete the procedures including assembly, inspection, painting, and packaging. VEs, the digital mirrors of the PEs, can operate in the virtual space to simulate assembly design, operation, and training. DD generates insights to support the assembly processes. Based on the functions and data of the PEs, VEs, and DD, Ss can provide encapsulated services to users via user-friendly interfaces for assembly design, guidance, and training.

With VR and AR, the DT-driven assembly process could achieve improvements to increase the assembly efficiency. With these

technologies, real data from the PEs (e.g., position, orientation, and motion) can be captured by devices such as gloves, cameras, handles, and RFIDs. Based on the data, the VEs in the assembly process will behave flexibly and provide multimodal feedback (e.g., visual, auditory, and tactile). In the process, VR enables the VEs to operate in an artificial and immersive environment, which is conducive to iterative and high-quality assembly planning, verification, and operator training, thus eliminating costly errors in the practical assembly process. AR can overlay graphical elements and insights from the VEs and DD on the PEs to provide just-in-time information rendering for assembly design, guidance, and training. AR helps to achieve more natural interaction and better user experience, such as manipulating virtual models in assembly via natural spoken language and operations.

11.6.2 Application of VR and AR in Assembly Based on Digital Twin

The assembly process for a complex aerospace product is taken as an example to illustrate the combination of five-dimension DT and VR and AR, as shown in [Fig. 11.2](#), which is being researched by Liu. Some key elements in the assembly process are introduced as follows:

1. *Physical assembly scene of the complex aerospace product*: It contains the real assembly machines, tools, operators, parts, and products, which are organized in an orderly fashion.
2. *Articulated arm and industrial personal computer (IPC)*: The articulated arm can scan medium or small parts to measure their geometric dimensions and tolerances. The obtained point cloud data will be transmitted to the IPC to support virtual modeling.
3. *Laser tracker and IPC*: The laser tracker can track a part and measure the part's three-dimension coordinates and key features in real-time to capture assembly details. The collected data will be stored in the IPC for virtual modeling, assembly accuracy prediction, and process control.
4. *Laser projector and IPC*: The laser projector can display outlines of the parts based on CAD data to guide assembly and support precise positioning.
5. *VR and AR devices and IPC*: Devices such as HMDs, gloves, and handles can capture data from the physical assembly scene and obtain feedback from the virtual world. The IPC can perform simulation,

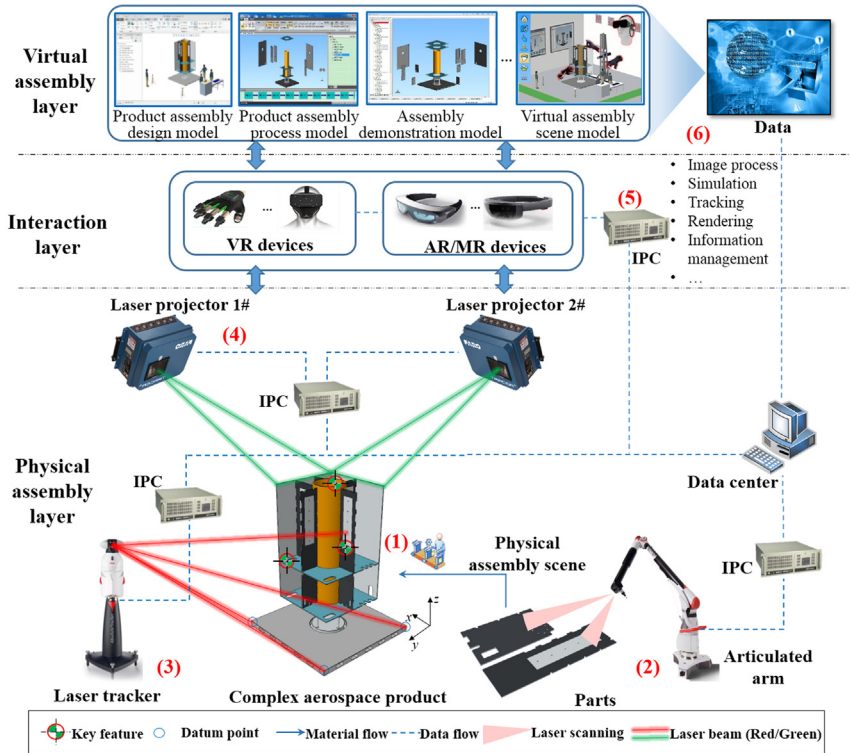


Figure 11.2 Application of VR and AR in assembly based on DT.

image processing, tracking, and information rendering to support the functions of the devices.

6. *Virtual models, data and services:* Based on the data collected from the physical assembly scene, high-fidelity virtual models can be built to describe and support the assembly process, including the assembly design model, process model, demonstration model, and scene model. Simulated data from the models and real data from the physical entities can be fused to generate valuable information for virtual model calibration and physical assembly optimization. The models and data are stored in the data center and can be invoked to support pure virtual or physical–virtual mixed assembly scenes enabled by the VR and AR technologies. Benefiting from these technologies, interactive and immersive services can be provided to assembly operators to improve the assembly efficiency.

In combination with VR and AR, the DT-based assembly process can be augmented in physical–virtual fusion. Assembly operators can be either

immersed in a completely virtual environment or in a mixed environment that integrates both the physical and virtual elements to perform assembly activities. In this way, the assembly process can be optimized through making full use of physical and virtual elements to improve the planning, guidance, and training.

11.7 SUMMARY

In this chapter, VR and AR (including MR) are applied in the DT to enhance the fusion of the physical and virtual worlds, thus further improving the efficiency of the DT. The applications of VR and AR in design, manufacturing, and service are introduced, and the differences and similarities of these technologies are compared from multiple aspects. Based on the existing and potential applications of VR and AR in the DT, a general framework is proposed to efficiently integrate these technologies into the five-dimension DT to provide users with more interactive and immersive services. Based on this framework, an assembly process is discussed. Given the benefits brought by VR and AR, combining with these technologies will be a prevalent and beneficial trend for the DT.

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CHAPTER 12

Digital Twin, Cyber—Physical System, and Internet of Things

Cyber—physical system (CPS) and digital twin (DT) share the same goal, that is, achieving seamless integration between the physical and cyber worlds. CPS provides an integrated and compatible framework with a comprehensive perspective, and DT can be considered as a focused application of it. The DT carries characteristics of CPS, such as cyber—physical mapping, closed-loop control, and three-level structure, while its elements (e.g., high-fidelity models, fused data, and on-demand services) can further enhance the cyber—physical integration based on CPS. In this chapter, according to relations between the two concepts, an implementation framework of DT-based CPS is proposed to efficiently couple the physical and cyber worlds. Further, the Internet of Things (IoT) is applied to the framework to realize the connections within and across the physical world and cyber world, thus keeping each element connected interactively with each other.

12.1 INTRODUCTION

Cyber—physical system (CPS) embeds computing, communication, and control capabilities into physical devices (e.g., cars, medical devices, and manufacturing equipment) to monitor, control, and coordinate the corresponding physical activities [1]. It couples cyber capabilities tightly with dynamics and physics of the physical space through technologies such as advanced sensors and actuators, embedded systems, computing, communication, networking, etc. The intimate coupling can be reflected in different scales, from a unit to a system and even to a system of systems (SoS) [2]. CPS is superior to the regular control or embedded system in that the combination of physical and cyber elements brings unprecedented improvements on interactivity, reconfigurability, scalability, and intelligence [3]. It is expected to yield revolutionary advances in different fields, such as health care, transportation, manufacturing, aerospace, defense, and buildings [4].

In CPS, the physical world includes physical objects that need to be controlled, while cyber systems conduct data analysis and decision making [5]. As the CPS is about the intersection, not the union, of the physical and cyber elements, interaction is therefore of great significance [6,7]. To realize the interaction, data from the physical objects can be collected using sensors in real time and transmitted to the cyber systems for processing and analysis, while real-time orders will be fed back to the actuators to regulate operations of the physical objects. CPS forms a closed loop between the physical world and cyber world, aiming at integrating the two worlds seamlessly.

Digital twin (DT) is another concept that can support cyber–physical integration. The DT creates high-fidelity virtual models for physical entities to simulate their states and behaviors and provide more insights. It keeps virtual models behaving realistically and allows physical processes to be detected and predicted efficiently, which can implement the coupled optimization for both physical and virtual objects.

Both CPS and DT aim to realize the seamless integration of the physical and cyber worlds. However, what is their relationship? How can they realize cyber–physical integration based on CPS and DT? To address these issues, this chapter examines CPS and DT from several aspects and proposes a DT-based CPS framework to tightly couple the physical and cyber worlds. As Internet of Things (IoT) technology is of key importance for connections in the DT-based CPS, the roles of IoT in the framework are also discussed.

12.2 CPS IN MANUFACTURING

12.2.1 Brief History and Concept of CPS

The term “CPS” stems from cybernetics [8], which was proposed by Wiener, an American mathematician who contributed greatly to the field of control theory. Although computer technology was not available at that time, the principles involved in Wiener’s research are similar to today’s computer-based feedback control systems [9]. Cybernetics could be considered as the combination of physical processes and capacities of computing and communication in the early days. Around 2006, the term CPS was coined by Gill at the National Science Foundation (NSF) in the United States to deeply synthesize the physical and cyber elements [10]. Since then, the NSF has awarded a large number of projects to support universities and institutes in conducting transformative research on CPS

[1]. In 2007, the US President's Council of Advisors on Science and Technology (PCAST) released a report to change the center of gravity in networking and information technology (NIT) and listed the development of CPS at the top [11]. Aware of the significance of CPS, researchers from other countries also paid substantial attention to it. In recent years, CPS has been regarded as key to supporting the smart manufacturing strategies proposed by different countries, such as Industrial Internet, Industry 4.0, Made in China 2025, cybermanufacturing, etc. Nowadays, CPS is a strong and growing area, and shows great potential in the near future.

There is no universal definition for CPS. For example, Lee described CPS as “integrations of computation with physical processes” [12]. Rajkumar defined CPS as “physical and engineered systems whose operations are monitored, coordinated, controlled, and integrated by a computing and communication core” [4]. Poovendran proposed that CPS is “a promising new class of systems that deeply embed cyber capabilities in the physical world, either on humans, infrastructure or platforms, to transform interactions with the physical world” [3]. Gunes et al. considered that CPS is “a term describing a broad range of complex, multidisciplinary, physically-aware next generation engineered system that integrates embedded computing technologies (cyber part) into the physical world” [5]. Although researchers define CPS differently, the core of these definitions is the tight coupling of cyber capabilities and physical processes.

12.2.2 CPS-Based Manufacturing

To date, CPS has attracted a great deal of attention from many different fields, especially the manufacturing field. Based on the CPS concept, cyber capabilities (i.e., computing, control, and communication) are expected to be embedded into manufacturing assets to form an intelligent distributed network, which can facilitate asset management, production optimization, and product quality control. The integration enabled by CPS is important for the implementation of smart manufacturing. The related research about "CPS-based manufacturing" has been explored in the authors' previous work [13], which is introduced as follows.

Monostori [14] introduced cyber—physical production systems (CPPS) into manufacturing to change the traditional hierarchical structure, including five levels from field to enterprise resource planning, to a distributed network for a more decentralized way of functioning. Setty et al. [15]

proposed a unified framework to design and deploy a CPS-based manufacturing system composed of networked embedded controllers and mechatronic devices, which provides a smart automation solution for production. Weyer et al. [16] presented a three-tier framework for seamless connection and interaction between CPS models and multidisciplinary simulation along the production lifecycle. The framework was validated in automotive production. Lee et al. [17] proposed a 5C architecture for the implementation of CPS and explored related applications and techniques. Based on the architecture, a prognostics and health management (PHM) approach for a machine tool was proposed to perform analytics at the cyber level and conduct just-in-time maintenance. Iarovyi et al. [18] created an open knowledge-driven manufacturing execution system (MES) based on CPS. As the shop-floor devices have increasing computing and communication capabilities, they can provide enhanced functions in the forms of data, services, and descriptions, so that higher-level knowledge can be obtained to support the MES. Leitão et al. [19] described four prototype implementations for industrial automation based on CPS and discussed the challenges. Besides, cloud technology is also applied to the CPS to provide powerful computing capability and on-demand services. For example, Wang et al. [20] designed an integrated CPS architecture including control tier, model tier, and view tier. Functionalities for remote monitoring and control, process planning, and remote assembly are supported by data, knowledge, and models stored in servers in the cloud and related services are provided to users through web browsers. Yue et al. [21] presented a cloud-based industrial CPS framework that took all infrastructures, platforms, and applications as services. Tao and Qi [22] proposed a service-oriented smart manufacturing (SoSM) framework integrating with the cloud, big data, IoT, and services based on CPS architecture.

In existing research on CPS-based manufacturing, there are many discussions on its concepts and frameworks, while few implementations have been reported. This is probably for two main reasons: (1) since manufacturing has high requirements on near real-time response, precision, and safety, applications of CPS-based manufacturing face enormous challenges and more technical details still need to be explored; and (2) current CPS is mainly conceptual aiming at inspiring new scientific foundations and technologies, rather than practical solutions to directly guide the development of engineered systems.

12.3 IOT IN MANUFACTURING

12.3.1 Brief History and Concept of IoT

IoT is an important driver for CPS. It can connect physical and cyber elements together to form a hybrid communication network where elements interact with each other in real-time, thus enabling the direct integration of the physical and cyber worlds. The brief history and concept of IoT have been studied by authors in their work [23], and the study is introduced as follows.

The origin of IoT can be traced back several decades ago. As early as 1982, students in Carnegie–Mellon’s Computer Science Department connected a Coca Cola vending machine to the Internet, which allowed customers to check the status of the drinks remotely [24]. It was the earliest prototype of the IoT system. In 1995, Gates mentioned that things and people could connect with each other in the future in the book *The Road Ahead* [25]. In 1999, the term IoT was originally proposed by Ashton at the Auto-ID center at the Massachusetts Institute of Technology (MIT). The center proposed to offer a unique e-tag to each product and connect the product to the Internet [23]. From then on, the IoT concept gradually came into view. In 2003, Walmart asked its 100 top suppliers to attach radio frequency identification (RFID) chips on products to support efficient supply chain management [26]. Having recognized the benefits brought by IoT technology, some reputable companies, for example, IBM, Tesco, Microsoft, and Philips, began to apply RFID tags in their businesses [27]. In 2005, the International Telecommunication Union (ITU) released the report “ITU Internet Reports 2005: The Internet of Things” [28] to introduce the characteristics, technologies, and challenges of IoT. In 2008, IBM launched “Smarter Planet” based on IoT to build smarter systems for health care, the power grid, water, public safety, traffic, and food [29]. Nowadays, IoT as global infrastructure for the information society has been widely introduced to logistics, manufacturing, health care, agriculture, smart cities, etc.

In the academic community, IoT has been defined from different perspectives and numerous definitions for IoT exist in the literature. These definitions have come along with the development of IoT. Ashton considered IoT as a technology mainly based on RFID to realize the interconnection of objects [23]. ITU proposed that IoT allows new forms of communication among people and things at any time and any place by

embedding short-distance mobile transceivers into parts and daily objects [28]. The European Technology Platform on Smart Systems Integration (EPoSS) defined IoT as “things having identities and virtual personalities operating in smart spaces using intelligent interfaces to connect and communicate within social, environmental, and user contexts” [30]. Up to now, although there is no unified definition for IoT, existing definitions generally focus on both object identification and communication. For example, Miragliotta et al. [31] illustrated that IoT is a paradigm realizing the interplay of smart objects and of smart communication networks. Atzori et al. [32] defined IoT as a pervasive presence of a variety of things or objects, such as RFID tags, sensors, actuators, and mobile phones, which are able to interact with each other through unique addressing schemes.

12.3.2 Applications of IoT Toward Smart Manufacturing

The authors have investigated the applications of IoT towards smart manufacturing in their work [33]. And the related investigations are introduced in this section. As IoT could provide a promising opportunity to build powerful manufacturing systems and applications by leveraging RFID, wireless and mobile sensory devices, embedded objects, ad hoc networks, and Internet-based information infrastructure, it has been applied to different fields toward smart manufacturing.

For example, in the field of aerospace industry, the application of IoT can effectively improve product safety and reliability by identifying the faulty and counterfeit parts or products. An investigation published by US Aviation declared that at least 28% of air accidents or incidents are caused by counterfeit products [34]. After employing IoT technology to identify and single out the subquality parts/products, the safety and operational reliability of aircrafts have been improved significantly. In the automotive industry, IoT is widely used in the production line, product quality monitor and control, assembly line, logistics, and product (or part) tracking. Among these procedures, intelligent labels are pasted on the components of every object, carrying the associated attribute information such as the manufacturer's name, serial number, product type, product code, as well as the exact location of the product [35,36]. The application of IoT can also help integrate different communication technologies used in manufacturing systems, such as global systems for mobile communications (GSM), near-field communication (NFC), low-power Bluetooth, ZigBee,

wireless local area networks (LANs), multilevel networks, and sensor network technology [37]. In addition, Xu et al. [38–41] have reviewed the advances of IoT in enterprise systems and industries. He et al. [42] have researched the application of IoT in the development of vehicular data cloud service. Tao et al. [33,43,44] explored methods to apply IoT to manufacturing resource intelligent perception and access to support energy consumption evaluation and analysis. They also reviewed the applications of IoT in product lifecycle energy management and discussed the potential applications and challenges [23].

With respect to IoT in manufacturing, much effort still needs to be made in the future, including design, production, and deployment of specific perception devices, equipment, and sensors; standards, protocols, safety and security, reliability and management technologies of applying IoT; and effective and reliable technologies and devices for mining and processing the data collected using IoT, etc. Nevertheless, IoT has shown great potential in device communication and networking, connections, and data collection, which has laid important foundations for cyber–physical integration.

12.4 DIGITAL TWIN AND CPS

12.4.1 Digital Twin and CPS

CPS involves a holistic and compatible architecture that can contain different technologies and spark innovations, while DT can be merged into the architecture and considered as a focused application of the CPS. Characteristics of CPS (e.g., cyber–physical mapping, closed-loop control, and three-level structure) can also be reflected in the DT.

1. *Cyber–physical mapping.* In CPS, each physical entity is linked with a cyber unit that is capable of computing, communication, and control, and deployed in different environments such as an embedded system or a cloud platform. As the cyber unit could include various algorithms, models, and rules to describe and analyze states and behaviors of the physical counterpart, a cyber–physical mapping exists between them. With respect to the DT, it emphasizes creating a high-fidelity virtual model for each physical entity to provide deeper insights. The model can operate synchronously with the corresponding physical one and perform judgment, analysis, evaluation, and prediction, thus forming an intuitive mapping to the physical entity.

2. *Closed-loop control.* For CPS, its essence is to construct a closed-loop control between the physical and cyber worlds to monitor and coordinate the physical processes efficiently. Through using multiple sensors mounted on physical devices and in the environment, large-scale distributed data acquisition can be enabled. After data processing and analysis, control orders can be generated and fed back to the actuators for execution. For the DT, the high-fidelity models in the cyber world can provide intuitive evaluation and prediction, then the fused data can capture comprehensive information to support more accurate decision making, and finally services will invoke functions of the DT based on demands to control corresponding physical processes.
3. *Three-level structure.* CPSs can be classified into three levels, including the unit level (e.g., a single machine), system level (e.g., a production line), and SoS level (e.g., a shop-floor) [2]. The front level is the implementation foundation for the latter. At each level, the CPS is deployed with sensors, actuators, and embedded systems and supports communication, computing, and control. DTs also include the three levels as stated in Chapter 8, Digital Twin and Cloud, Fog, Edge Computing. For DT in each level, besides physical entities with sensors, actuators, and embedded systems, it also requires high-fidelity virtual models, fused data, on-demand services as well as connections among these elements. The DTs attach great importance on models and data, and are easy to use.

DT carries the characteristics of CPS, and DT elements (e.g., high-fidelity models, fused data, and on-demand services) also make the DT have its own advantages. Combining with these elements, the CPS can be implemented in a more intuitive and efficient way. Therefore, in the next section, an implementation framework of DT-based CPS is proposed.

12.4.2 Implementation of Digital Twin-Based CPS

As shown in Fig. 12.1, in the CPS framework, a closed loop can be formed between the physical and cyber worlds through data collection, management and analysis, decision making, and control order execution. Elements of DT such as the physical entities (PEs), virtual entities (VEs), services (Ss), and DT data (DD) are merged with the CPS framework and enhance the functions of CPS, especially on the cyber side.

The physical world mainly consists of PEs, which refer to the physical entities with sensory devices, RFID tags, embedded systems, and

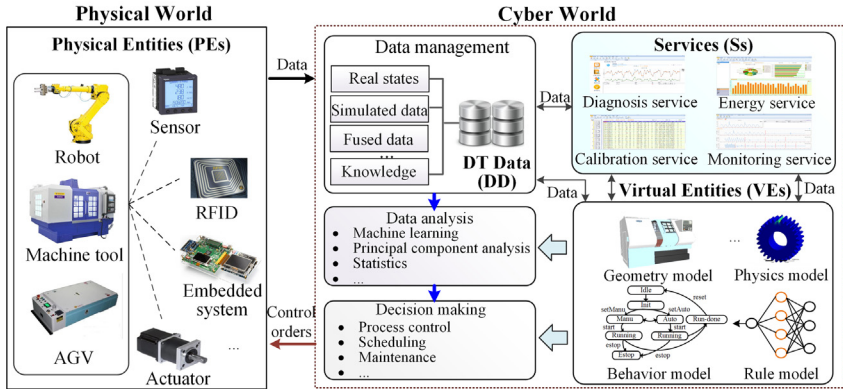


Figure 12.1 Implementation of DT-based CPS.

actuators. The PEs can be connected with each other to form a distributed network. Data from the network can be collected and transmitted to the cyber side for analysis, while orders will be fed back to control and coordinate the operations of the PEs.

In the cyber world, the main aim of the DT is to create virtual mirrors (i.e., VEs) to simulate states and behaviors of the corresponding PEs. The VEs operate synchronously with the physical counterparts, record real-time data, and generate a large amount of simulated data on the geometric and physical properties, behaviors, and rules. With VEs, insights into the PEs can be obtained without disturbing the normal operations, and data from different stages (e.g., design, manufacturing, and maintenance) can be integrated. Simulated data from VEs and real data from PEs are stored in DD and managed by the database tools. Besides, the DD also includes data from services and domain knowledge. These data are fused and analyzed with different algorithms, such as machine learning, principal component analysis, and statistics. Combining with data from multiple sources, data analysis can be performed to capture more accurate and comprehensive information. Based on the analysis results, decision making will be made for process control, scheduling, maintenance, etc. Before actual execution, VEs can be used to validate the decisions in an intuitive way and provide optimization advice through simulation to avoid costly mistakes. Finally, control orders can be generated and provided to the physical world to guide the operations of PEs. Services supported by DD and VEs can be deployed on computers, tablets, and mobile phones, while encapsulating related functions into standard forms with user-friendly interfaces.

By using the services, operators can monitor the physical processes in real-time, obtain relevant engineering information and knowledge, and make necessary judgments through manual intervention.

With the DT, CPS can be enhanced using realistic simulation, fused data, and services, which could bring efficiency improvements and economic benefits. In the DT-based CPS framework, connections exist within the physical world and cyber world, and even across the two worlds. Realization of the ubiquitous connections for the framework is a key issue. In the following section, IoT is introduced to address this.

12.5 IOT IN DIGITAL TWIN-BASED CPS

As shown in [Fig. 12.2](#), IoT connects PEs in the physical world, and VEs, DD, and Ss in the cyber world, while coupling the physical and cyber elements tightly.

In the physical world, sensors and actuators deployed on different PEs are connected. Data from machine tools, robots, materials, and the environment can be collected in real-time by sensory devices, such as RFID tags, temperature and humidity sensors, electric current meters, flow meters, cameras, etc. As the data are transmitted with different communication protocols (e.g., RS232/485, ZigBee, Wi-Fi, and Bluetooth), data exchange is difficult. To address this problem, gateway devices are deployed to analyze the communication protocols, transform the data into standard encapsulation, and support standard data exchange. Besides, gateway devices such as Industrial IoT Hub (II-hub) developed by Tao et al. [\[45\]](#) can also perform data computing, so that conditions of different PEs can be converged and analyzed in real-time, and the corresponding control orders will be generated according to the analysis results. The orders can be transmitted to computers, tablets, or phones through Wi-Fi, thus guiding operators to carry out adjustments on the PEs, or be directly transmitted to the actuators of the equipment to control their operations. For example, according to the collected machining speeds of different machine tools, the transport time, frequency, and path of automated guided vehicles (AGVs) can be regulated accordingly.

However, as data processing and analysis performed in the physical world are relatively limited, large-scale computing and simulation still need to be carried out in the cloud environment deployed in the cyber world. In this world, cyber elements of DT (i.e., DD, VEs, and Ss) exist and connect with each other. VEs operate in professional software such as

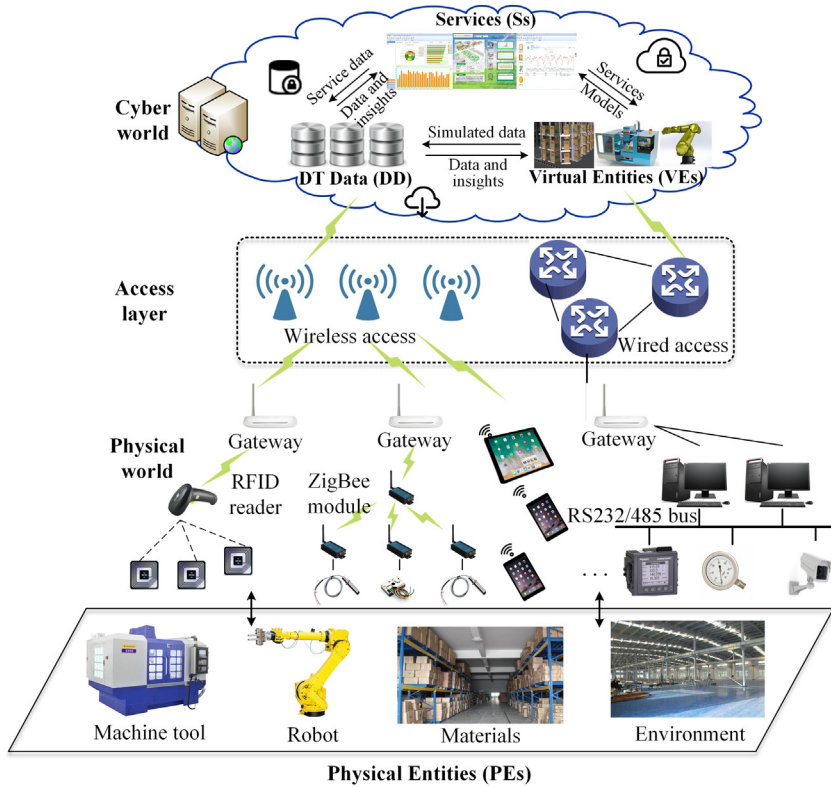


Figure 12.2 IoT in DT-based CPS.

Auto CAD, ADAMS, and MATLAB, and S_s mainly consists of application software developed using standard programming languages such as Java and C. The two elements can build connections with databases and excel spreadsheets in DD through ready-made encapsulated interfaces, such as Open Database Connectivity (ODBC). Based on the connections, data in DD can be read to support the operations of VEs and S_s , while simulated data and service data can be injected into DD continuously. Connections between VEs and S_s can be implemented in different ways, such as building shared documents or realizing bidirectional communication via socket technology. Through the connections, VEs can be provided to S_s to make services more intuitive and interactive, and services in S_s can be invoked to evaluate and calibrate the VEs .

Connections between the physical and cyber worlds enable the closed-loop control. Through conversion in the gateway devices, real-time data from PEs are transformed into a unified format and transmitted

to the cloud through wired (e.g., Ethernet) or wireless (e.g., 2G, 3G, and 4G) access modes. The data can be stored in DD and fused with data from VEs and Ss. Based on DD, VEs will perform accurate analysis and prediction of the physical counterparts, and Ss can provide encapsulated services. The simulation results and services can be presented to operators in the physical world through human–machine interfaces or transformed into control orders and then delivered to actuators of the physical equipment directly.

With IoT technology, physical and cyber elements in DT-based CPS form a hybrid communication network that enables the elements to connect and exchange data with each other. In the network, interactions within the physical and cyber worlds are enabled, which allows the physical world to be monitored and controlled by the computer-based cyber systems.

12.6 SUMMARY

DT can be considered as a focused application of CPS. On one side, the DT inherits characteristics of CPS such as cyber–physical mapping and closed-loop control to make the physical and cyber worlds interactive and integrated. On the other side, elements of DT can be combined with the CPS framework to support the high-fidelity simulation, data fusion, and service invocation, thus providing the CPS with more efficient data analysis and decision making. In this chapter, the DT and CPS are viewed from several aspects and a framework of DT-based CPS is proposed. IoT technology is applied to the framework to realize the ubiquitous connections within and across the physical and cyber worlds.

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Digital Twin Driven Smart Manufacturing

Fei Tao, Meng Zhang, A.Y.C. Nee

This book provides the ideal entry point to this subject for readers in industry and academia, answering questions such as “What is a digital twin, and how is it constructed?” “How can a digital twin be used to improve manufacturing efficiency?” “What are the essential activities in the implementation of a digital twin?” “What are the most important obstacles to overcome for the successful deployment of a digital twin?” “What are the relations between digital twin and new technologies?” and “How to combine digital twin with the new technologies to achieve high efficiency and smartness in manufacturing?”

Following an introduction to the background, concept and applications of the digital twin, initial chapters discuss the five-dimensional digital twin modeling and its key technologies, digital twin shop-floor, the equipment energy consumption management involved, and cyber-physical fusion in digital twin shop-floor. The book then proceeds with the relationships between digital twin and cloud, fog and edge computing, big data, services, virtual and augmented/mixed reality, as well as cyber-physical system, concluding with how an Internet of Things is applied to the physical and cyber worlds.

Digital Twin-Driven Smart Manufacturing enables both engineers and academics with an interest in smart manufacturing to make the best use of digital twin technology, and analyze differences, synergies, and possibilities for integration between digital twin technology and other technologies such as Internet of Things and big data.

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